Oracle Fusion Middleware 12*c*: Multitenancy Administration

Activity Guide
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Author

Tom Eliason

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Course Practice Environment: Security Credentials

For OS usernames and passwords, see the following:

- If you are attending a classroom-based or live virtual class, ask your instructor or LVC producer for OS credential information.
- If you are using a self-study format, refer to the communication that you received from Oracle University for this course.

For product-specific credentials used in this course, see the following table:

Product-Specific Credentials		
Product/Application	**Username**	**Password**
Oracle WebLogic Server	weblogic	Welcome1
Oracle WebLogic Server – bayland Security Realm	administrator	bayland123
Oracle WebLogic Server – valley Security Ream	administrator	valley123
Oracle WebLogic Server	wlstadmin	Welcome1
Oracle Database	system	welcome1
Oracle Database	oracle	welcome1
Oracle Database	sys	welcome1
Oracle Database	MEDREC	MEDREC
Oracle Database	sys as sysdba	welcome1
Oracle Database	FULL_STB	welcome1
Oracle Database	FULL	welcome1
Oracle Database	Administrative User	welcome1
Oracle Database	trade	trade

Practices for Lesson 1: Course Overview

Chapter 1

Practices for Lesson 1

There are no practices for this lesson.

Practices for Lesson 2: Multitenancy Concepts

Chapter 2

Practices for Lesson 2

There are no practices for this lesson.

Practices for Lesson 3 - Domain Partitions

Chapter 3

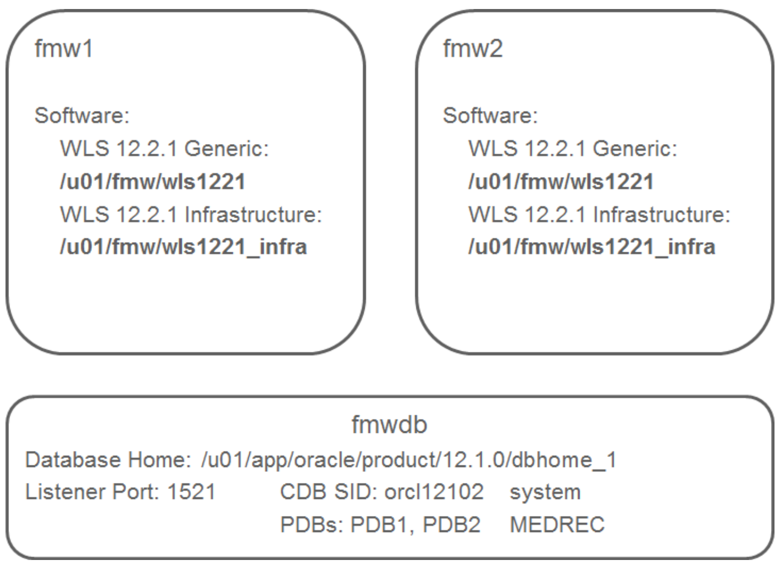
Practices for Lesson 3: Overview

Overview

In these practices, you create WebLogic Server domain partitions for two scenarios:

- Single Application Domain Partition using the WebLogic Server Administration Console
  - A scenario where an application is deployed in a domain partition and uses a separate security realm from the WebLogic Server domain
  - Also, exporting and importing the domain partition with a single application is useful in development-to-test and test-to-production deployment scenarios
- Multiple Applications in a domain partition using Fusion Middleware Control
  - This scenario deploys the same application (the MedRec sample) using different partitions and resources, such as JDBC Data Sources, to a WebLogic Server domain

The practice environment is configured with two fusion middleware hosts and one database host:



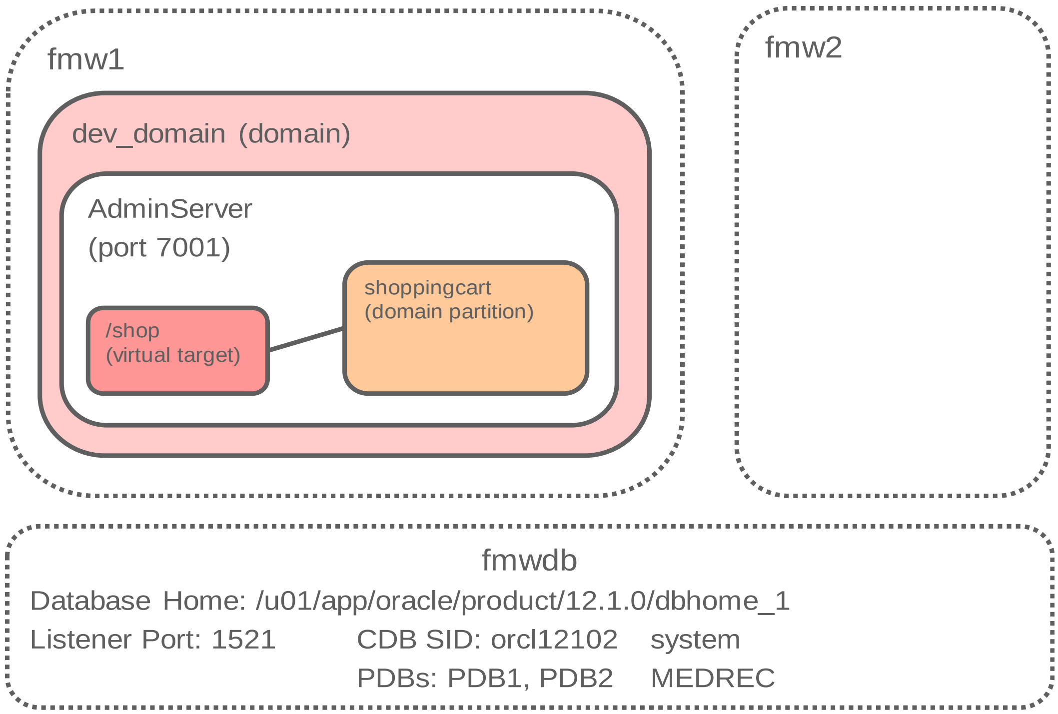
Practice 3-1: Creating a Single Application Domain Partition

Overview

In this practice, using the WebLogic Server Administration Console, you create a new WebLogic Server domain, virtual target, and domain partition for the MedRec sample application. You also create a new security realm for the domain partition and deploy the sample application. Finally, you test the sample application to ensure it functions as expected.

Because many WebLogic Server customers have requirements for this type of configuration—a Java application deployed to a single partition in WebLogic Server—this is a useful practice to execute the steps to create and use this configuration.

The diagram below illustrates the configuration:



Assumptions

- WebLogic Server 12.2.1 generic is installed in `/u01/fmw/wls1221`.

Tasks

1. Connect to fmw1 as the oracle user.
   a. Refer to the instructions provided by your instructor to connect to your machine.
   b. Unless stated otherwise, you will be working within fmw1 for the remainder of this practice.
2. Create the WebLogic Server Domain.
   a. Open a Terminal window as the oracle user, and navigate to `/u01/fmw/wls1221/`.

   Tip: There is a launcher for a Terminal window in the panel at the top of the desktop.

   ```
   $ cd /u01/fmw/wls1221/
   ```

   b. Execute the Fusion Middleware Configuration Wizard.

   ```
   $ oracle_common/common/bin/config.sh
   ```

3. After the command, the graphical Fusion Middleware Configuration Wizard appears. Use the guidelines in the following table to configure the WebLogic Server Domain:

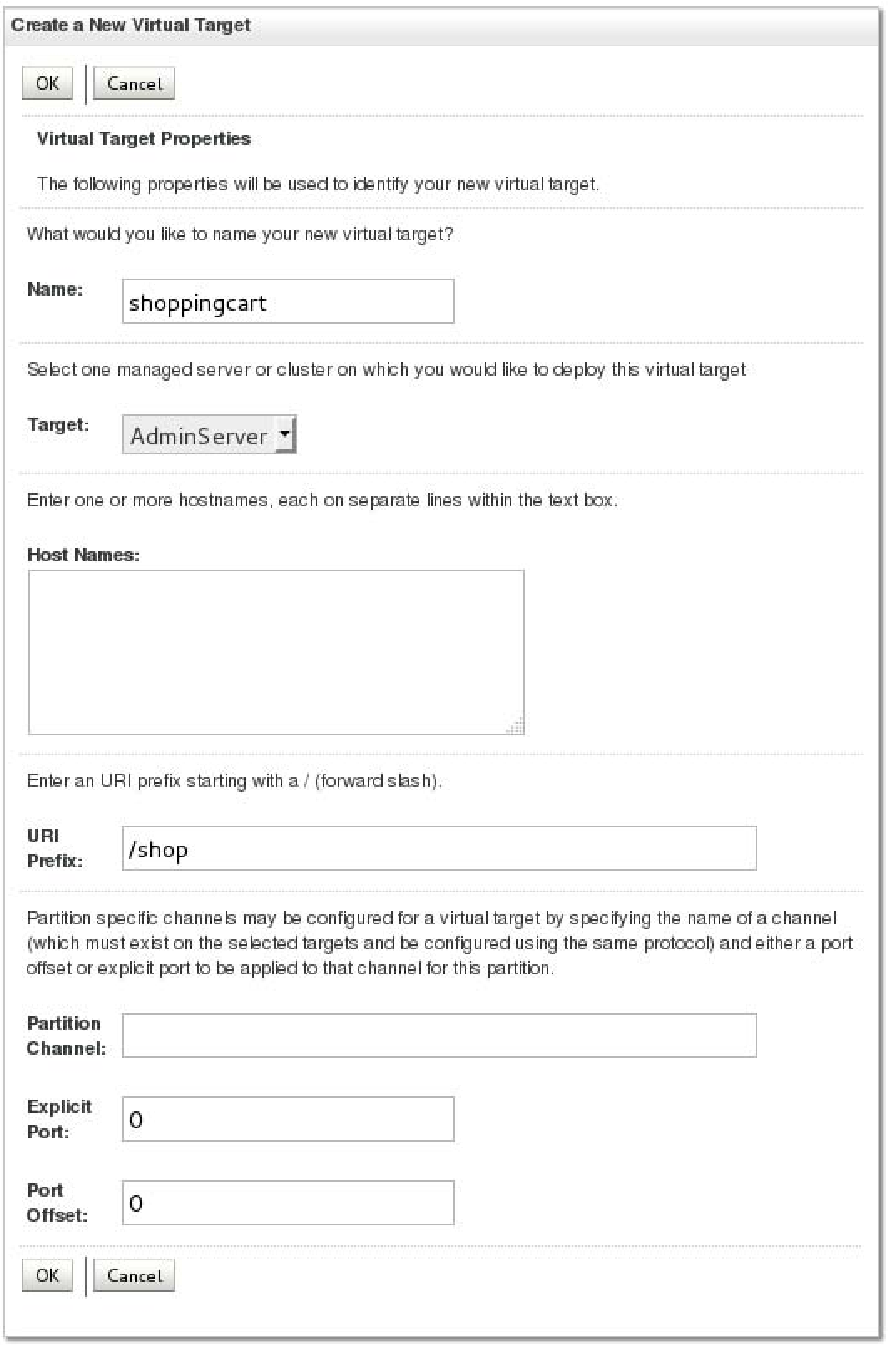
Step	Window/Page Description	Choices or Values
a.	Page 1 of 8 - Create Domain	Click **Create a new domain.** For **Domain Location,** enter `/u01/fmw/wls1221/user_projects/domains/dev_domain` and click **Next**.
b.	Page 2 of 8 - Templates	The Basic Template is preselected: • Basic WebLogic Server Domain – 12.2.1 [wlserver] Click **Next**.
c.	Page 3 of 8 – Administrator Account	For **Name,** enter `weblogic`. For **Password,** enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. For **Confirm Password,** enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click **Next**.
d.	Page 4 of 8 – Domain Mode and JDK	For **Domain Mode** select **Development,** and click **Next**.
e.	Page 5 of 8 – Advanced Configuration	Click **Next**.
f.	Page 6 of 8 – Configuration Summary	Click **Create**.
g.	Page 7 of 8 – Configuration Progress	Click **Next**.
h.	Page 8 of 8 – End of Configuration	Click **Finish**.

4. Log in to the WebLogic Server domain using the WebLogic Server Administration Console.
   a. In the Terminal window, navigate to `/u01/fmw/wls1221/user_projects/domains/dev_domain`
   ```
   $ cd /u01/fmw/wls1221/user_projects/domains/dev_domain/
   ```
   b. Execute `startWebLogic.sh` to start the WebLogic Administration Server.
   ```
   $ ./startWebLogic.sh
   ```
   c. Open a Browser window and navigate to **http://fmw1:7001/console.**

d. For **User Name,** enter `weblogic` and for **Password,** enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click **Login.**

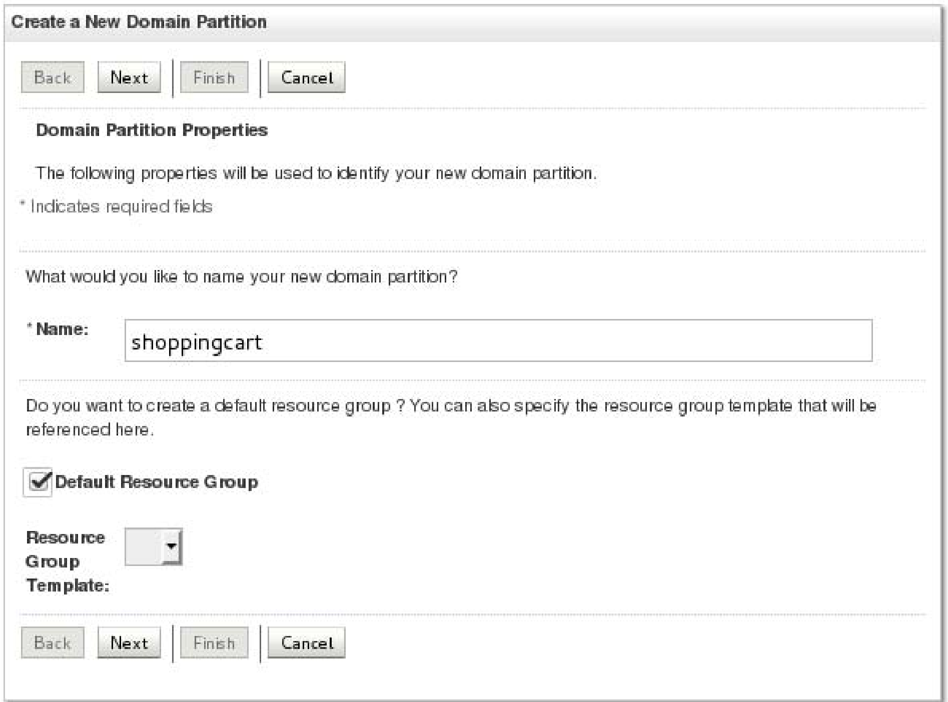
   The WebLogic Server Administration Console home page for **dev_domain is** displayed.

5. Create a Virtual Target for shoppingcart.
   a. From the **Domain Structure** panel menu, click **Environment** and then click **Virtual Targets.**
   b. Click **New.**
   c. For **Name,** enter `shoppingcart` and for **URI Prefix** enter /`shop`.



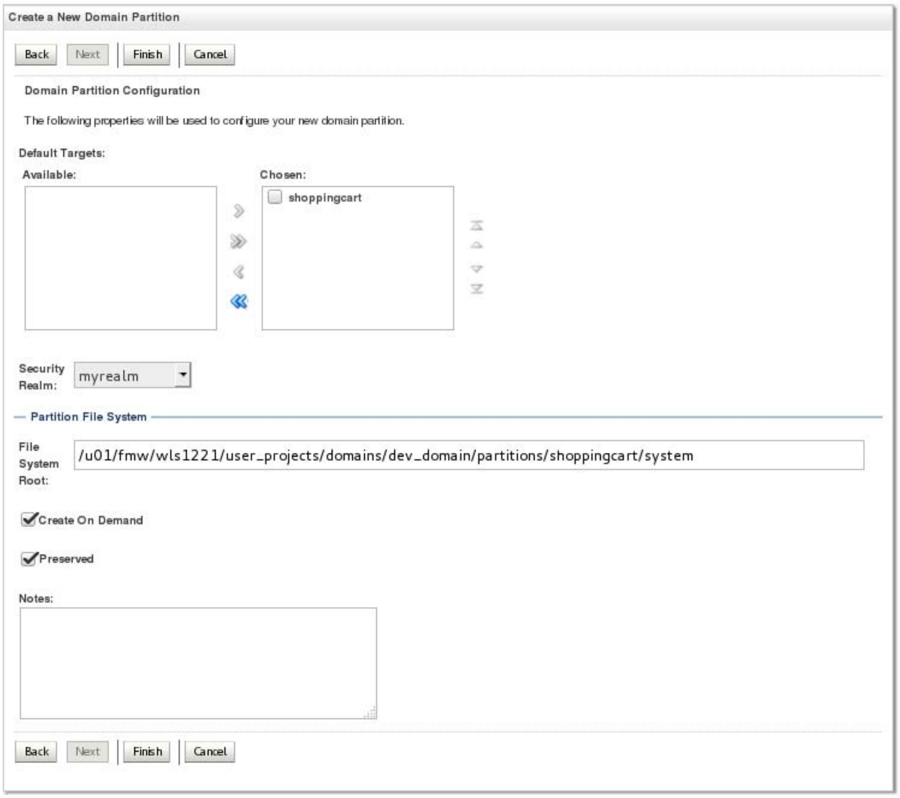
d. For **Target,** select AdminServer and Click **OK.**
   The Virtual Target **shoppingcart** is created successfully.

6. Create a domain partition for shoppingcart.
   a. In the Browser window, from the **Domain Structure** panel, click **Domain Partitions.**
   b. Click **New.**
   c. For **Name,** enter `shoppingcart` and click **Next.**



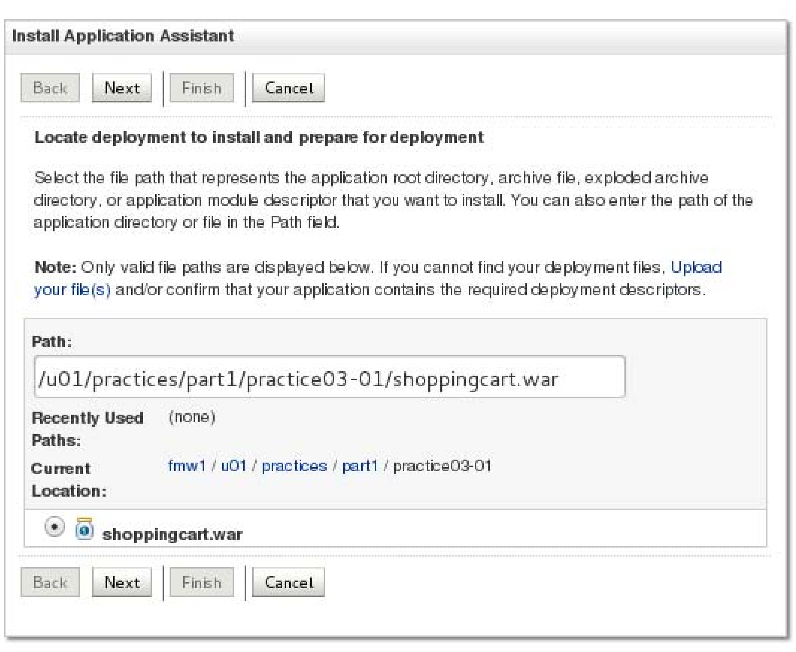
   d. Select the check box for the **Virtual Target shoppingcart**. Click **Next.**

   e. In the **Available Targets** page, select the **shoppingcart** target from **Available** to **Chosen.**
   f. From the **Security Realm** drop-down menu, select **myrealm.** Click **Finish.**

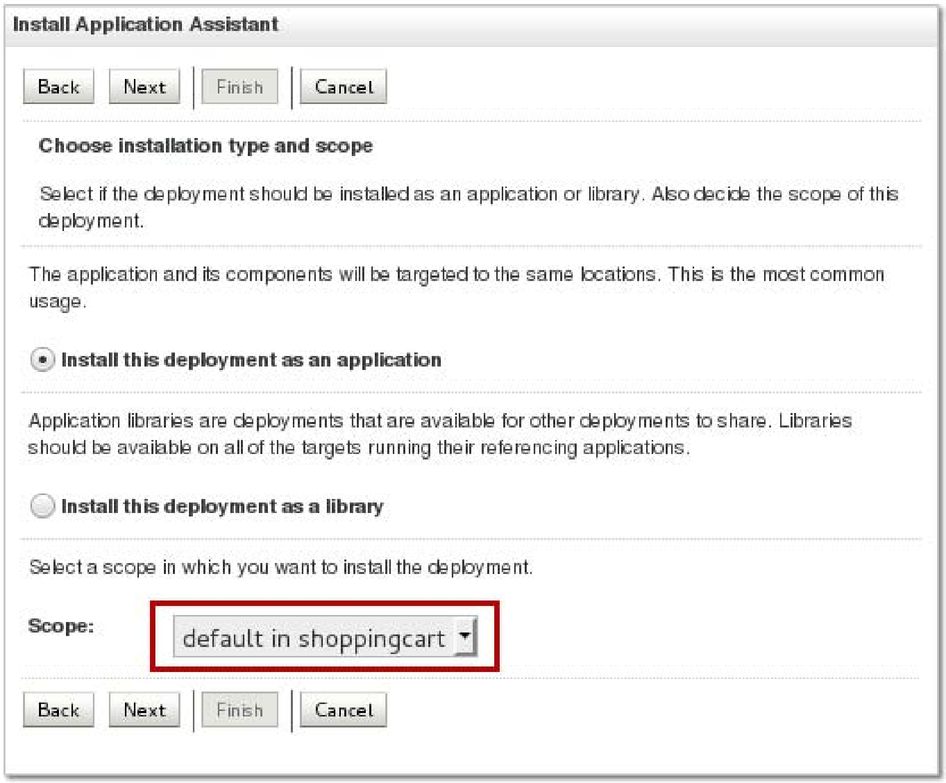


The Domain Partition **shoppingcart** is created successfully.

7. Deploy applications to the shoppingcart domain partition.
   a. From the **Domain Structure** pane, click **Deployments.**
   b. Click **Install.**
   c. In the **Locate deployment to install and prepare for deployment** step, navigate to the `/u01/practices/part1/practice03-01/` directory, select the `shoppingcart.war` file and click **Next.**



d. In the **Choose installation type and scope** step, click **Install this deployment as an application,** select **default in shoppingcart** for **Scope** and click **Next.**



e. Accept the defaults for Optional Settings and click **Next.**

f. Review the deployment configuration and click **Finish.**

8. Start the shoppingcart domain partition.

   a. From the **Domain Structure** pane, click **Domain Partitions.**

   b. Click the **Control** tab.

   c. Select the check box for the shoppingcart domain partition and click **Start.**

   d. Click the refresh icon until the domain partition state changes to RUNNING.

9. Test the shoppingcart Virtual Target.

   a. Open a new Browser tab, navigate to **http://fmw1:7001/shop/shoppingcart.**

   b. Click Browse the Store, Go Shopping or View Shopping Cart.

The shopping cart application has been deployed to a domain partition. The next practice will export this domain partition and perform a domain partition import.

Practice Solution: Creating a Single Application Domain Partition

Perform the following tasks if you did not complete this practice and want to use the finished solution.

> **Note:** If you started creating the dev_domain, this script will not work. To run the script in that case: delete the `/u01/fmw/wls1221/user_projects/domains/dev_domain` directory.

Assumptions

- WebLogic Server 12.2.1 generic is installed in `/u01/fmw/wls1221`.

Solution Tasks

1. Connect to fmw1 as the oracle user.
2. Run the solution script.
   a. Open a Terminal window as the `oracle` user and navigate to the practice directory.
   ```
   $ cd /u01/practices/part1/practice03-01
   ```
   b. Run the solution script.
   ```
   $ ./solution.sh
   ```
   c. Enter any passwords as you are prompted.

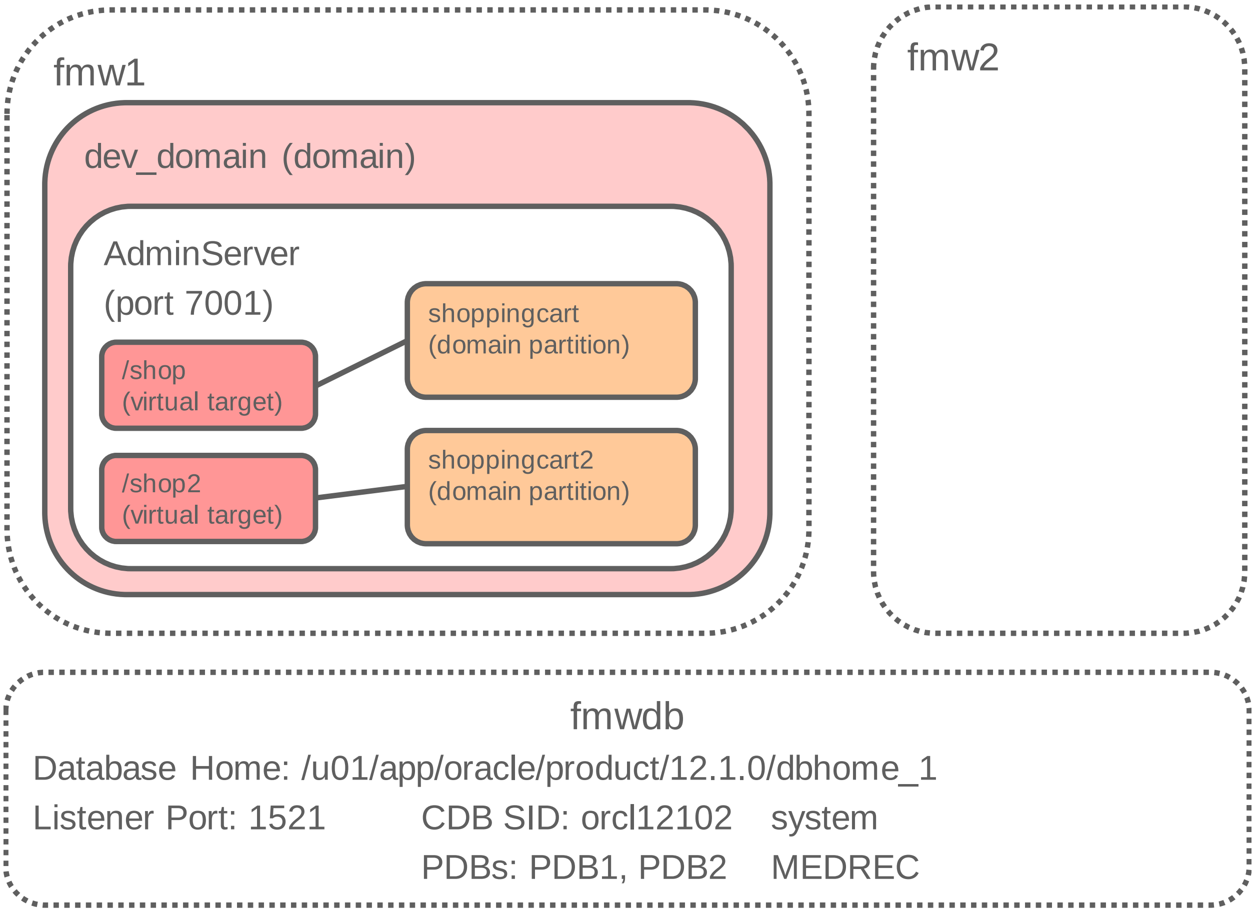
   Note: This script starts the WebLogic Scripting Tool (WLST) passing it the WLST script `create_domain.py`, which creates the domain, partition and starts the partition for you. The WLST script prints out messages. Be patient and do not close the Terminal window until you see these messages:
   ```
   ...
   >>>Practice 03-01 solution applied

   Exiting WebLogic Scripting Tool.
   ```
3. Close the Terminal window.

Practice 3-2: Exporting and Importing a Single Application Domain Partition

Overview

In this practice, you export the domain partition created in the previous practice, modify the domain partition archive attributes and then import the domain partition archive into a new domain partition.

The following diagram illustrates the configuration:



Assumptions

- WebLogic Server 12.2.1 generic is installed in `/u01/fmw/wls1221`.
- The dev_domain administration server is running.
- The shoppingcart domain partition has been created, started, and tested successfully.

Tasks

1. Log in to the WebLogic Server Domain using the WebLogic Server Administration Console.
   a. In the Browser window, navigate to **http://fmw1:7001/console.**
   b. For **User Name,** enter `weblogic` and for **Password,** enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click **Login.**

      The WebLogic Server Administration Console home page for **dev_domain** is displayed.

2. Export the shoppingcart domain partition.
   a. In the **Domain Structure** pane, click **Domain Partitions.**
   b. Select the check box next to the **shoppingcart** domain partition and click **Export.**
   c. Select **Include Application Bits.**
   d. For **Path,** enter `/u01/fmw/wls1221/user_projects` and click **OK.**
3. Create a virtual target named shoppingcart2 for the new partition.
   a. In the **Domain Structure** pane, click **Environment** and then click **Virtual Targets.**
   b. Click **New.**
   c. For **Name,** enter `shoppingcart2` and for **URI Prefix** enter /`shop2`.
   d. For Target, select AdminServer and Click **OK.**
      The Virtual Target **shoppingcart2** is created successfully.
4. Edit the export file to use the new Virtual Target avitekclinic2.
   a. Open a Terminal window on host `fmw1` as the `oracle` user, and navigate to `/u01/fmw/wls1221/user_projects`.
      ```
      $ cd /u01/fmw/wls1221/user_projects
      ```
   b. Edit the `shoppingcart-attributes.json` file.
      ```
      $ vi shoppingcart-attributes.json
      ```
   c. Set the default-target to shoppingcart2:
      ```
      "default-target": [{"virtual-target": {"name": "
      shoppingcart2"}}],
      ```
   d. Set the available-target to shoppingcart2:
      ```
      "available-target": [{"virtual-target": {"name": "
      shoppingcart2"}}]
      ```
   e. Save and close the `shoppingcart -attributes.json` file:
      ```
      :wq
      ```
5. Import the domain partition.
   a. In the **Domain Structure** pane, click **Domain Partitions.**
   b. Click **Import.**
   c. In the **Import a Domain Partition** page, select **Create New,** enter `shoppingcart2` for **Domain Partition Name,** enter `/u01/fmw/wls1221/user_projects` for **Path,** and select the **shoppingcart.zip** file. Click **OK.**
6. Start the imported shoppingcart2 Domain Partition.
   a. From the **Domain Structure** pane, click **Domain Partitions.**
   b. Click the **Control** tab.
   c. Select the check box for the shoppingcart2 domain partition and click **Start.**
   d. Click the refresh icon until the domain partition state changes to **RUNNING.**
7. Test the shoppingcart2 Virtual Target.
   a. Open a new Browser tab, navigate to **http://fmw1:7001/shop2/shoppingcart.**
   b. Click Browse the Store, Go Shopping or View Shopping Cart.

8. Stop the AdminServer for dev_domain.
   a. In Browser tab, navigate to **http://fmw1:7001/console.**
   b. For **User Name,** enter `weblogic` and for **Password,** enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click **Login.**

      The WebLogic Server Administration Console home page for the **dev_domain** is displayed.
   c. In the WebLogic Server Administration Console, from the **WebLogic Domain Structure** panel, click **Environment** and click **Servers.**
   d. Click the **Control** tab.
   e. Select the **AdminServer.**
   f. Click **Shutdown** and select **Force Shutdown Now.**

Practice Solution: Exporting and Importing a Single Application Domain Partition

Perform the following tasks if you did not complete this practice and want to use the finished solution.

> **Note:** If you started creating the shoppingcart2 domain partition or virtual target this script will not work. To run the script in that case: delete the shoppingcart2 domain partition and virtual target.

Assumptions

- WebLogic Server 12c (12.2.1) generic is installed in /`u01/fmw/wls1221`.
- The dev_domain from the previous practice has been created.
- The AdminServer from dev_domain is running.
- The avitekclinic domain partition has been created, started, and tested successfully.

Solution Tasks

1. Connect to fmw1 as the `oracle` user.
2. Run the solution script.
   a. Open a Terminal window as the `oracle` user and navigate to the practice directory.
      `$ `**`cd /u01/practices/part1/practice03-02`**
   b. Run the solution script.
      `$ `**`./solution.sh`**
   c. Enter any passwords as you are prompted.

   Note: This script starts the WebLogic Scripting Tool (WLST) passing it the WLST script `export_import.py`, which creates the new generic data source for you. The WLST script prints out messages. Be patient and do not close the Terminal window until you see these messages:

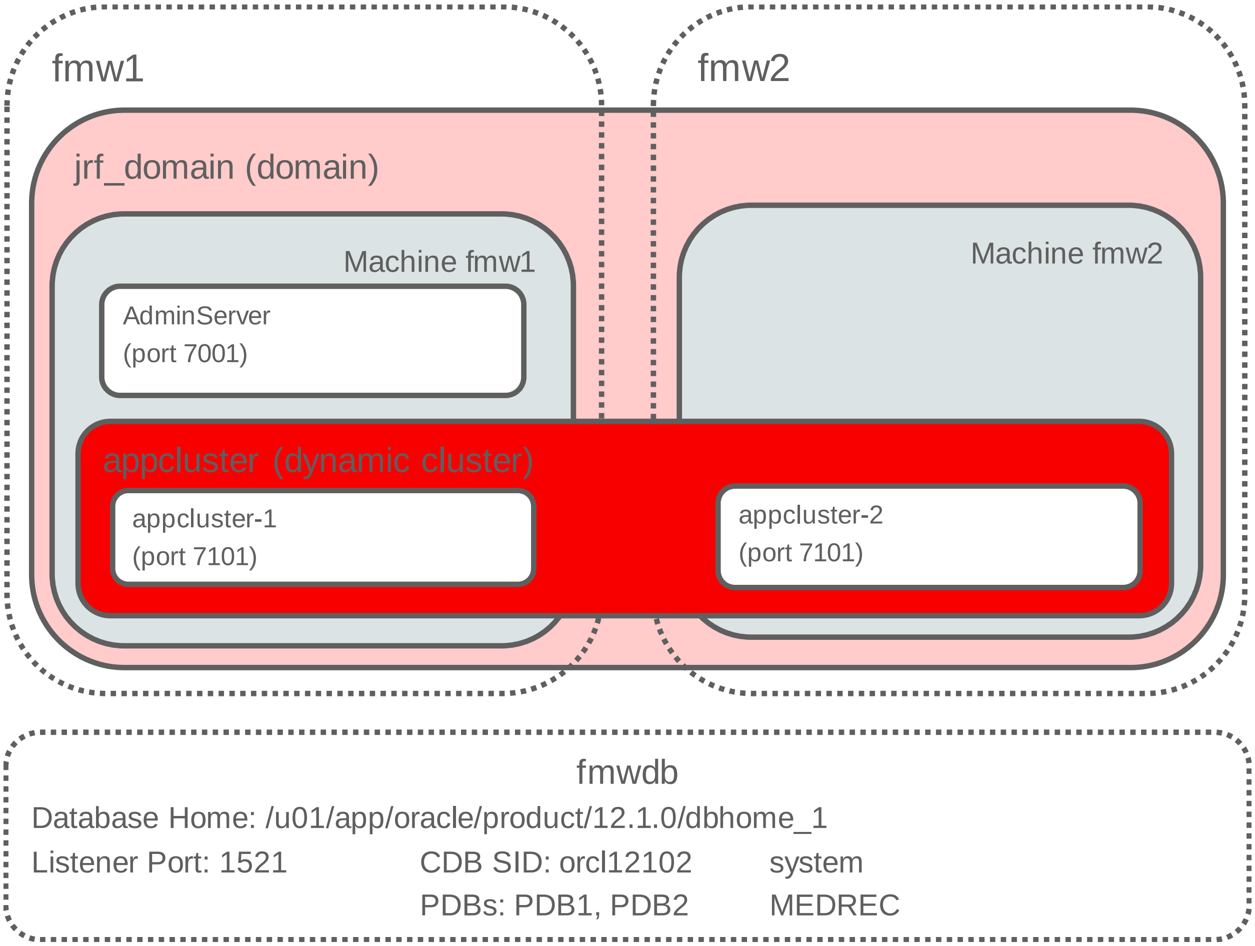
   ```
   ...
   >>>Practice 03-02 solution applied

   Exiting WebLogic Scripting Tool.
   ```
3. Run the solution script.

Practice 3-3: Creating a Domain to Support Multitenancy Using Fusion Middleware Control

Overview

In this practice, you create a WebLogic Server domain using the Fusion Middleware Configuration Wizard – this practice creates a domain that uses Fusion Middleware Control for administration, Oracle Traffic Director and Oracle Coherence.

The following diagram illustrates the configuration:



Assumptions

- WebLogic Server 12*c* (12.2.1) infrastructure is installed in `/u01/fmw/wls1221_infra`.
- The AdminServer from dev_domain has been shut down.

Tasks

1. Connect to fmw1 as the `oracle` user.
   a. Refer to the instructions provided by your instructor to connect to your machine.
   b. Unless stated otherwise, you will be working within fmw1 for the remainder of this practice.
2. Create the WebLogic Server Domain.

a. Open a Terminal window as the `oracle` user, and navigate to `/u01/fmw/wls1221_infra/`.

Tip: There is a launcher for a Terminal window in the panel at the top of the desktop.

```
$ cd /u01/fmw/wls1221_infra/
```

b. Execute the Fusion Middleware Configuration Wizard.

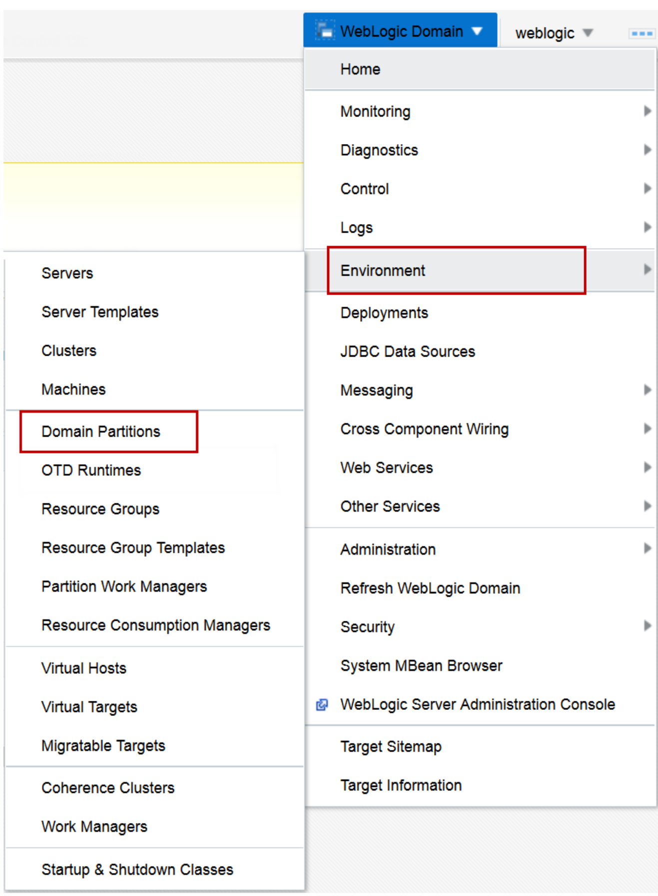
```
$ oracle_common/common/bin/config.sh
```

3. After the command, the graphical Fusion Middleware Configuration Wizard appears. Use the guidelines in the following table to configure the WebLogic Server Domain:

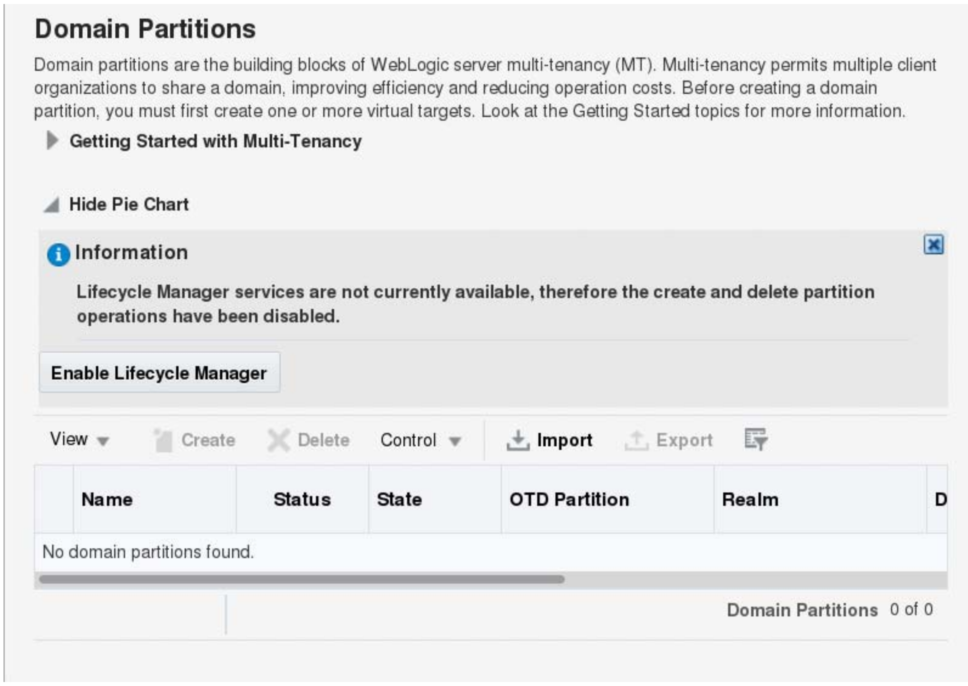
Step	Window/Page Description	Choices or Values
a.	Page 1 of 8 - Create Domain	Click **Create a new domain.** For **Domain Location,** enter `/u01/fmw/wls1221_infra/user_projects/domains/jrf_domain` and click **Next**.
b.	Page 2 of 8 - Templates	Select these **Available Templates:** • Oracle Traffic Director – Restricted JRF – 12.2.1 [otd] • Oracle Enterprise Manager – Restricted JRF – 12.2.1 [em] • Oracle Restricted JRF – 12.2.1 [oracle_common] • WebLogic Coherence Cluster Extension – 12.2.1 [wlserver] Click **Next**.
c.	Page 3 of 9 – Application Location	For **Application Location**, enter `/u01/fmw/wls1221_infra/user_projects/applications/jrf_domain` and click **Next**.
d.	Page 4 of 9 – Administrator Account	For **Name,** enter `weblogic`. For **Password,** enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. For **Confirm Password,** enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click **Next**.
e.	Page 5 of 9 – Domain Mode and JDK	For **Domain Mode** select **Development,** and click **Next**.
f.	Page 6 of 9 – Advanced Configuration	Click **Next**.
g.	Page 7 of 9 – Configuration Summary	Click **Create**.
h.	Page 8 of 9 – Node Manager	Click **Next**.
i.	Page 9 of 9 –	Click **Finish**.

Step	Window/Page Description	Choices or Values
	End of Configuration	

4. Log in to the WebLogic Server domain using Fusion Middleware Control.
   a. In the Terminal window, navigate to `/u01/fmw/wls1221_infra/user_projects/domains/jrf_domain`.
      $ **cd /u01/fmw/wls1221_infra/user_projects/domains/jrf_domain**
   b. Set the maximum memory for WebLogic Server to 512m.
      $ **sed -i 's/1024/512/g' bin/setStartupEnv.sh**
   c. Execute `startWebLogic.sh` to start the WebLogic Administration Server.
      $ **./startWebLogic.sh**
   d. Open a Browser window and navigate to **http://fmw1:7001/em.**
   e. For **User Name,** enter `weblogic` and for **Password,** enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click **Login.**
      The home page for the **jrf_domain** is displayed.
5. Enable the Lifecycle Manager.
   a. From the **WebLogic Domain** drop-down menu, select **Environment** and then select **Domain Partitions.**



b. Click **Enable Lifecycle Manager.**



c. In the terminal window where `startWebLogic.sh`, is running, enter `control-c` to stop the WebLogic Server.

d. After the WebLogic Server has stopped, execute `startWebLogic.sh` to start the WebLogic Administration Server.

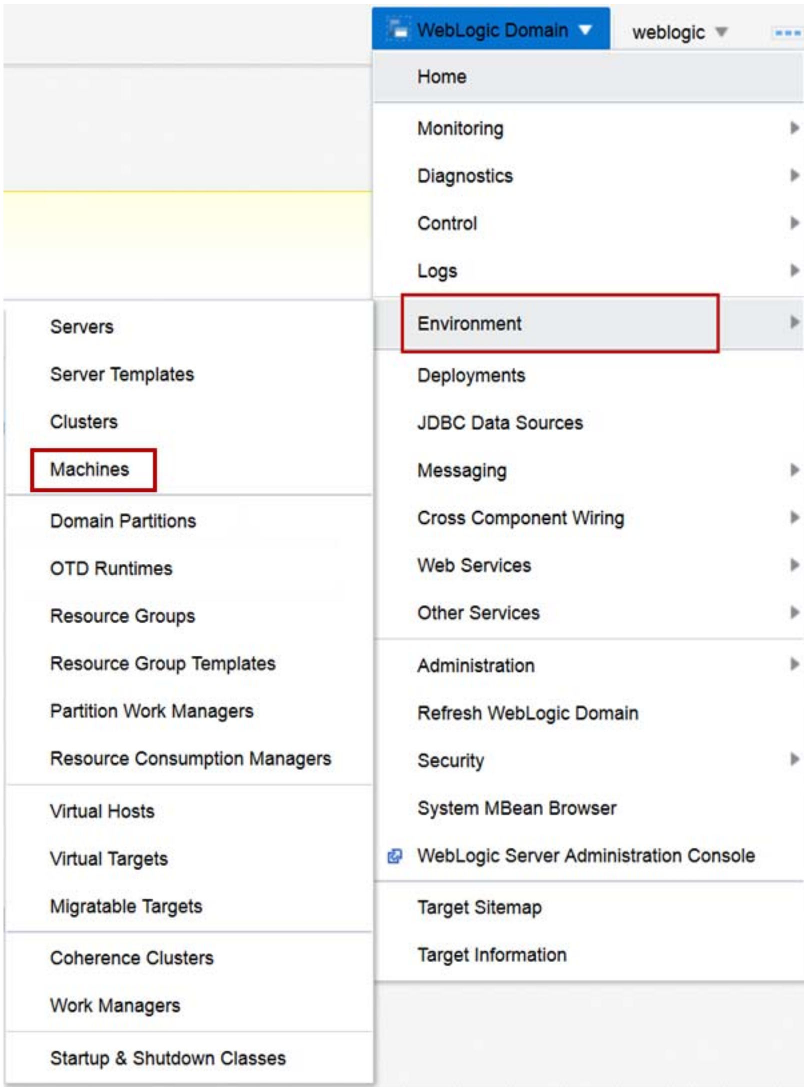
```
$ ./startWebLogic.sh
```

e. In the Browser window navigate to **http://fmw1:7001/em**.

f. For **User Name,** enter `weblogic` and for **Password,** enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click **Login.**

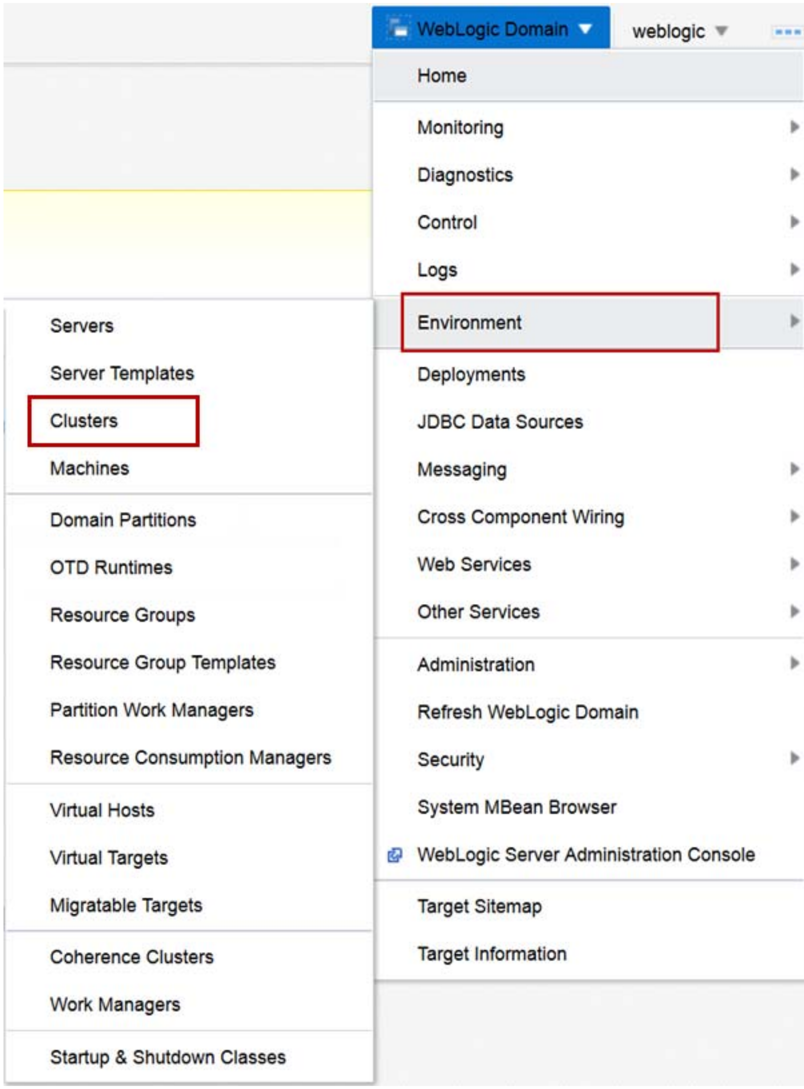
The home page for the **jrf_domain** displays and the Lifecycle Manager has been enabled.

6. Configure the WebLogic Server domain for machines for hosts fmw1 and fmw2.
   a. From the **WebLogic Domain** drop-down menu, select **Environment** and then select **Machines.**



   b. Click **Create.**
   c. For **Name,** enter `fmw1` and for **Machine OS** select **Other.** Click **Next.**
   d. For **Listen Address,** enter `fmw1` and for **Listen Port** enter `5556.` Click **Next.**
   e. Review the settings and click **Create.**
      The machine **fmw1** is created successfully.
   f. Click **Create.**
   g. For **Name,** enter `fmw2` and for **Machine OS** select **Other.** Click **Next.**
   h. For **Listen Address,** enter `fmw2` and for **Listen Port** enter `5556.` Click **Next.**
   i. Review the settings and click **Create.**
      The Machine **fmw2** is created successfully.

7. Configure the WebLogic Server domain with a Dynamic Cluster named appcluster.
   a. From the **WebLogic Domain** drop-down menu, select **Environment** and then select **Clusters.**



   b. Click **Create** and select **Dynamic Cluster.**
   c. For **Name,** enter `appcluster`**.** Click **Next.**
   d. For **Dynamic Cluster Size,** enter `2` and for **Max Dynamic Cluster Size** enter `4`. Click **Next.**
   e. Select **Use any machine configured in this domain** and click Next.
   f. Select **Assign each dynamic server unique listen ports,** and for **Base Listen Port** enter `7100`, and for **SSL Base Listen Port** enter `8100`. Click **Next.**
   g. Review the configuration and click Create.
      The cluster **appcluster** creates successfully.
8. Pack and unpack the jrf_domain to deploy it to the fmw2 host.
   a. In a new Terminal window navigate to `/u01/fmw/wls1221_infra/`.
      ```
      $ cd /u01/fmw/wls1221_infra/
      ```
   b. Execute `pack.sh` to pack the jrf_domain.

```
$ oracle_common/common/bin/pack.sh -domain
/u01/fmw/wls1221_infra/user_projects/domains/jrf_domain/ -
template /u01/fmw/wls1221_infra/user_projects/jrf_domain.jar -
template_name jrf_domain –managed true
```

c. Create the `user_projects` directory on the fmw2 machine.

```
$ ssh fmw2 "mkdir –p
/u01/fmw/wls1221_infra/user_projects/domains"
```

d. Enter the `oracle` user password when prompted.

e. Copy the `jrf_domain.jar` to fmw2 machine.

```
$ scp user_projects/jrf_domain.jar
fmw2:/u01/fmw/wls1221_infra/user_projects/jrf_domain.jar
```

f. Enter the `oracle` user password when prompted.

g. Connect to the fmw2 machine as the `oracle` user.

h. Open a new Terminal window on fmw2 as the `oracle` user and navigate to `/u01/fmw/wls1221_infra/`.

```
$ cd /u01/fmw/wls1221_infra/
```

i. Execute `unpack.sh` to unpack the jrf_domain.

```
$ /u01/fmw/wls1221_infra/oracle_common/common/bin/unpack.sh -
domain /u01/fmw/wls1221_infra/user_projects/domains/jrf_domain -
template /u01/fmw/wls1221_infra/user_projects/jrf_domain.jar
```

The jrf_domain is now unpacked and ready to use NodeManager and start managed servers on host fmw2.

9. Start NodeManager on host fmw1.

a. Connect to the fmw1 machine as the `oracle` user.

b. Open a new Terminal window on fmw1 as the `oracle` user and navigate to `/u01/fmw/wls1221_infra/user_projects/domains/jrf_domain/`.

```
$ cd /u01/fmw/wls1221_infra/user_projects/domains/jrf_domain/
```

c. Copy the nodemanager directory to the `nodemanager-fmw1` directory.

```
$ cp –r nodemanager nodemanager-fmw1
```

d. Copy the startNodeManager.sh script to the `nodemanager-fmw1` directory.

```
$ cp bin/startNodeManager.sh nodemanager-fmw1
```

e. Edit the nodemanager.properties file.

```
$ vi nodemanager-fmw1/nodemanager.properties
```

f. Set the NodeManagerHome property:

```
NodeNamangerHome=/u01/fmw/wls1221_infra/user_projects/domains/jr
f_domain/nodemanager-fmw1
```

g. Set the ListenAddress property:

```
ListenAddress=fmw1
```

h. Set the NodeManagerHome property:

```
LogFile=/u01/fmw/wls1221_infra/user_projects/domains/jrf_domain/
nodemanager-fmw1/nodemanager.log
```

i. Save and close the `nodemanager.properties` file:

```
:wq
```

j. Edit the `startNodeManager.sh` file.

$ **vi nodemanager-fmw1/startNodeManager.sh**

k. Set the `NODEMGR_HOME` property:

```
NODEMGR_HOME="/u01/fmw/wls1221_infra/user_projects/domains/jrf_d
omain/nodemanager-fmw1"
```

l. Save and close the `nodemanager.properties` file:

:wq

m. Navigate to the nodemanager-fmw1 directory.

$ **cd nodemanager-fmw1**

n. Execute the `startNodeManager.sh` script.

$ **./startNodeManager.sh**

The NodeManager process starts successfully – leave the Terminal window open.

10. Start NodeManager on host fmw2.

a. Connect to the fmw2 machine as the `oracle` user.

b. In the Terminal window on fmw2 as the `oracle` user navigate to `/u01/fmw/wls1221_infra/user_projects/domains/jrf_domain/`.

$ **cd /u01/fmw/wls1221_infra/user_projects/domains/jrf_domain/**

c. Copy the nodemanager directory to the `nodemanager-fmw2` directory.

$ **cp –r nodemanager nodemanager-fmw2**

d. Copy the `startNodeManager.sh` script to the `nodemanager-fmw2` directory.

$ **cp bin/startNodeManager.sh nodemanager-fmw2**

e. Edit the nodemanager.properties file.

$ **vi nodemanager-fmw2/nodemanager.properties**

f. Set the `NodeManagerHome` property:

```
NodeNamangerHome=/u01/fmw/wls1221_infra/user_projects/domains/jr
f_domain/nodemanager-fmw2
```

g. Set the `ListenAddress` property:

```
ListenAddress=fmw2
```

h. Set the `NodeManagerHome` property:

```
LogFile=/u01/fmw/wls1221_infra/user_projects/domains/jrf_domain/
nodemanager-fmw2/nodemanager.log
```

i. Save and close the `nodemanager.properties` file:

:wq

j. Edit the `startNodeManager.sh` file.

$ **vi nodemanager-fmw2/startNodeManager.sh**

k. Set the NODEMGR_HOME property:

```
NODEMGR_HOME="/u01/fmw/wls1221_infra/user_projects/domains/jrf_d
omain/nodemanager-fmw2"
```

l. Save and close the `nodemanager.properties` file:

:wq

m. Navigate to the `nodemanager-fmw2` directory.

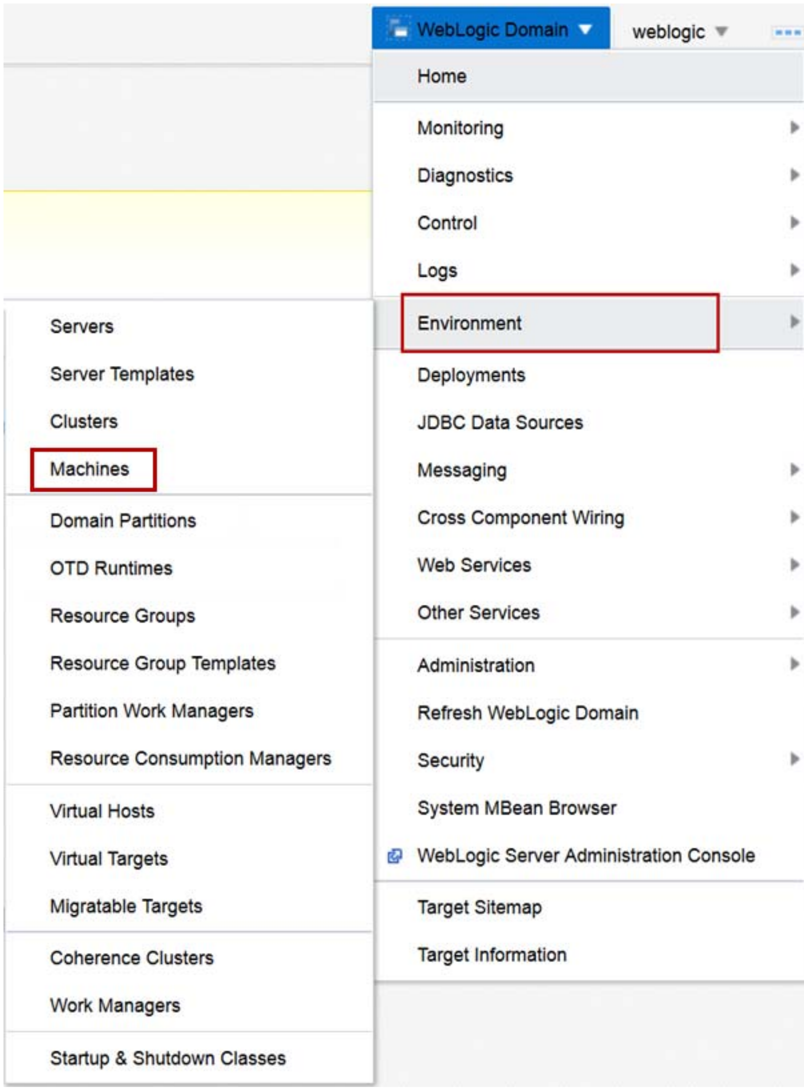
$ **cd nodemanager-fmw2**

n. Execute the `startNodeManager.sh` script.

```
$ ./startNodeManager.sh
```

The NodeManager process starts successfully – leave the Terminal window open.

11. Validate that both machines are reachable from the WebLogic Administration Server.
    a. From the **WebLogic Domain** drop-down menu, select **Environment** and then select **Machines.**



    b. Click **fmw1.**
    c. Click the **Monitoring** tab.
    d. The **Node Manager Status** reports **Reachable.**
    e. Click **fmw2.**
    f. Click the **Monitoring** tab.
    g. The **Node Manager Status** reports **Reachable.**

The jrf_domain is created with a dynamic cluster and NodeManager running on both hosts to support a multitenancy deployment.

Practice Solution: Creating a Domain to Support Multitenancy using Fusion Middleware Control

Perform the following tasks if you did not complete this practice and want to use the finished solution.

> **Note:** If you started creating the jrf_domain, this script will not work. To run the script in that case: delete the `/u01/fmw/wls1221_infra/user_projects/domains/jrf_domain` directory and delete the `/u01/fmw/wls1221_infra/user_projects/applications/jrf_domain` directory.

Assumptions

WebLogic Server 12*c* (12.2.1) infrastructure is installed in `/u01/fmw/wls1221_infra`.

Solution Tasks

1. Connect to fmw1 as the `oracle` user.
2. Run the solution script.
   a. Open a Terminal window as the oracle user and navigate to the practice directory.
   ```
   $ cd /u01/practices/part1/practice03-03
   ```
   b. Run the solution script.
   ```
   $ ./solution.sh
   ```
   c. Enter any passwords as you are prompted.

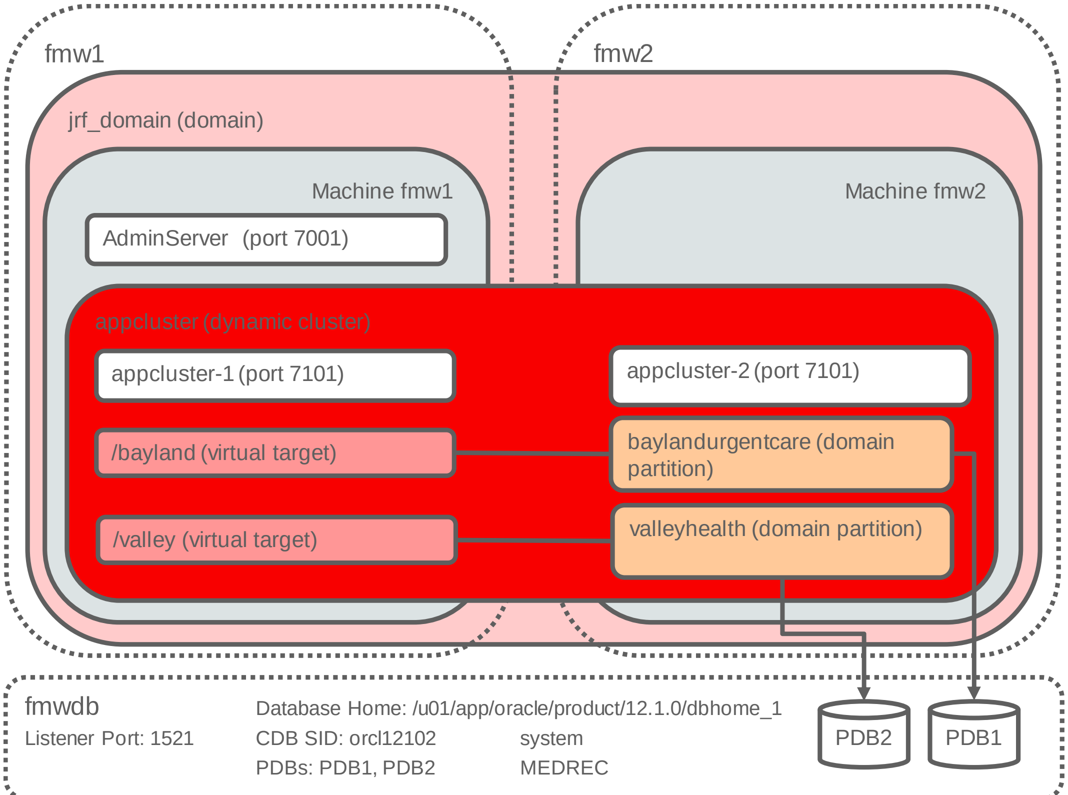
   Note: This script starts the WebLogic Scripting Tool (WLST) passing it the WLST script `create_domain.py`, which creates the new domain for you. The WLST script prints out messages. Be patient and do not close the Terminal window until you see these messages:
   ```
   ...
   >>>Practice 03-03 solution applied successfully!

   Exiting WebLogic Scripting Tool.
   ```
3. **Note:** this script starts a WebLogic Server domain in the background and may take several minutes start and make the console and Fusion Middleware Control available.
4. Close the Terminal window.

Practice 3-4: Creating Domain Partitions Using Fusion Middleware Control

Overview

In this practice, you create Domain Partitions for the Avitek Medical Records (MedRec) sample application using Fusion Middleware Control.

The following diagram illustrates the configuration:



Assumptions

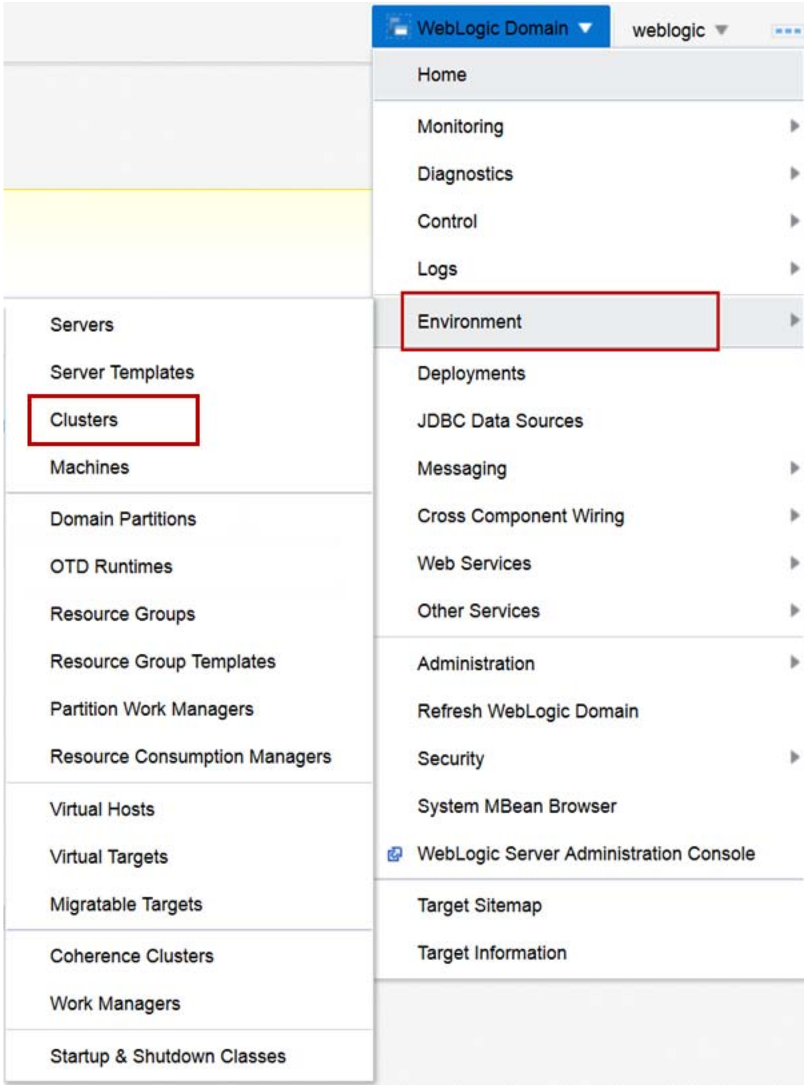
- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf_domain admin server is running.
- The nodemanager for host fmw1, nodemanager-fmw1 is running.
- The nodemanager for host fmw2, nodemanager-fmw2 is running.
- The machines fm1 and fmw2 have been created.
- The dynamic cluster appcluster has been created.

Tasks

1. Start the Dynamic Cluster appcluster.
   a. Open a Browser window and navigate to **http://fmw1:7001/em.**
   b. For **User Name,** enter `weblogic` and for **Password,** enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click **Login.**

      The Fusion Middleware Control home page for the **jrf_domain** is displayed.

c. From the **WebLogic Domain** drop-down menu, select **Environment** and then select **Clusters.**



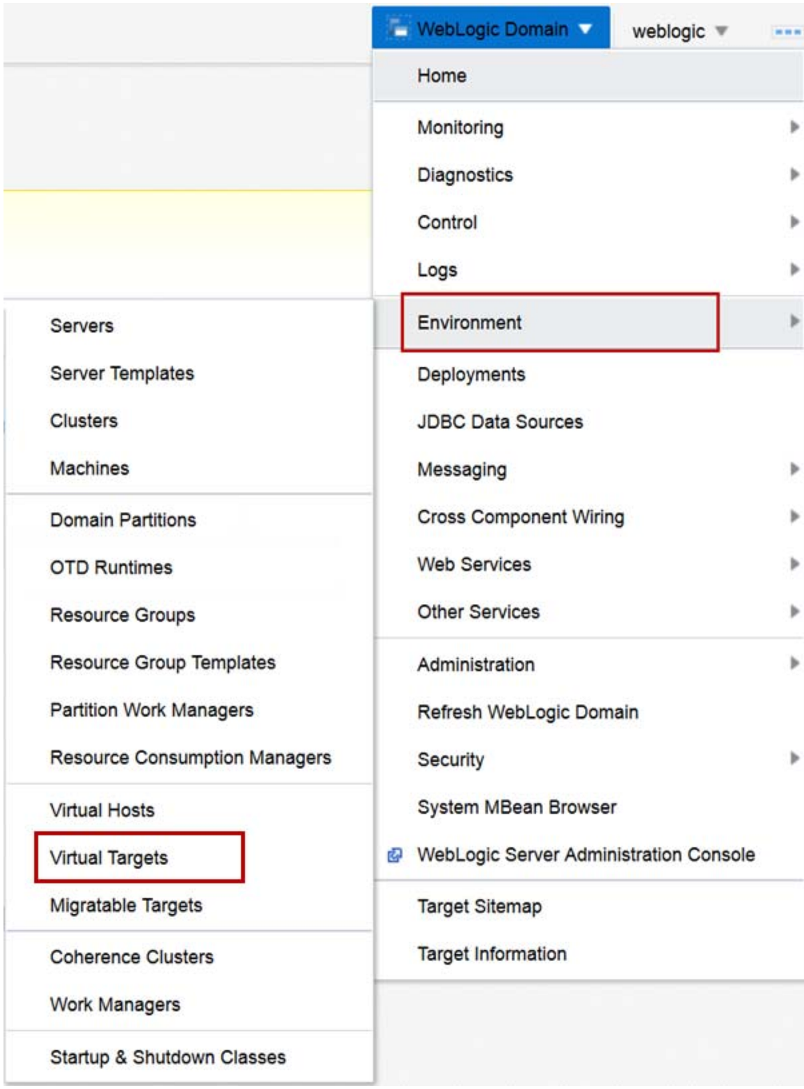
d. Select the **appcluster** entry, click **Control** and select **Start.**

e. Click **Start Servers.**

The request is sent to Node Manager to start the servers in appcluster.

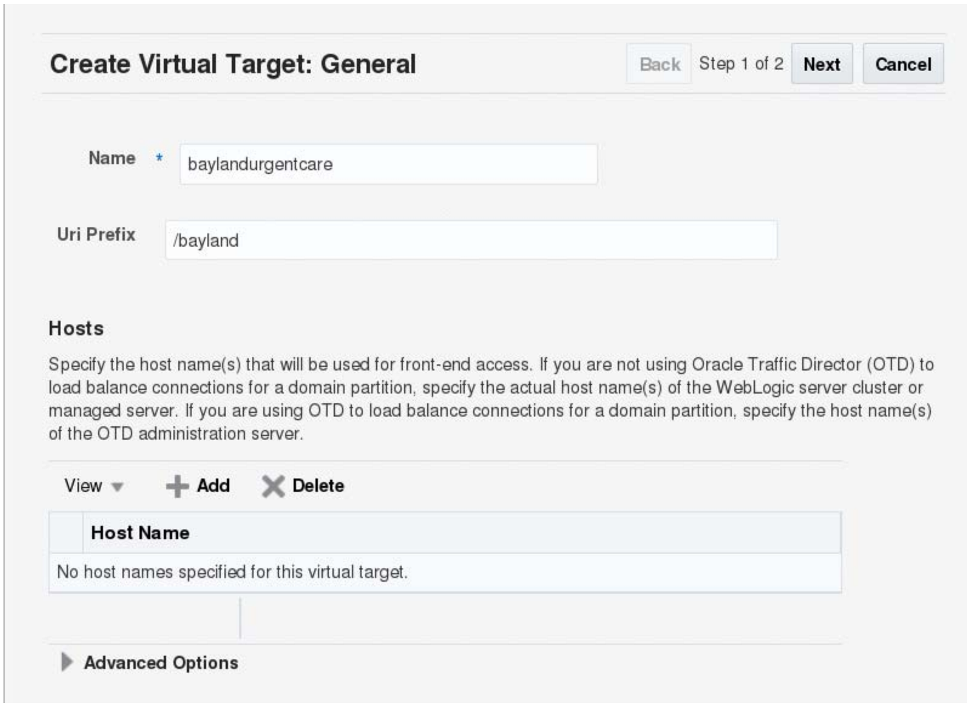
2. Create a Virtual Target for baylandurgentcare.

   a. From the **WebLogic Domain** drop-down menu, select **Environment** and then select **Virtual Targets.**



   b. Click **Create.**

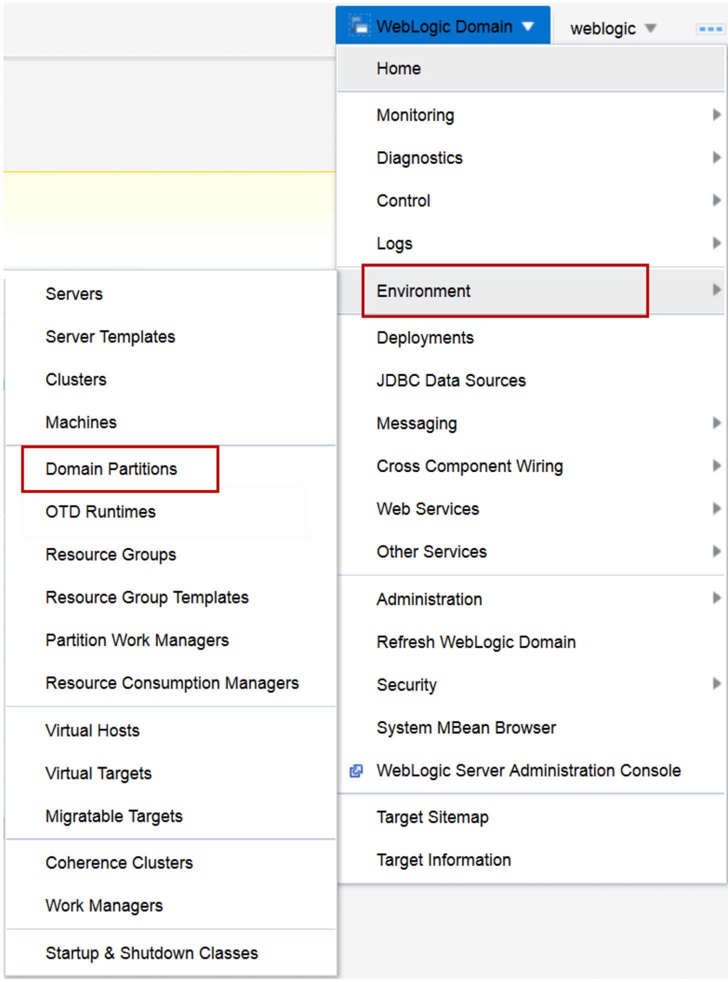
   c. For **Name,** enter `baylandurgentcare` and for **URI Prefix** enter /`bayland`. Click **Next.**



d. From the **Cluster** menu select **appcluster** . Click **Create.**
The Virtual Target **baylandurgentcare** creates successfully.

3. Create a Domain Partition for baylandurgentcare.

   a. In the Browser window, from the **WebLogic Domain** drop-down menu, select **Environment** and then select **Domain Partitions.**



   Tip: If the Lifecycle Manager is not enabled, click the button to enable it for this domain.

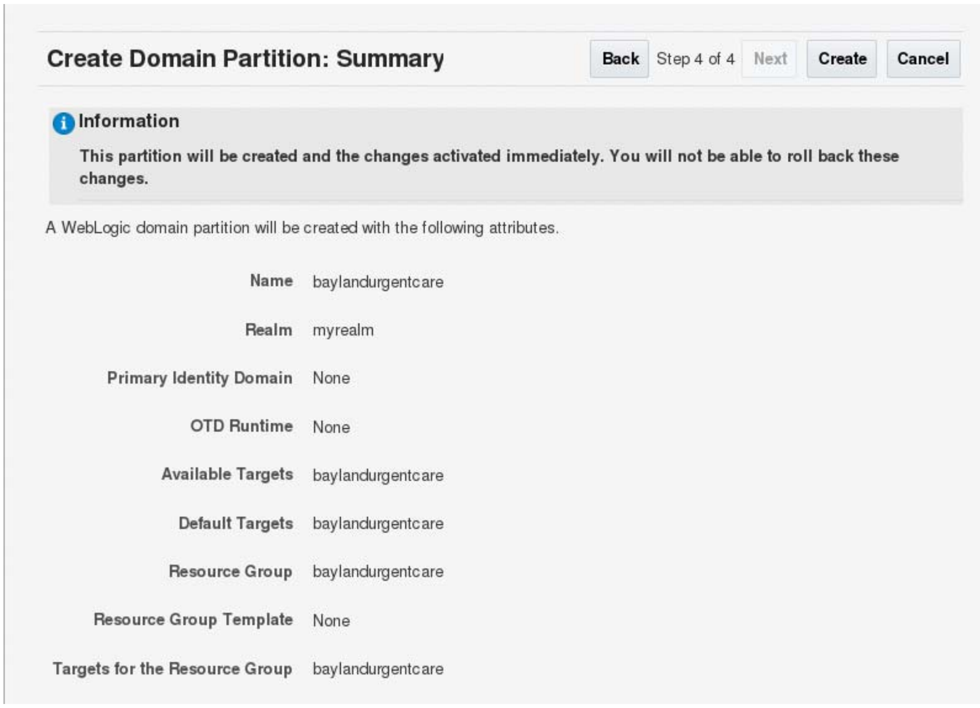
   b. Click **Create.**

   c. For **Name,** enter `baylandurgentcare` and for **Security Realm** select **myrealm.** Click **Next.**

   d. Select the check boxes for the **Virtual Target baylandurgentcare** in the **Select** column and in the **Set as Default** column. Click **Next.**

   e. For **Resource Group Name,** enter `baylandurgentcare`.

   f. Move the **baylandurgentcare** target from **Available Targets** to **Selected Targets.** Click **Next.**



g. Review the summary and click **Create.**

   The Domain Partition **baylandurgentcare** creates successfully.

4. Create a Data Source for the baylandurgentcare Domain Partition.
   a. In the Browser window, click the **baylandurgentcare** Domain Partition.
   b. From the **Domain Partition** menu, select **JDBC Data Sources.**
   c. Click **Create** and select **Generic Data Source.**
   d. For **Data Source Name,** enter `MedRecGlobalDataSourceXA`.
   e. For **JNDI Name,** enter `jdbc/MedRecGlobalDataSourceXA.`
   f. Click **Next.**
   g. Click **Generate URL and Properties…**
   h. For **Database Host Name,** enter `fmwdb`.
   i. For **Database Listener Port,** enter `1521.`
   j. For **Database Name,** enter `PDB1.`
   k. For **Database User Name,** enter `MEDREC.`
   l. For **Password,** enter the password from the Course Practice Environment: Security Credentials document for the MEDREC user for the Oracle Database product.
   m. For **Confirm Password,** enter the password from the Course Practice Environment: Security Credentials document for the MEDREC user for the Database product.
   n. Click **OK.**
   o. Click **Test Database Connection.**

      The test should succeed.

p. If the test fails, validate that the listener is started, the database is started, and the pluggable databases are open on host fmwdb.

1) Connect to fmwdb as the `oracle` user.

   a) Refer to the instructions provided by your instructor to connect to your machine.

   b) Open a Terminal window to the oracle user home directory.

2) Execute the `lsnrctl` utility with the `start` command.

```
$ lsnrctl start
```

   Note: your output may vary – if the listener is running you will receive a message that it's already started.

3) Execute the `sqlplus` utility, using `sys as sysdba` for user-name. For password, enter the password from the Course Practice Environment: Security Credentials document for the sys user for the Oracle Database product.

```
$ sqlplus
Enter user-name: sys as sysdba
Enter password:
```

4) If the connection message to the database includes:

```
Connected to an idle instance.
```

   then execute startup to start the database:

```
SQL> startup
```

   the output includes database memory usage and:

```
Database mounted.
Database opened.
```

5) Execute the command to open the PDB1 database.

```
SQL> ALTER PLUGGABLE DATABASE PDB1 OPEN READ WRITE;
```

6) Execute the command to open the PDB2 database.

```
SQL> ALTER PLUGGABLE DATABASE PDB1 OPEN READ WRITE;
```

7) Execute the query to list the mode for pluggable databases and validate that PDB1 and PDB2 are open for READ WRITE.

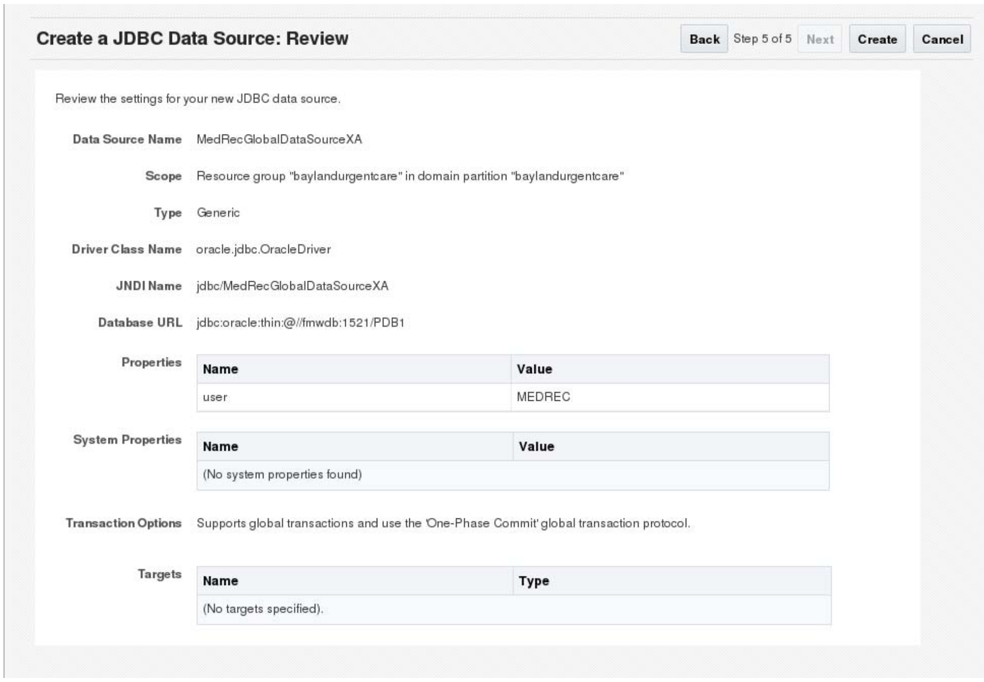
```
SQL> SELECT NAME, OPEN_MODE, RESTRICTED FROM v$pdbS;

NAME                           OPEN_MODE  RES
------------------------------ ---------- ---
PDB$SEED                       READ ONLY  NO
PDBORCL                        MOUNTED
NEW_PDB                        MOUNTED
PDB0                           MOUNTED
PDB1                           READ WRITE NO
PDB2                           READ WRITE NO

6 rows selected.
```

8) In the Browser, click **Test Database Connection.**

   The test should succeed.

q. In the Browser, click **Next.**

r. Accept the **Transaction Properties** defaults and click **Next.**

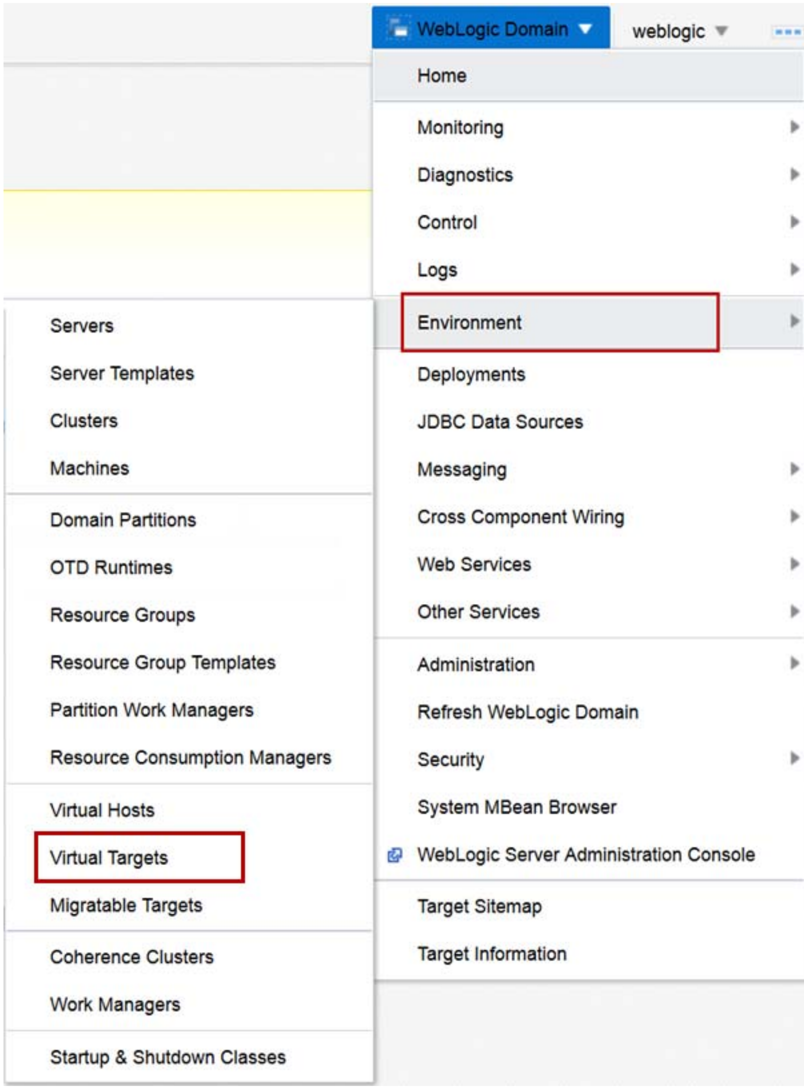


s. Review the settings and click **Create.**

5. Deploy applications to the baylandurgentcare Domain Partition.
   a. From the **Domain Partition** menu, select **Deployments.**
   b. From the **Deployment** menu, select **Deploy.**
   c. In the **Deploy Java EE Application: Select Archive** page, select **Archive or exploded directory is on the server where Enterprise Manager is running** and click **Browse…**
   d. Navigate to the `/u01/fmw/wls1221_infra/wlserver/samples/server/medrec/dist/standalone` directory, select the `medrec.ear` file and click **OK.**
   e. Click **Next.**
   f. In the **Distribution** section, select **Install only. Do not start.**
   g. Click **Next.**
   h. Review the Deployment Settings and click **Deploy.**
   i. After the Deployment succeeds, click **Close.**
6. Start the baylandurgentcare Domain Partition.
   a. Next to the **Domain Partition** menu, click **Start Up.**
   b. After the **Domain Partition** starts, click **Close.**
7. Test the baylandurgentcare Virtual Target.
   a. Open a new Browser tab, navigate to **http://fmw1:7101/bayland/medrec.**

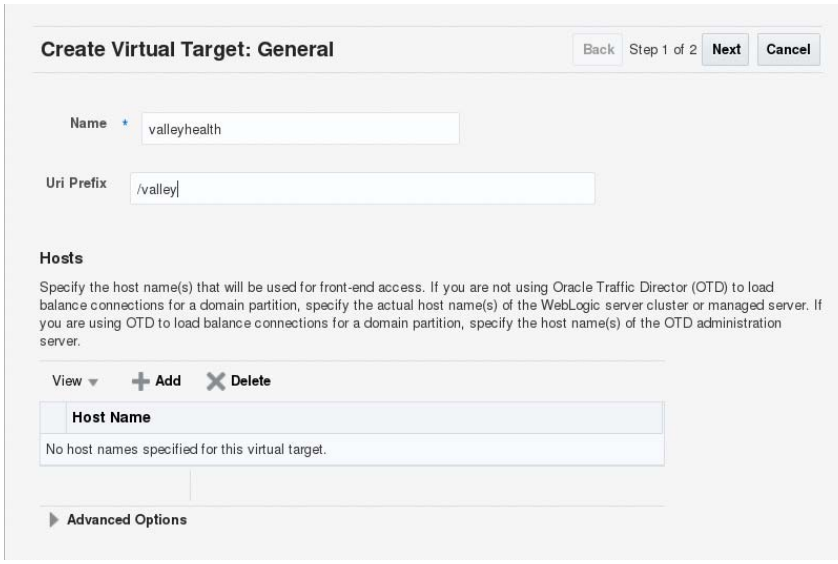
8. Create a Virtual Target for valleyhealth.

   a. In the Browser window, from the **WebLogic Domain** drop-down menu, select **Environment** and then select **Virtual Targets.**



   b. Click **Create.**

c. For **Name,** enter `valleyhealth` and for **URI Prefix** enter `/valley`. Click **Next.**

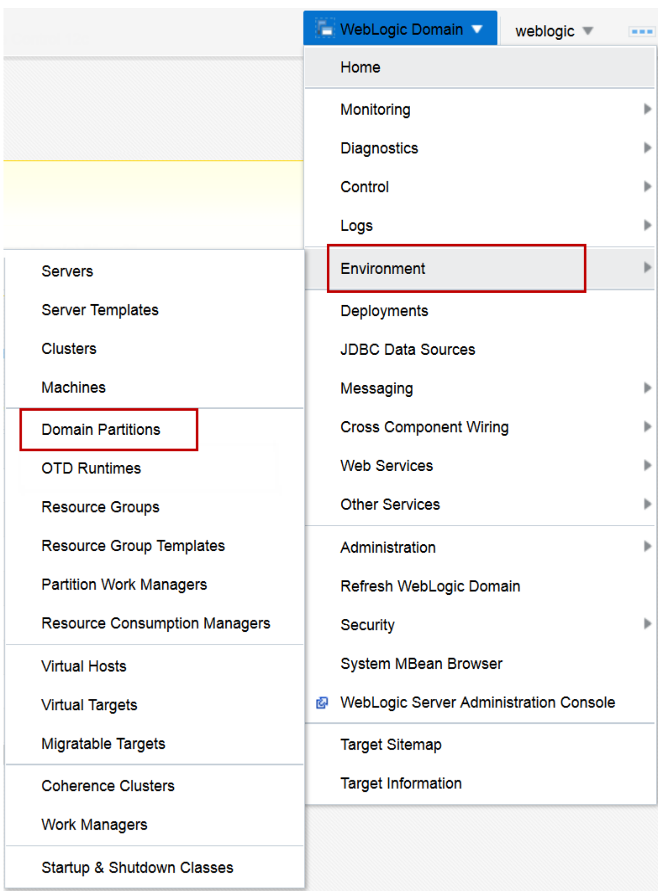


d. From the **Cluster** menu select **appcluster**. Click **Create.**

The Virtual Target **valleyhealth** creates successfully.

9. Create a Domain Partition for valleyhealth.

a. In the Browser window, from the **WebLogic Domain** dropdown menu, select **Environment** and then select **Domain Partitions.**



Tip: If the Lifecycle Manager is not enabled, click the button to enable it for this domain.

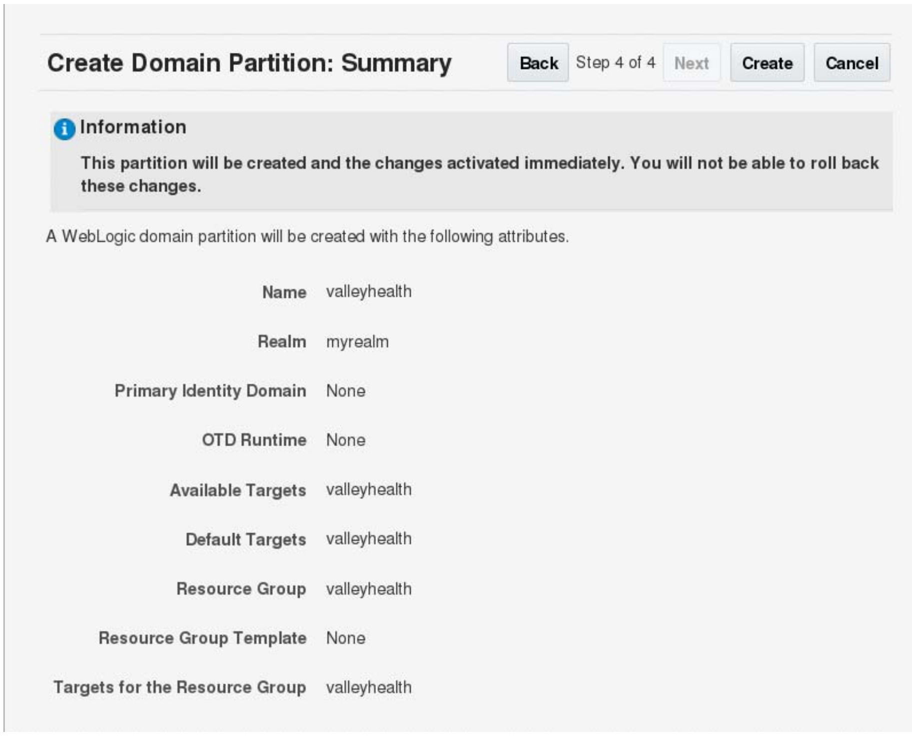
b. Click **Create.**

c. For **Name,** enter `valleyhealth` and for **Security Realm** select **myrealm.** Click **Next.**

d. Select the check boxes for the **Virtual Target valleyhealth** in the **Select** column and in the **Set as Default** column. Click **Next.**

e. For Resource Group Name, enter `valleyhealth`.

f. Move the **valleyhealth** target from **Available Targets** to **Selected Targets.** Click **Next.**



g. Review the summary and click **Create.**

The domain partition **valleyhealth** creates successfully.

10. Create a Data Source for the valleyhealth domain partition.

a. In the Browser window, click the **valleyhealth** domain partition.

b. From the **Domain Partition** menu, select **JDBC Data Sources.**

c. Click **Create** and select **Generic Data Source.**

d. For **Data Source Name**, enter `MedRecGlobalDataSourceXA`.

e. For **JNDI Name,** enter `jdbc/MedRecGlobalDataSourceXA`.

f. Click **Next.**

g. Click **Generate URL and Properties…**

h. For **Database Host Name,** enter `fmwdb`.

i. For **Database Listener Port,** enter `1521`.

j. For **Database Name,** enter `PDB2`.

k. For **Database User Name,** enter `MEDREC`.

l. For **Password,** enter the password from the Course Practice Environment: Security Credentials document for the MEDREC user for the Oracle Database product.

m. For **Confirm Password,** enter the password from the Course Practice Environment: Security Credentials document for the MEDREC user for the Oracle Database product.

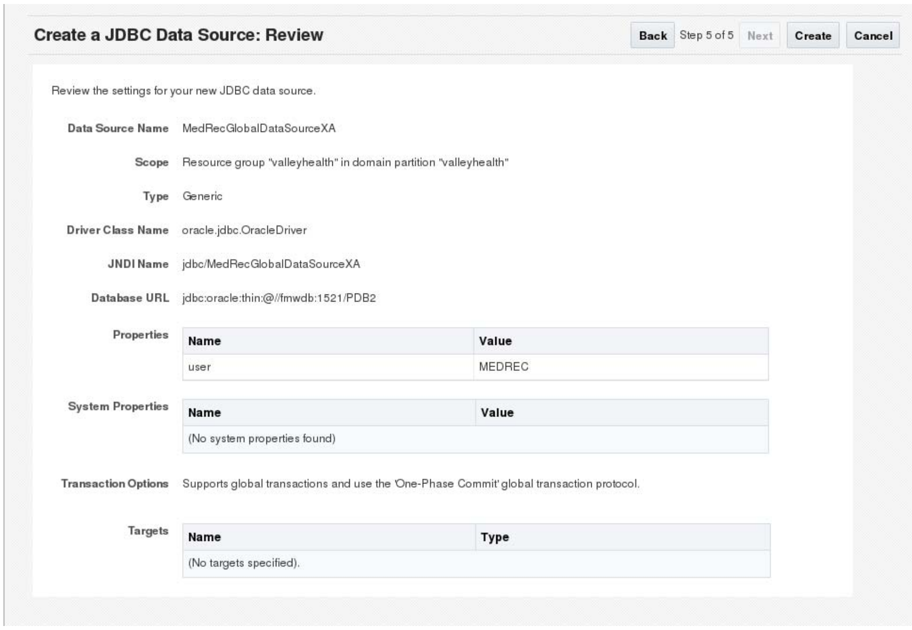
n. Click **OK.**

o. Click **Test Database Connection.**

The test should succeed.

p. Click **Next.**

q. Accept the **Transaction Properties** defaults and click **Next.**



r. Review the settings and click **Create.**

11. Deploy applications to the valleyhealth Domain Partition.

a. From the **Domain Partition** menu, select **Deployments.**

b. From the **Deployment** menu, select **Deploy.**

c. In the **Deploy Java EE Application: Select Archive** page, select **Archive is on the machine where this Web browser is running** and click **Browse…**

d. Navigate to the `/u01/fmw/wls1221_infra/wlserver/samples/server/medrec/dist/standalone` directory, select the `medrec.ear` file and click **OK.**

e. Click **Next.**

f. In the **Distribution** section, select **Install only. Do not start.**

g. Click **Next.**

h. Review the Deployment Settings and click **Deploy.**

i. After the Deployment succeeds, click **Close.**

12. Start the valleyhealth domain partition.

a. Next to the **Domain Partition** menu, click **Start Up.**

b. After the **Domain Partition** starts, click **Close.**

13. Test the valleyhealth Virtual Target.

a. Open a new Browser tab, navigate to **http://fmw1:7101/valley/medrec.**

Two tenants – baylandurgentcare and valleyhealth – have been created with domain partitions – using the MedRec sample application deployed to both partitions isolated by URI.

Practice Solution: Creating Domain Partitions Using Fusion Middleware Control

Perform the following tasks if you did not complete this practice and want to use the finished solution.

Note: If you started creating the domain partitions or virtual targets, this script will not work. To run the script in that case: delete any domain partitions or virtual targets.

Assumptions

- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf_domain admin server is running.
- The nodemanager for host fmw1, nodemanager-fmw1 is running.
- The nodemanager for host fmw2, nodemanager-fmw2 is running.
- The machines fm1 and fmw2 have been created.
- The dynamic cluster appcluster has been created.

Solution Tasks

1. Connect to fmw1 as the `oracle` user.
2. Run the solution script.
   a. Open a Terminal window as the oracle user and navigate to the practice directory.
      ```
      $ cd /u01/practices/part1/practice03-04
      ```
   b. Run the solution script.
      ```
      $ ./solution.sh
      ```
   c. Enter any passwords as you are prompted.

   Note: This script starts the WebLogic Scripting Tool (WLST) passing it the WLST script `create_partitions.py`, which creates the new generic data source for you. The WLST script prints out messages. Be patient and do not close the Terminal window until you see these messages:

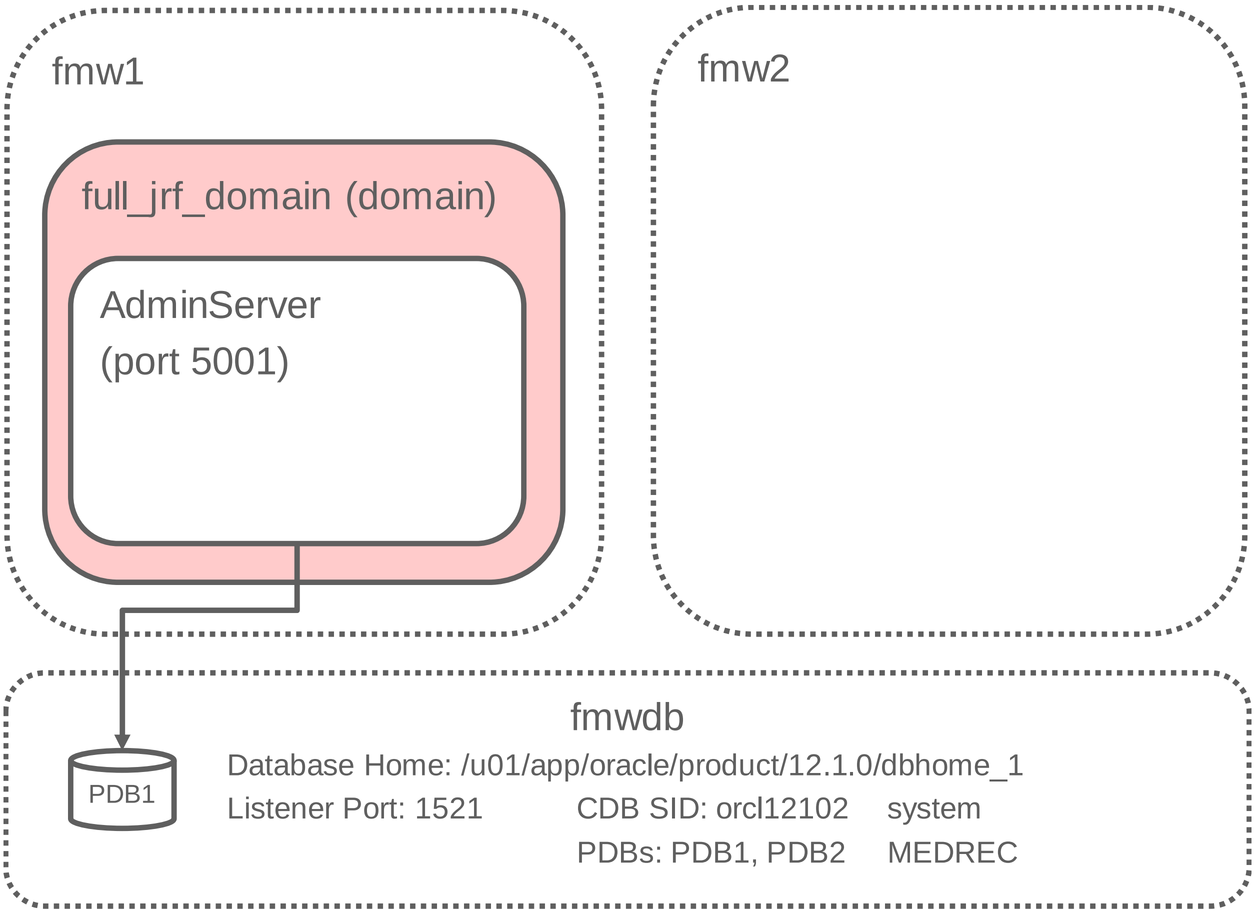
      ```
      ...
      >>>Practice 03-04 solution applied successfully!

      Exiting WebLogic Scripting Tool.
      ```
3. Close the Terminal window.

Practice 3-5: Creating a Full JRF Domain to Support Multitenancy using Fusion Middleware Control

Overview

In this practice, you create a WebLogic Server Domain that supports the full set of JRF functionality using the Fusion Middleware Configuration Wizard. To support full JRF, the WebLogic Server Domain requires a database and the Repository Creation Utility is used to create the required tables in a pluggable database. This practice creates a domain that uses Fusion Middleware Control for administration, Oracle Traffic Director and Oracle Coherence.

The following diagram illustrates the configuration:



Assumptions

- WebLogic Server 12.2.1 Infrastructure is installed at `/u01/fmw/wls1221_infra/`.
- Oracle Database on host fmwdb is configured for JRF and is available.

Tasks

1. Connect to fmw1 as the `oracle` user.
   a. Refer to the instructions provided by your instructor to connect to your machine.
   b. Unless stated otherwise, you will be working within fmw1 for the remainder of this practice.
2. Create the Full JRF schemas with the Repository Creation Utility.

a. Open a Terminal window as the `oracle` user, and navigate to `/u01/fmw/wls1221_infra/`.

Tip: There is a launcher for a Terminal window in the panel at the top of the desktop.

```
$ cd /u01/fmw/wls1221_infra/
```

b. Execute the Repository Creation Utility.

```
$ oracle_common/bin/rcu
```

3. After the command, the graphical Repository Creation Utility appears. Use the guidelines in the following table to configure the repository:

Step	Window/Page Description	Choices or Values
a.	Step 1 of 8 – Welcome	Click **Next.**
b.	Step 2 of 8 – Create Repository	Select **Create Repository** and select **System Load and Product Load.** Click **Next**.
c.	Step 3 of 8 – Database Connection Details	For **Database Type,** select **Oracle Database.** For **Host Name,** enter `fmwdb`. For **Port,** enter `1521`. For **Service Name,** enter `PDB1`. For **Username,** enter `sys`. For **Password,** enter the password from the Course Practice Environment: Security Credentials document for the sys user for the Oracle Database product. Click **Next**.
d.	Repository Creation Utility – Checking Prerequisites	After the operation has completed, click **OK.**
e.	Step 4 of 8 – Select Components	Select **Create new prefix.** For **prefix,** enter `FULL`. Check **Oracle Platform Security Services.** Click **Next.**
f.	Repository Creation Utility – Checking Prerequisites	After the operation has completed, click **OK.**
g.	Step 5 of 8 – Schema Passwords	Select **Use same passwords for all schemas.** For **Password,** enter the password from the Course Practice Environment: Security Credentials document for the FULL_STB user for the Oracle Database product.. For **Confirm Password,** enter the password from the Course Practice Environment: Security Credentials document for the FULL_STB user for the Oracle Database product. Click **Next.**

Step	Window/Page Description	Choices or Values
h.	Step 6 of 8 – Map Tablespaces	Accept the default tablespace mappings and click **Next.**
i.	Repository Creation Utility – Confirm	Click **OK** to create tablespaces.
j.	Repository Creation Utility – Creating Tablespaces	After the operation has completed, click **OK.**
k.	Step 7 of 8 - Summary	Click **Create.**
l.	Step 8 of 8 – Completion Summary	Click **Close.**

4. Create the Full JRF WebLogic Server Domain.
   a. Open a Terminal window, and navigate to `/u01/fmw/wls1221_infra/`.

   Tip: There is a launcher for a Terminal window in the panel at the top of the desktop.

   ```
   $ cd /u01/fmw/wls1221_infra/
   ```

   b. Execute the Fusion Middleware Configuration Wizard.

   ```
   $ oracle_common/common/bin/config.sh
   ```

5. After the command, the graphical Fusion Middleware Configuration Wizard appears. Use the guidelines in the following table to configure the WebLogic Server Domain:

Step	Window/Page Description	Choices or Values
a.	Page 1 of 8 - Create Domain	Click **Create a new domain.** For **Domain Location,** enter `/u01/fmw/wls1221_infra/user_projects/domains/full_jrf_domain` and click **Next**.
b.	Page 2 of 8 - Templates	Select these **Available Templates:** • Oracle Traffic Director – 12.2.1 [otd] • Oracle Enterprise Manager – 12.2.1 [em] • Oracle JRF – 12.2.1 [oracle_common] • WebLogic Coherence Cluster Extension – 12.2.1 [wlserver] Click **Next**.
c.	Page 3 of 12 – Application Location	For **Application Location**, enter `/u01/fmw/wls1221_infra/user_projects/applications/full_jrf_domain` and click **Next**.
d.	Page 4 of 12 – Administrator Account	For **Name,** enter `weblogic`. For **Password,** enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product.

Step	Window/Page Description	Choices or Values
		For **Confirm Password,** enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click **Next**.
e.	Page 5 of 12 – Domain Mode and JDK	For **Domain Mode** select **Development,** and click **Next**.
f.	Page 6 of 12 – Database Configuration Type	Select **RCU Data.** For **Vendor,** select **Oracle.** For **Driver,** select **Oracle's Driver (Thin) for Service connections.** For **DMBS/Servers,** enter `PDB1`. For **Host Name,** enter `fmwdb`. For **Port,** enter `1521`. For **Schema Owner,** enter `FULL_STB`. For **Schema Password,** enter the password from the Course Practice Environment: Security Credentials document for the FULL_STB user for the Oracle Database product. Click **Get RCU Configuration.** Click **Next**.
g.	Page 7 of 12 – Component Datasources	Click **Next**.
h.	Page 8 of 12 – JDBC Test	Click **Next**.
i.	Page 9 of 13 – Advanced Configuration	Select **Administration Server.** Click **Next**.
j.	Page 10 of 13 – Administration Server	For **Listen Port,** enter `5001`. Click **Next**.
k.	Page 11 of 13 – Configuration Summary	Click **Create**.
l.	Page 12 of 13 – Configuration Progress	Click **Next.**
m.	Page 13 of 13 – End of Configuration	Click **Finish.**

6. Log in to the WebLogic Server Domain using Fusion Middleware Control.
   a. In the Terminal window navigate to `/u01/fmw/wls1221_infra/user_projects/domains/full_jrf_domain`.
      ```
      $ cd
      /u01/fmw/wls1221_infra/user_projects/domains/full_jrf_domain
      ```
   b. Execute `startWebLogic.sh` to start the WebLogic Administration Server.
      ```
      $ ./startWebLogic.sh
      ```
   c. Open a Browser window and navigate to **http://fmw1:5001/em.**
   d. For **User Name,** enter `weblogic` and for **Password,** enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click **Login.**

      The home page for the **full_jrf_domain** is displayed.

The full_jrf_domain is created and running.

7. Stop the WebLogic Server Administration Server because the restricted JRF domain will be used for later practices.
   a. From the **WebLogic Domain** menu, select **Environment** and then select **Servers.**
   b. Click **AdminServer.**
   c. Click **Control,** click **Shutdown,** and select **Force Shutdown Now.**
   d. Click **Forcibly Shutdown Servers.**

Practices for Lesson 4 - Resource Groups

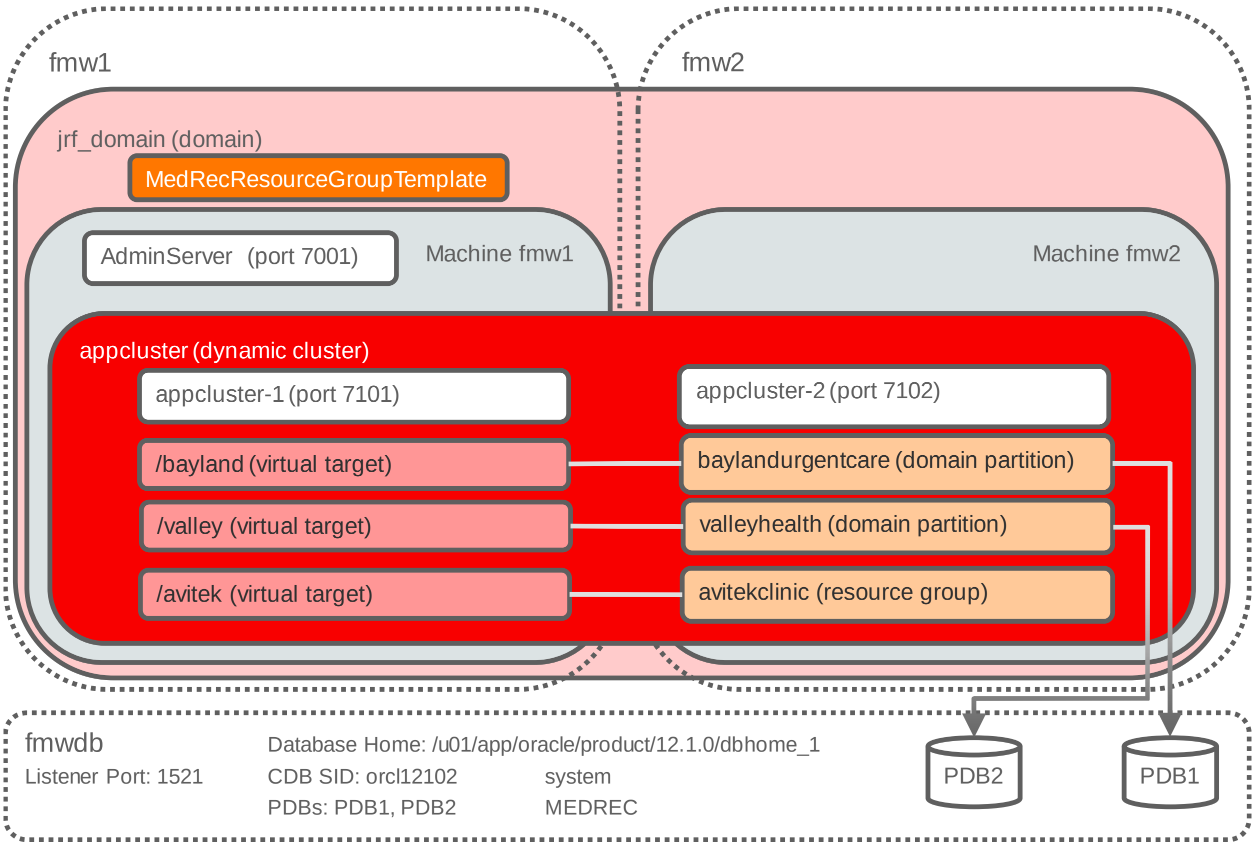
Chapter 4

Practices for Lesson 4: Overview

Overview

In these practices, you create a Resource Group Template with a data source, JMS module and applications for the Avitek Medical Records (MedRec) sample application. Then, you create a Domain Partition that uses the Resource Group Template to create a Resource Group and override properties to meet the tenant requirements.

The following diagram illustrates the configuration:



Practice 4-1: Creating a Resource Group Template

Overview

In this practice, using Fusion Middleware Control, you create a new Resource Group Template and Java application resources including JDBC Data Sources, JMS resources and mail sessions for MedRec sample application deployments.

Because many WebLogic Server customers have requirements for this type of configuration—a Java application with a similar set of resources for each deployment configuration—this is a useful practice to execute the steps to create a Resource Group Template.

Assumptions

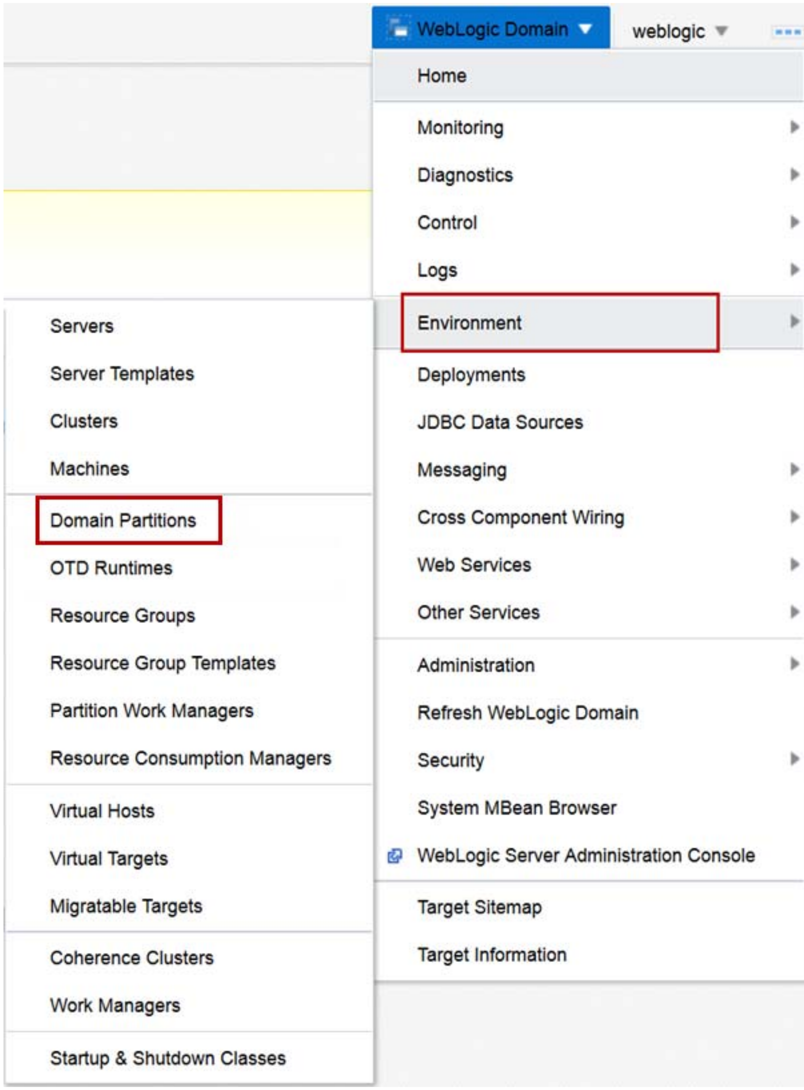
- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf_domain admin server is running.
- The nodemanager for host fmw1, nodemanager-fmw1 is running.
- The nodemanager for host fmw2, nodemanager-fmw2 is running.
- The machines fmw1 and fmw2 have been created.
- The dynamic cluster appcluster has been created.

Tasks

1. Log in to the jrf_domain WebLogic Server Domain using Fusion Middleware Control.
   a. Open a Browser window and navigate to **http://fmw1:7001/em.**
   b. For **User Name,** enter `weblogic` and for **Password,** enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click **Login.**

      The Fusion Middleware Control home page for the **jrf_domain** is displayed.

2. If the domain partitions baylandurgentcare and valleyhealth are still configured and running, stop and delete both partitions.
   a. From the **WebLogic Domain** menu, select **Environment** and then click **Domain Partitions.**

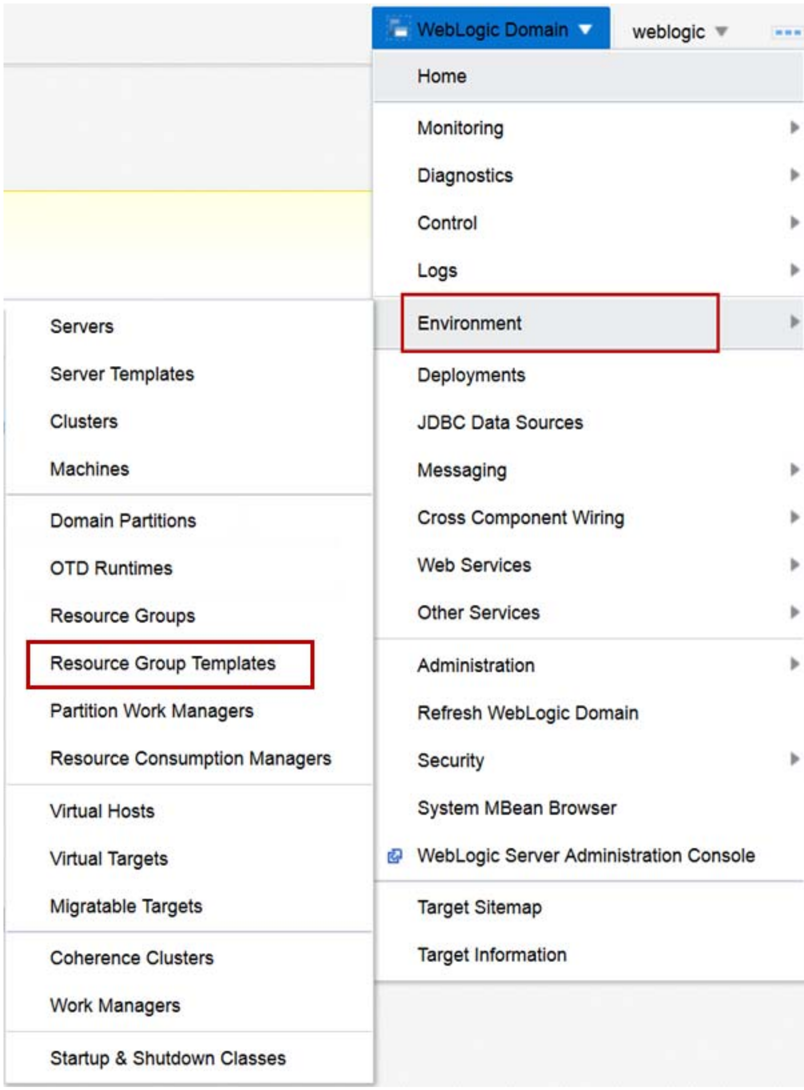


   b. Click the table entry for the **baylandurgentcare** domain partition.
   c. Click **Control,** select **Stop,** and select **Force stop now.**
   d. Click **Close.**
   e. Click the table entry for the **baylandurgentcare** domain partition.
   f. Click **Delete.**
   g. In the **Delete Domain Partition** dialog box, click **OK.**
   h. Click the table entry for the **valleyhealth** domain partition.
   i. Click **Control,** select **Stop,** and select **Force stop now.**
   j. Click **Close.**
   k. Click the table entry for the **valleyhealth** domain partition.
   l. Click **Delete.**
   m. In the **Delete Domain Partition** dialog box, click **OK.**

The baylandurgentcare and valleyhealth domain partitions are deleted—note that you will create these partitions again in these practices—but with a Resource Group Template. Also note that the Virtual Targets for both domain partitions will be used again in these practices.

3. Create a Resource Group Template for a MedRec type application.

   a. From the **WebLogic Domain** menu, select **Environment** and then click **Resource Group Templates.**



   b. Click **Create.**

   c. For **Name,** enter `MedRecResourceGroupTemplate` and click **Create.**

      The Resource Group Template **MedRecResourceGroupTemplate** creates successfully.

4. Create the JDBC Data Source for the MedRecResourceGroupTemplate.

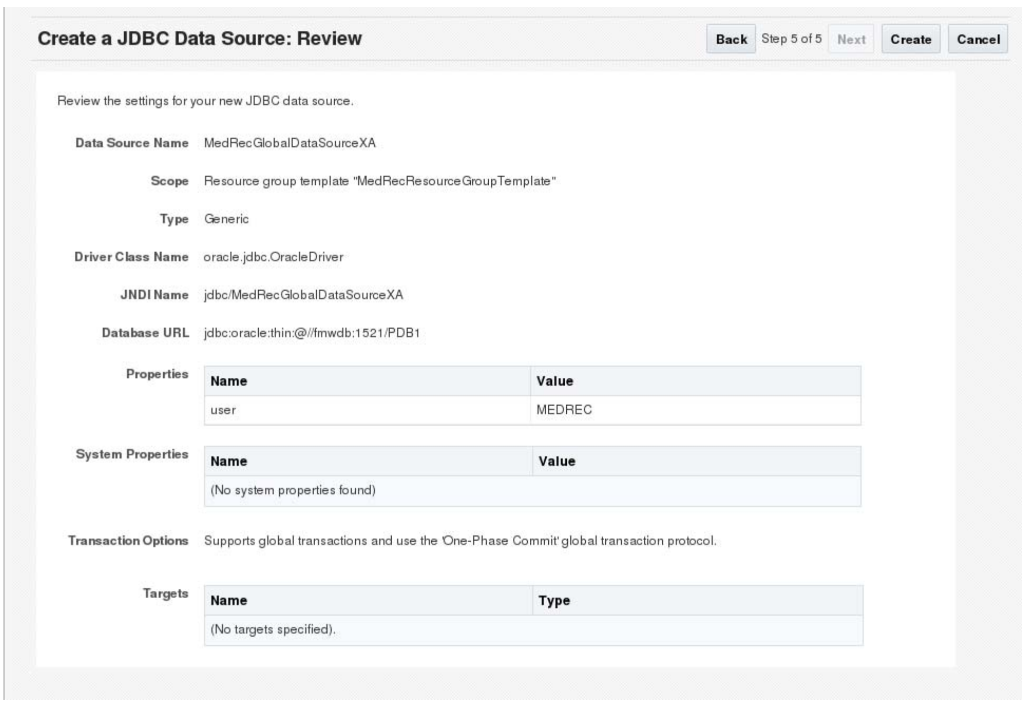
   a. In the Fusion Middleware Control window, click the **MedRecResourceGroupTemplate.**

   b. Click **Services.**

   c. In the **JDBC** tab, click **Create** and select **Generic Data Source.**

   d. For **Name,** enter `MedRecGlobalDataSourceXA`.

e. For **JNDI Name,** enter `jdbc/MedRecGlobalDataSourceXA`.
f. For **Driver Class Name,** select **oracle.jdbc.OracleDriver.**
g. Click **Next.**
h. Click **Generate URL and Properties…**
i. For **Database Host Name,** enter `fmwdb`.
j. For **Database Listener Port,** enter `1521`.
k. For **Database Name,** enter `PDB1`.
l. For **Database User Name,** enter `MEDREC`.
m. For **Password,** enter the password from the Course Practice Environment: Security Credentials document for the MEDREC user for the Oracle Database product.
n. For **Confirm Password,** enter the password from the Course Practice Environment: Security Credentials document for the MEDREC user for the Oracle Database product.
o. Click **OK.**
p. Click **Test Database Connection.**
   The test should succeed.
q. Click **Next.**
r. Accept the **Transaction Properties** defaults and click **Next.**



s. Review the settings and click **Create.**
   The JDBC Data Source MedRecGlobalDataSourceXA creates successfully.

5. Create the JMS Messaging resources for the MedRecResourceGroupTemplate.
   a. In the Fusion Middleware Control window, click the **Messaging** subtab in the **Services** tab of the **MedRecResourceGroupTemplate.**
   b. Click the **JMS Servers** tab.
   c. Click **Create.**
   d. For **Name,** enter `MedRecJMSServer` and click **Create a Store** and select **File Store.**

e. For **Name,** enter `MedRec-fs` and click **Next.**

f. Click **Create.**

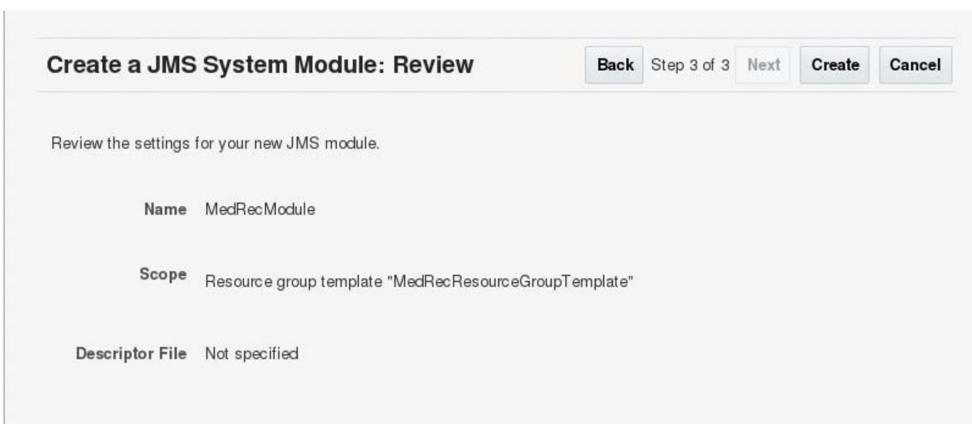
g. Select the **MedRec-fs(File Store)** as the **Persistent Store** and click **Next.**

h. Review the JMS Server settings and click **Create.**

i. Click the **JMS Modules** tab.

j. Click **Create.**

k. For Name, enter `MedRecModule` and click **Next.**



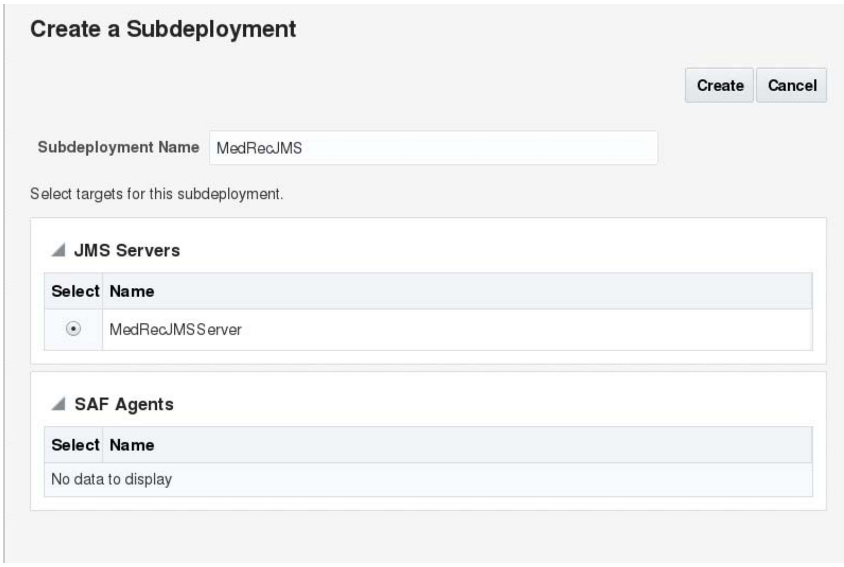
l. Click **Create.**

m. Click the JMS Module MedRecModule.

n. Click the **Subdeployments** tab.

o. Click **Create.**

p. Enter `MedRecJMS` for **SubDeployment Name,** select the **MedRecJMSServer** and click **Create.**



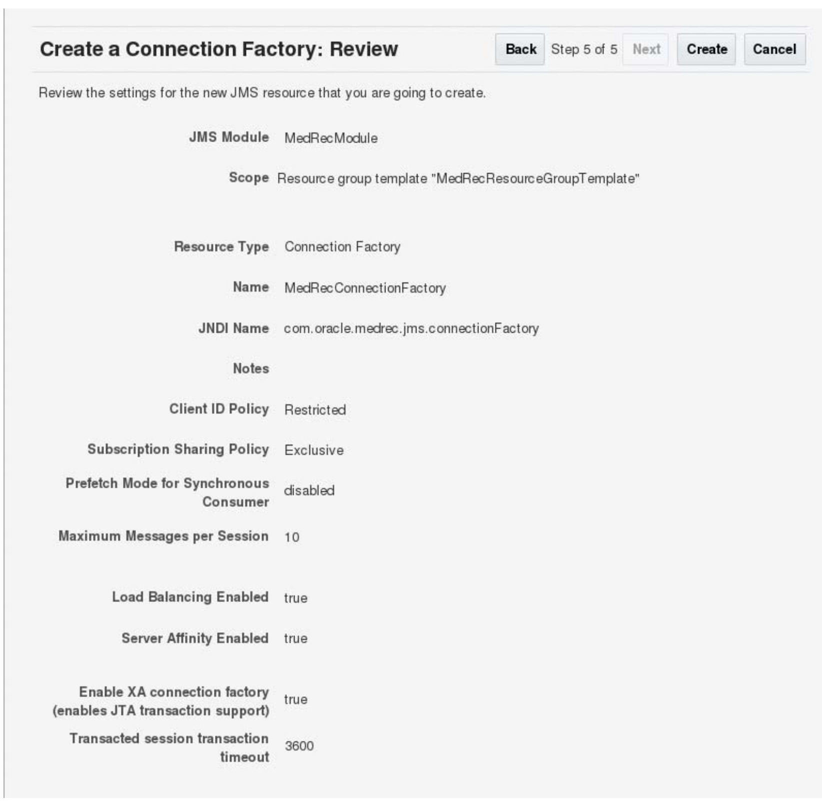
q. Click the **General** tab in the JMS Module **MedRecModule.**

r. Click **Create.**

s. Select **Connection Factory** and click **Next.**

t. Click **Next.**

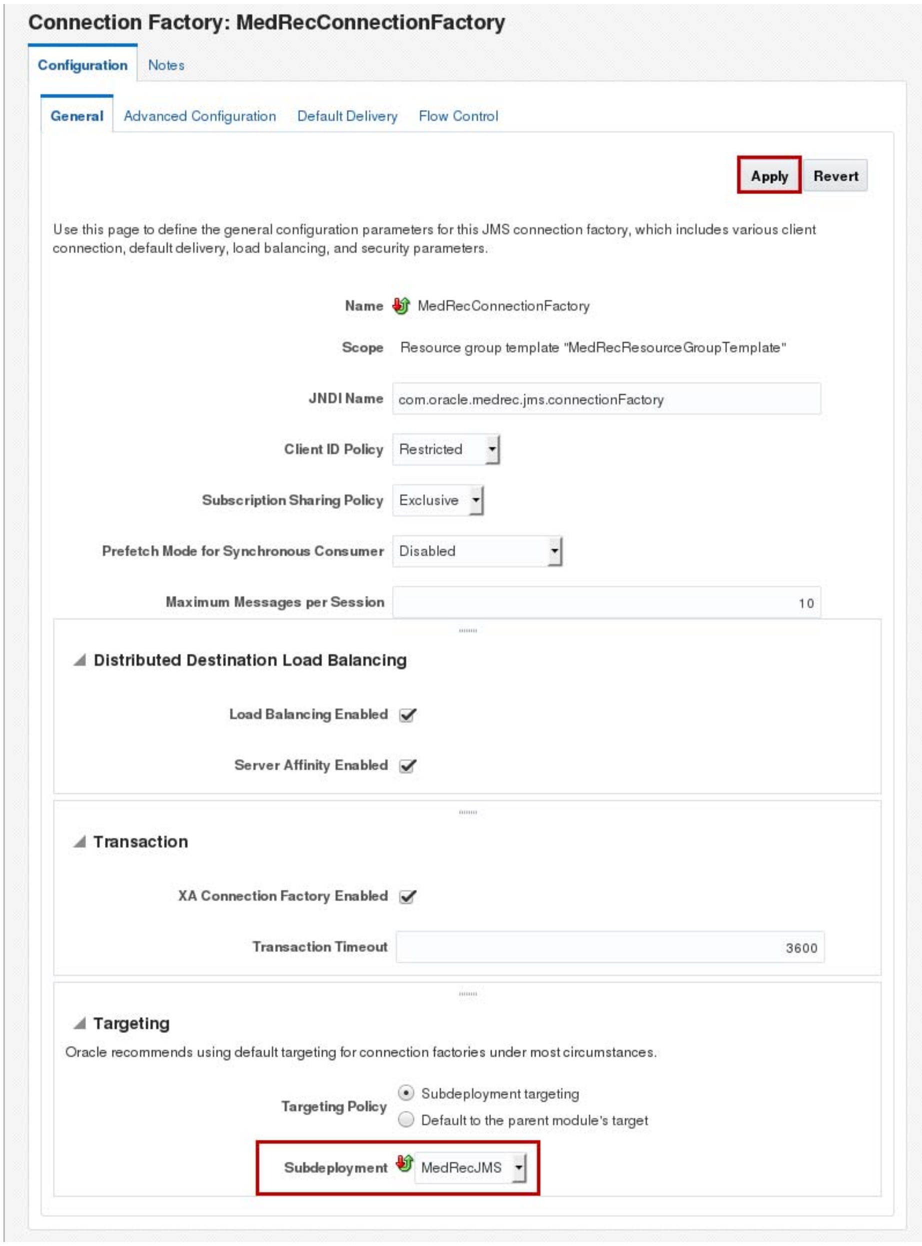
u. Enter `MedRecConnectionFactory` for **Name** and `com.oracle.medrec.jms.connectionFactory` for **JNDI Name.** Click **Next.**



v. Review the Connection Factory settings and click **Create.**

w. Click the **MedRecConnectionFactory.**

x. In the **General** tab, scroll down to the **Targeting** section. Select **Subdeployment targeting** for **Targeting Policy** and select the **MedRecJMS** for **Subdeployment** and click **Apply.**



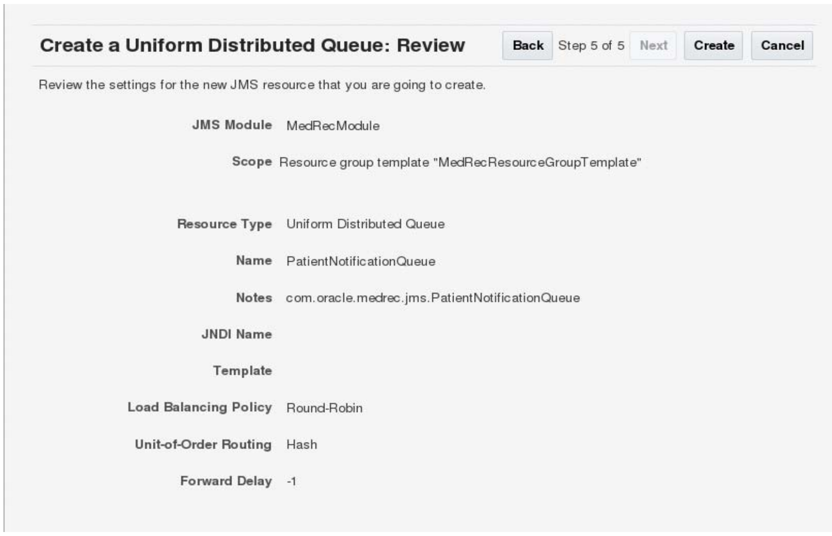
y. Click the **General** tab in the JMS Module **MedRecModule.**

z. Click **Create.**

aa. Select **Uniform Distributed Queue** and click **Next.**

bb. Click Next.

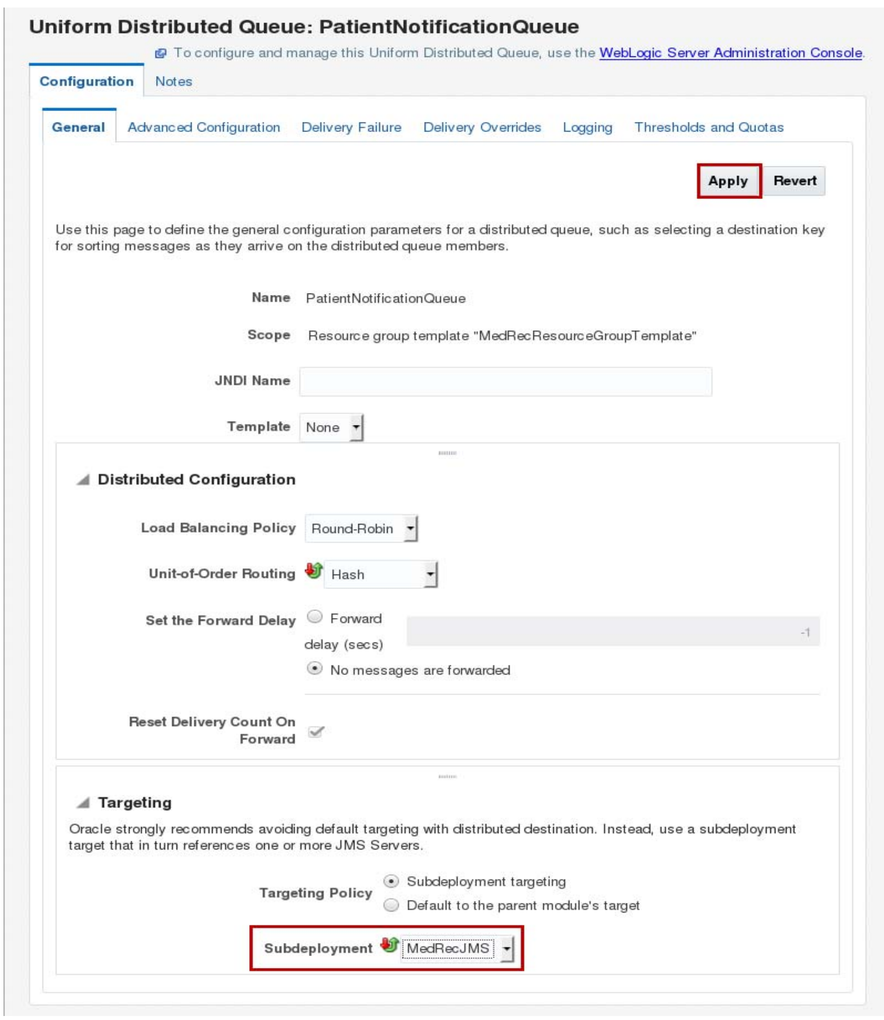
cc. Enter `PatientNotificationQueue` for **Name** and `com.oracle.medrec.jms.PatientNotificationQueue` for **JNDI Name.** Click **Next.**



dd. Review the Uniform Distributed Queue settings and click **Create.**

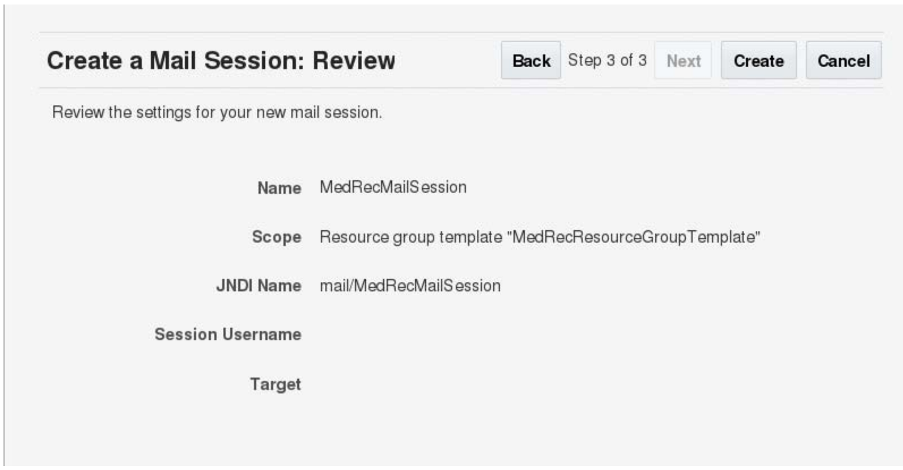
ee. Click the **PatientNotificationQueue.**

ff. Scroll down to the **Targeting** section, select **Subdeployment targeting** for **Targeting Policy,** and select the **MedRedJMS** for **Subdeployment** and click **Apply.**



The JMS Server and resources have been created.

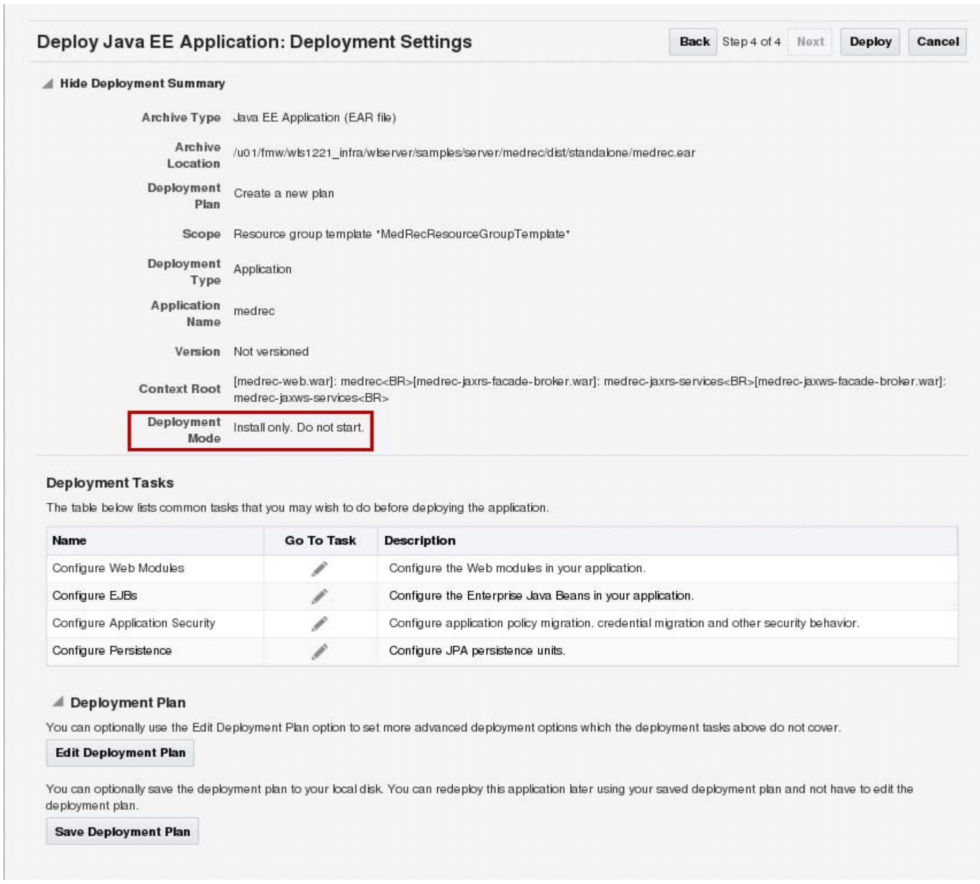
6. Create the Mail session for the MedRecResourceGroupTemplate.
   a. In the Fusion Middleware Control window, click the **Mail** subtab in the **Services** tab of the **MedRecResourceGroupTemplate.**
   b. Click **Create.**
   c. Enter `MedRecMailSession` for **Name** and enter `mail/MedRecMailSession` for **JNDI Name**.
   d. Select **Resource Group Template** for **Scope** and select the **MedRecResourceGroupTemplate.**
   e. Click **Next.**



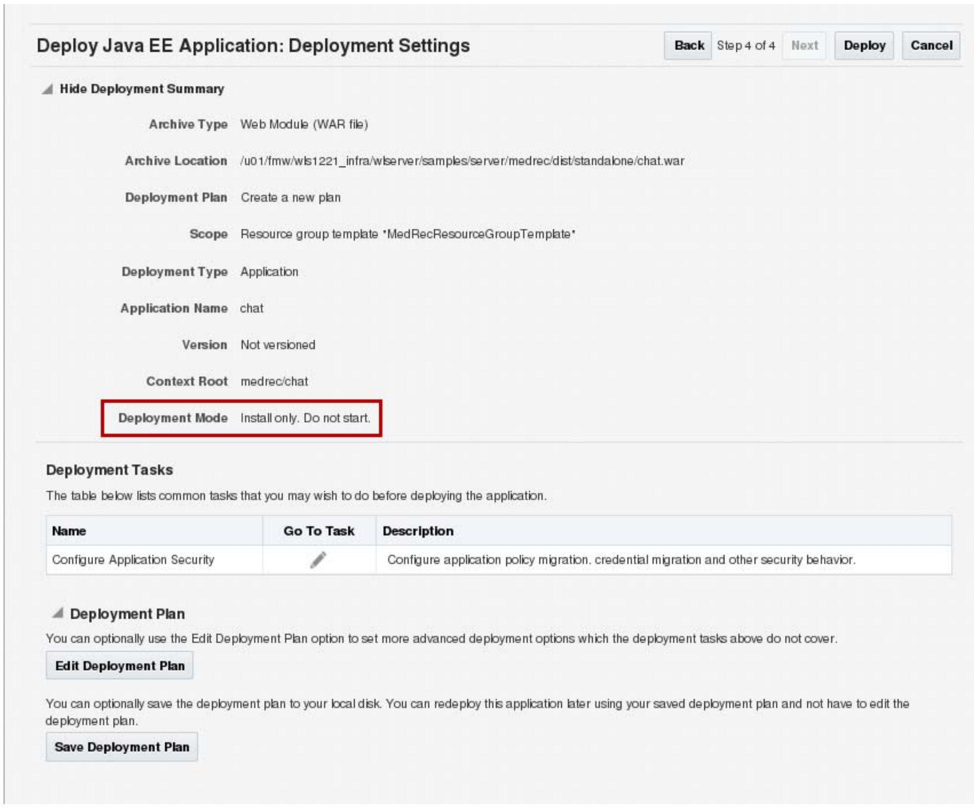
   f. Click **Create.**

      The mail session for the MedRecResourceGroupTemplate is created.

7. Deploy the MedRec Java applications to the MedRecResourceGroupTemplate.
   a. In the Fusion Middleware Control window, click the **Deployments** tab of the **MedRecResourceGroupTemplate.**
   b. Click **Deployment** and select **Deploy.**
   c. Click **Archive is on the machine where this Web browser is running.**
   d. Click **Browse…**
   e. Navigate to the **`/u01/fmw/wls1221_infra/wlserver/samples/server/medrec/dist/standalone`** directory, select the **`medrec.ear`** file and click **Open.**
   f. Click **Next.**



g. Review the Application Attributes and click **Deploy.**
h. After the application successfully deploys, click **Close.**
i. Click **Deployment** and select **Deploy.**
j. Click **Archive is on the machine where this Web browser is running.**
k. Click **Browse…**
l. Navigate to the **`/u01/fmw/wls1221_infra/wlserver/samples/server/medrec/dist/standalone`** directory, select the **`chat.war`** file and click **Open.**
m. Click **Next.**



n. Review the Application Attributes and click **Deploy.**

o. After the application is successfully deployed, click **Close.**

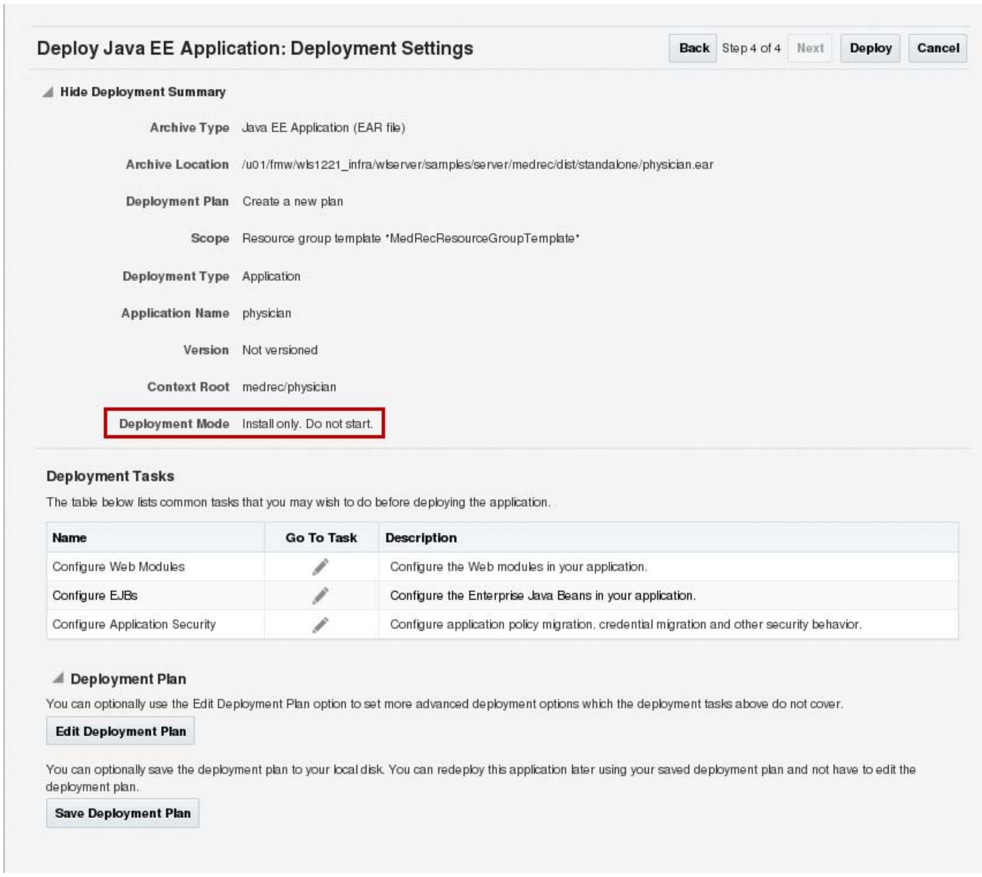
p. Click **Deployment** and select **Deploy.**

q. Click **Archive is on the machine where this Web browser is running.**

r. Click **Browse…**

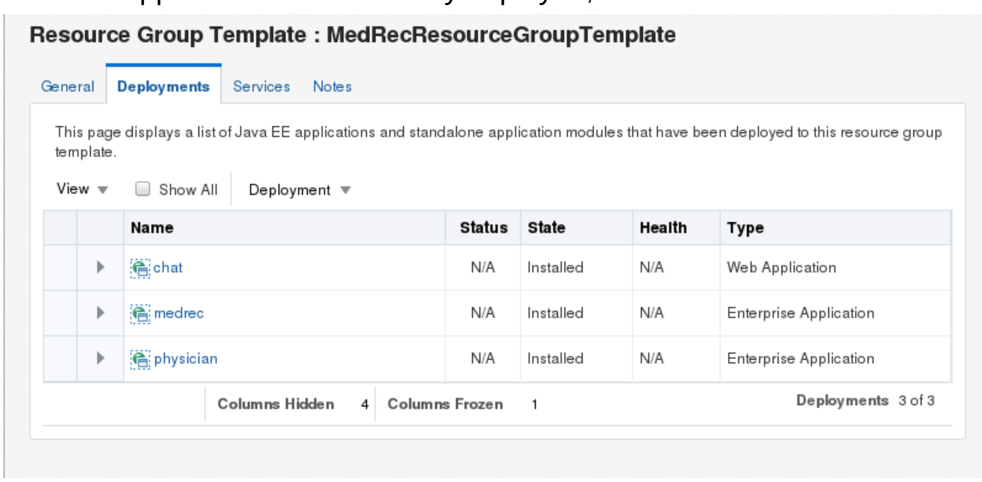
s. Navigate to the **/u01/fmw/wls1221_infra/wlserver/samples/server/medrec/dist/standalone** directory, select the **physician.ear** file and click **Open.**

t. Click **Next.**



u. Review the Application Attributes and click **Deploy.**

v. After the application is successfully deployed, click **Close.**



The MedRec applications are successfully deployed to the MedRecGroupTemplate.

Practice Solution: Creating a Resource Group Template

Perform the following tasks if you did not complete this practice and want to use the finished solution.

Note: If you started to create the Resource Group Template, delete the partially created Resource Group Template before executing the solution.

Assumptions

- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf_domain admin server is running.
- The nodemanager for host fmw1, nodemanager-fmw1 is running.
- The nodemanager for host fmw2, nodemanager-fmw2 is running.
- The machines fm1 and fmw2 have been created.
- The dynamic cluster appcluster has been created.

Solution Tasks

1. Connect to fmw1 as the `oracle` user.
2. Run the solution script.
   a. Open a Terminal window as the oracle user and navigate to the practice directory.
      ```
      $ cd /u01/practices/part1/practice04-01
      ```
   b. Run the solution script.
      ```
      $ ./solution.sh
      ```
   Note: This script starts the WebLogic Scripting Tool (WLST) passing it the WLST script `create_rgt.py`, which creates the resource group template for you. The WLST script prints out messages. Be patient and do not close the Terminal window until you see these messages:
      ```
      ...
      >>>Solution for Practice 04-01 applied successfully!

      Exiting WebLogic Scripting Tool.
      ```
3. Close the Terminal window.

Practice 4-2: Creating a Domain Partition from a Resource Group Template

Overview

In this practice, you create the security realms for baylandurgentcare and valleyhealth – because it is required that different administrators work with each application. Because the virtual targets for baylandurgentcare and valleyhealth were created in the practices for the lesson titled “Domain Partitions,” they are reused here. Finally, you create the domain partitions using the MedRecResourceGroupTemplate.

Assumptions

- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf_domain admin server is running.
- The nodemanager for host fmw1, nodemanager-fmw1 is running.
- The nodemanager for host fmw2, nodemanager-fmw2 is running.
- The machines fm1 and fmw2 have been created.
- The dynamic cluster appcluster has been created.
- The resource group template MedRecResourceGroupTemplate has been created.

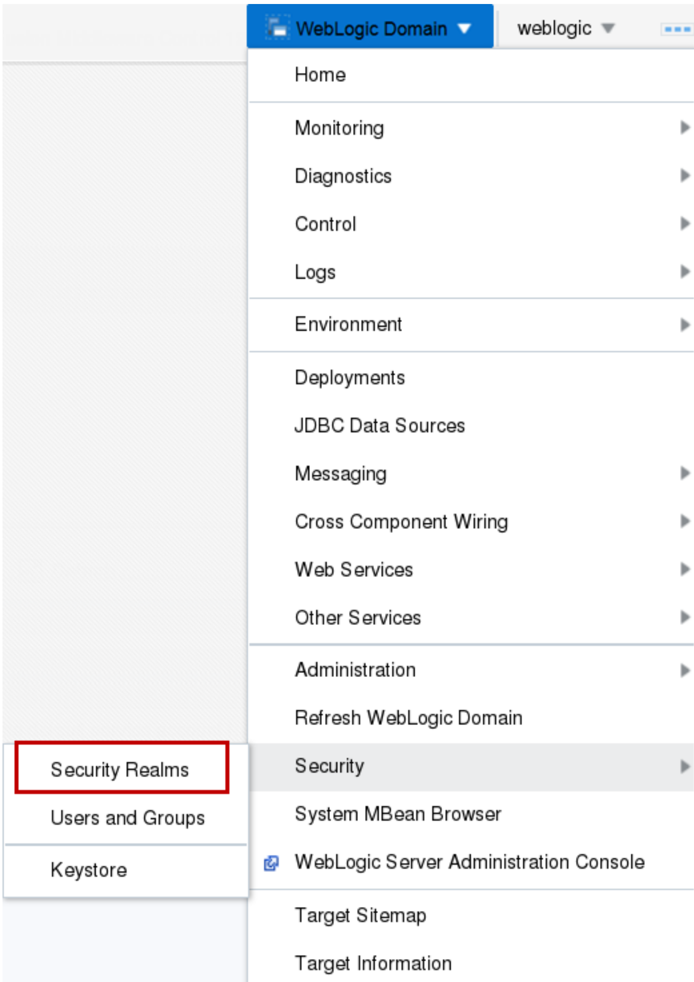
Tasks

1. Log in to the jrf_domain WebLogic Server domain using Fusion Middleware Control.
   a. Open a Browser window and navigate to **http://fmw1:7001/em.**
   b. For **User Name,** enter `weblogic` and for **Password,** enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click **Login.**

      The Fusion Middleware Control home page for the **jrf_domain** is displayed.

2. Create a Security Realm for baylandurgentcare.

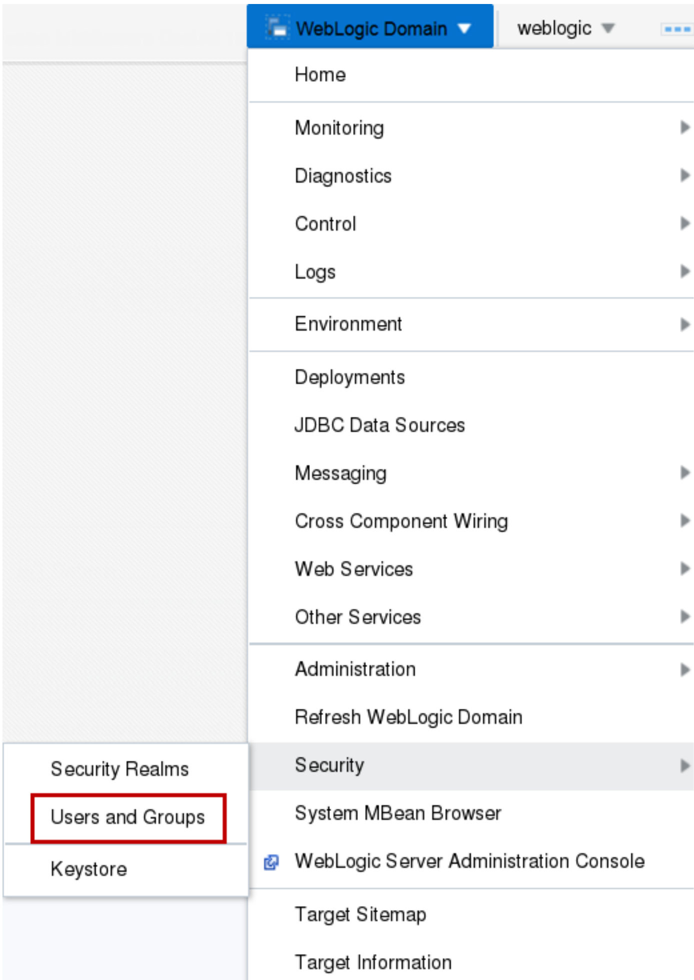
   a. In Fusion Middleware Control, from the **WebLogic Domain** menu, select **Security** and click **Security Realms.**



   b. Click **Create.**

   c. For **Name,** enter `bayland` and click **Create.**

d. From the **WebLogic Domain** menu, select **Security** and click **Users and Groups.**



e. In the **Select a Realm** drop-down menu, select the **bayland** realm.

f. Click **Create.**

g. For **Name,** enter `administrator`.

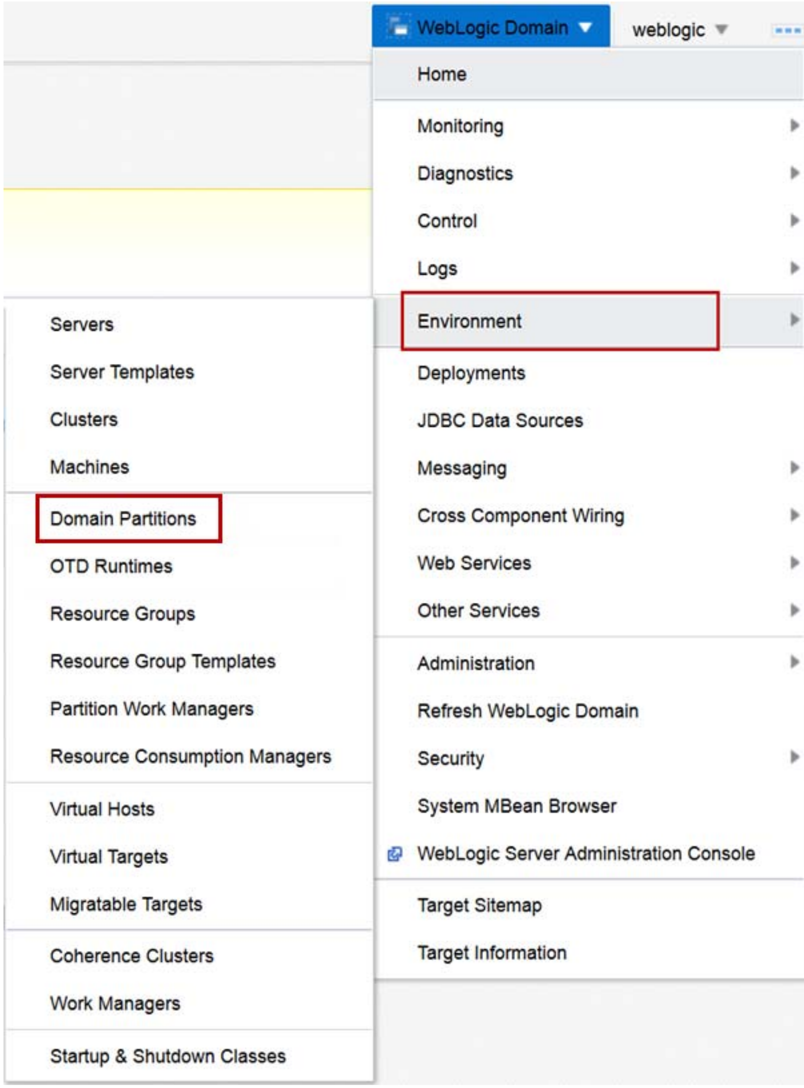
h. For **Password** and **Confirm Password**, enter the password from the Course Practice Environment: Security Credentials document for the administrator user for the Oracle WebLogic Server product (bayland Security Realm).

i. Click **Create.**

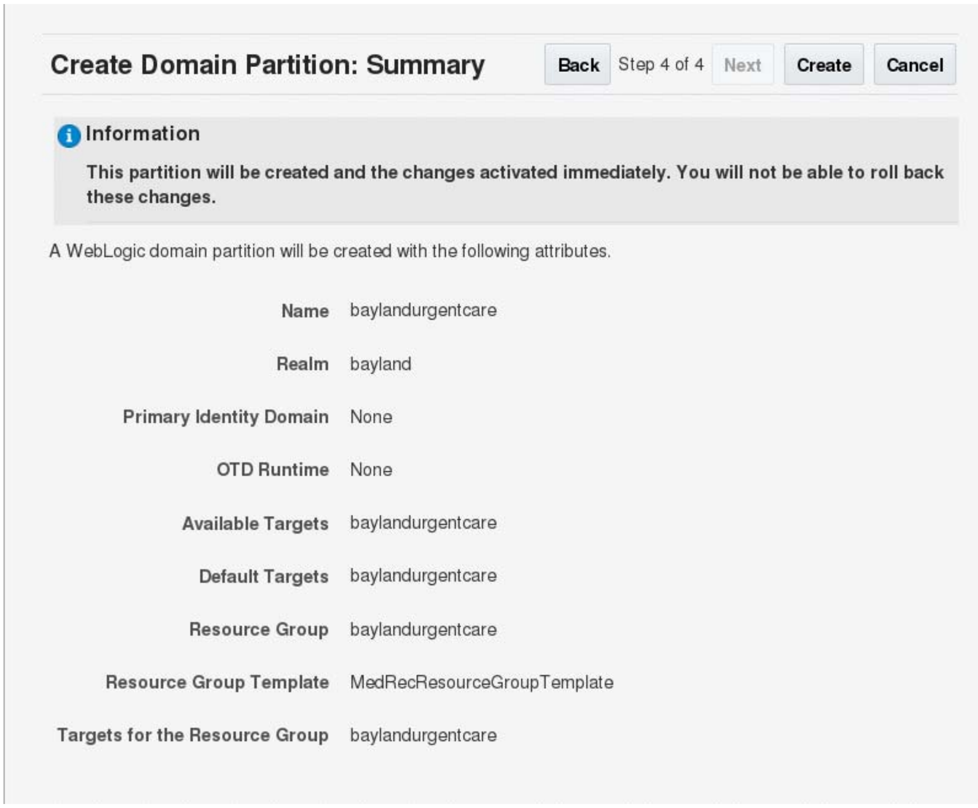
The Security Realm **bayland** and the user **administrator** for the MedRec are created successfully.

3. Create the baylandurgentcare domain partition.

   a. From the **WebLogic Domain** menu, select **Environment** and then click **Domain Partitions.**



   b. Click **Create.**
   c. For **Name,** enter `baylandurgentcare`.
   d. For **Security Realm,** select bayland.
   e. Click **Next.**
   f. For the Virtual Target baylandurgentcare, choose both **Select** and **Set as Default** and click **Next.**
   g. For **Resource Group Name,** enter `baylandurgetcare`.
   h. For **Resource Group Template,** select **MecRedResourceGroupTemplate.**
   i. Move the **baylandurgentcare** target to **Selected Targets.**
   j. Click **Next.**



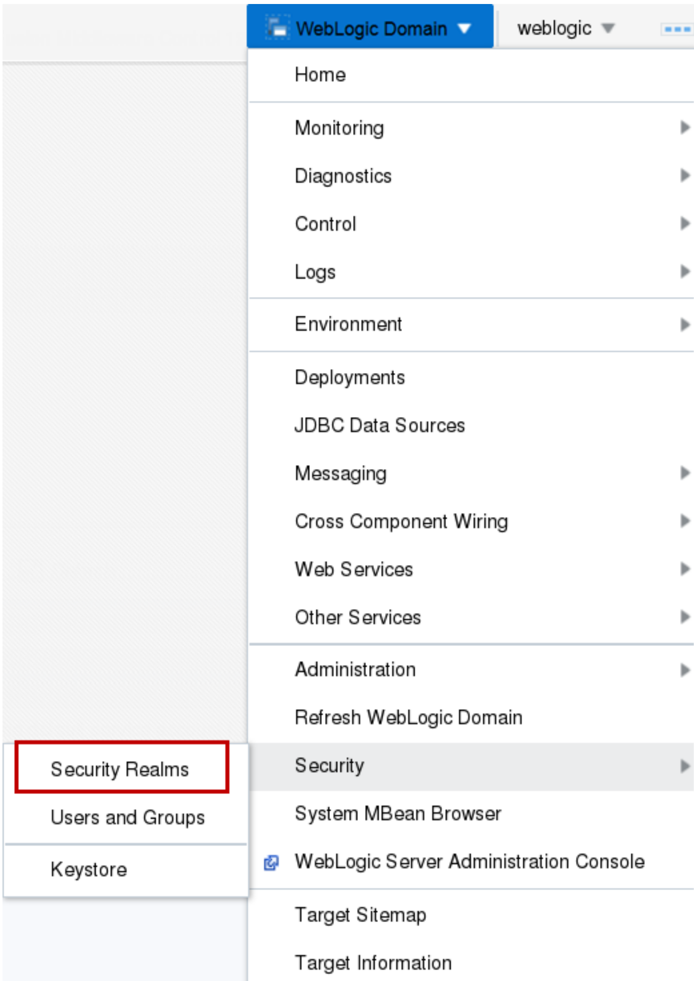
k. Review the domain partition summary and click **Create.**

The baylandurgentcare domain partition is created.

4. Start the baylandurgentcare domain partition.
   a. In the list of Domain Partitions, click the **baylandurgentcare** entry.
   b. Click **Control** and click **Start**.
   c. When the Confirmation dialog box indicates that the Start Operation is successful, click **Close.**

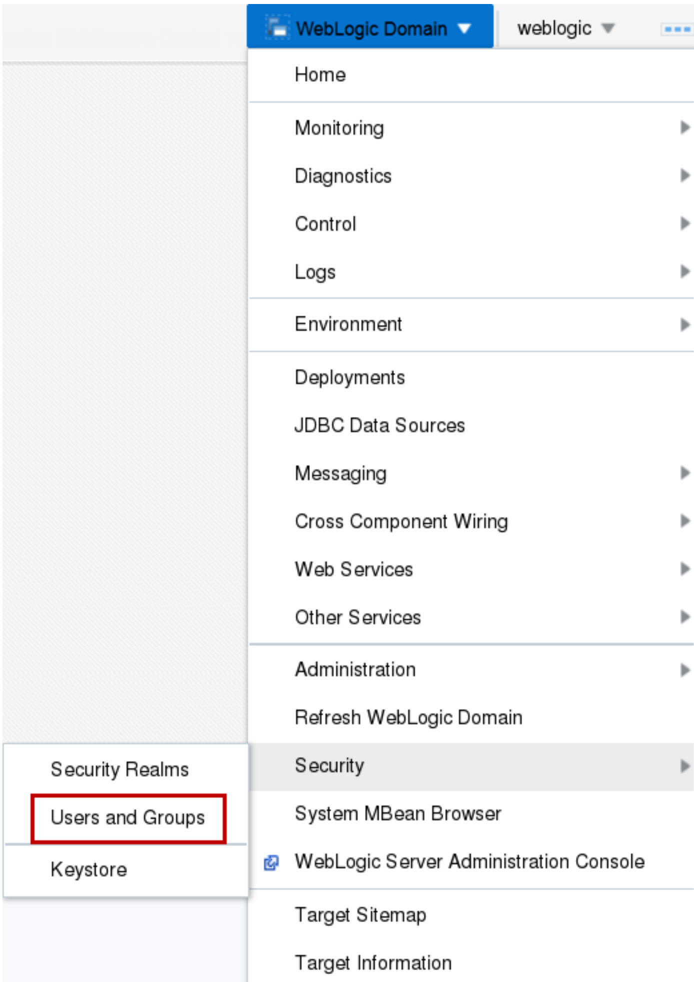
      The baylandurgentcare domain partition successfully starts.
5. Test the baylandurgentcare domain partition.
   a. In the Browser window, navigate to **http://fmw1:7101/bayland/medrec**.

6. Create a Security Realm for valleyhealth.
   a. In Fusion Middleware Control, from the **WebLogic Domain** menu, select **Security** and click **Security Realms.**

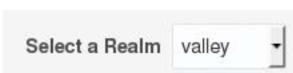


   b. Click **Create.**
   c. For **Name,** enter `valley` and click **Create.**

d. From the **WebLogic Domain** menu, select **Security** and click **Users and Groups.**



e. In the **Select a Realm** dropdown, select the **valley** realm.



f. Click **Create.**

g. For **Name,** enter `administrator`.

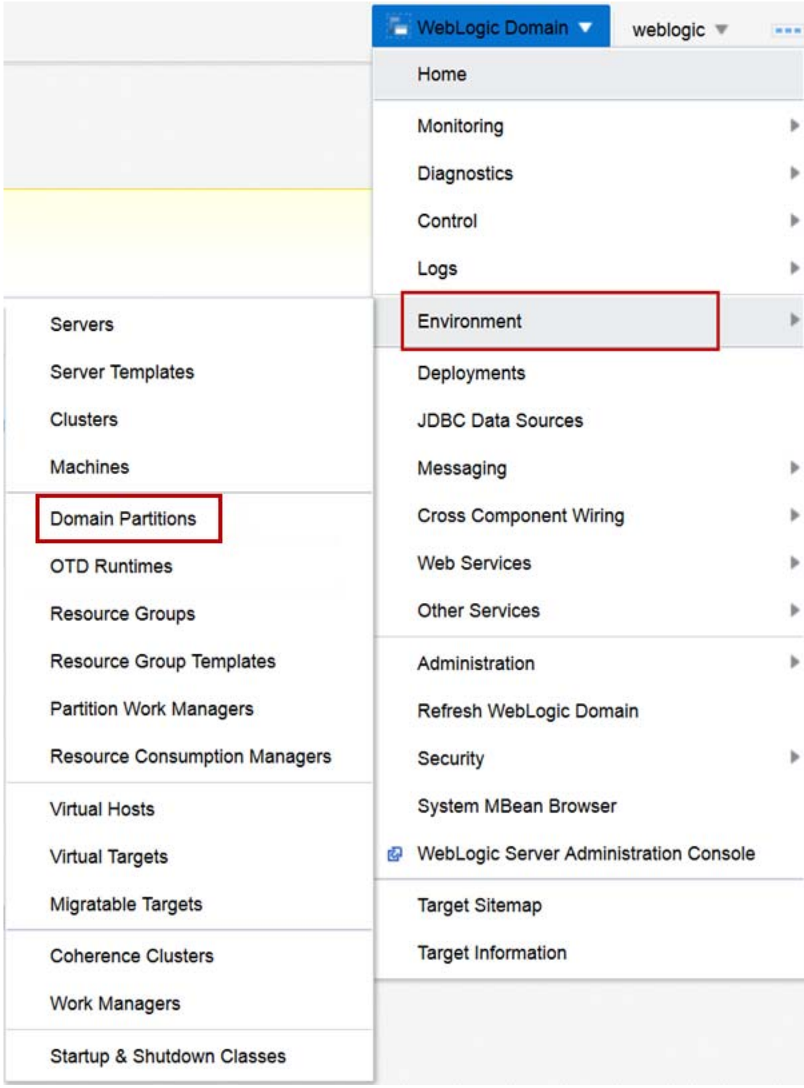
h. For **Password** and **Confirm Password**, enter the password from the Course Practice Environment: Security Credentials document for the administrator user for the Oracle WebLogic Server product (valley Security Realm).

i. Click **Create.**

The Security Realm **valley** and the user **administrator** for the MedRec application are created successfully.

7. Create the valleyhealth domain partition.

   a. From the **WebLogic Domain** menu, select **Environment** and then click **Domain Partitions.**



   b. Click **Create.**

   c. For **Name,** enter `valleyhealth`.

   d. For **Security Realm,** select **valley.**

   e. Click **Next.**

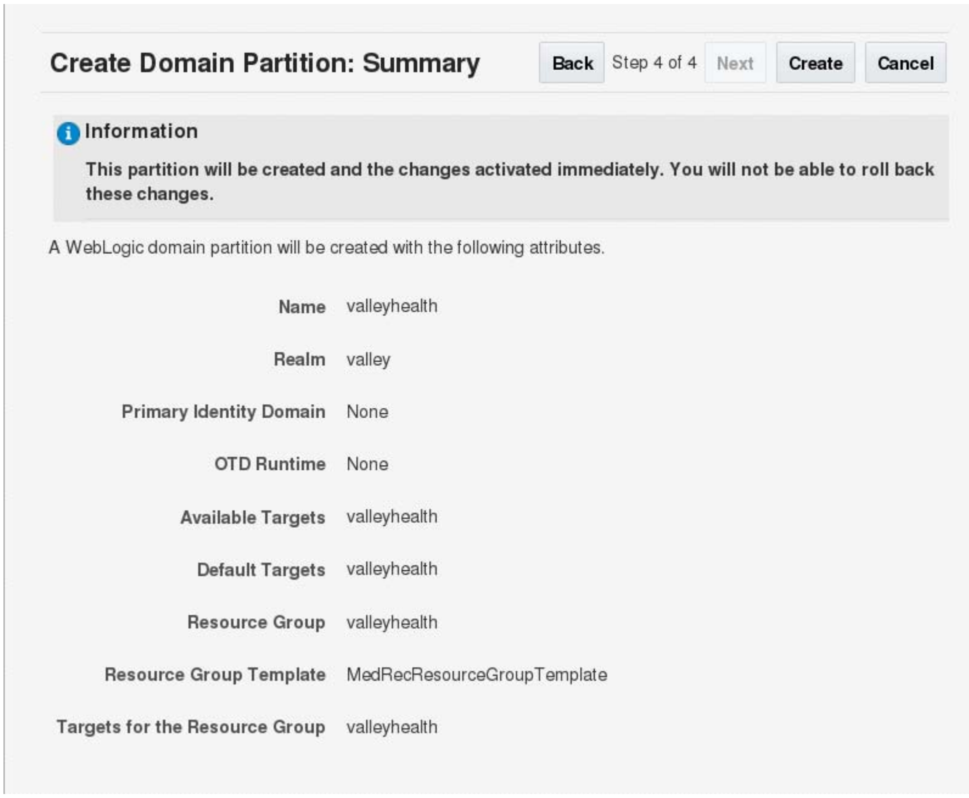
   f. For the Virtual Target **valleyhealth**, check both **Select** and **Set as Default** and click **Next.**

   g. For **Resource Group Name,** enter `valleyhealth`.

   h. For **Resource Group Template,** select **MecRedResourceGroupTemplate.**

   i. Move the **valleyhealth** target to **Selected Targets.**

   j. Click **Next.**



k. Review the domain partition summary and click **Create.**

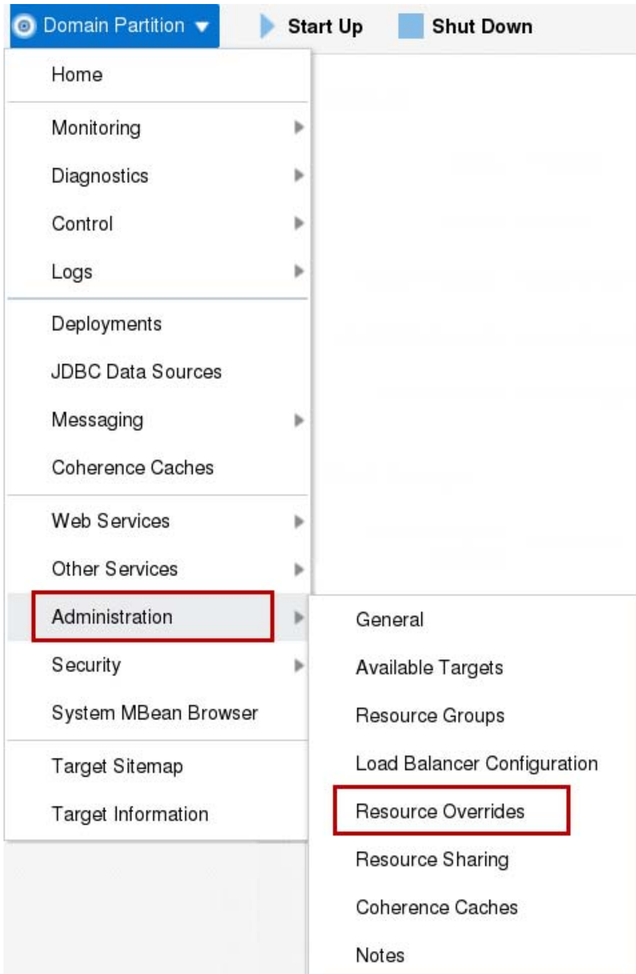
   The valleyhealth domain partition is created.

   However, before starting the valleyhealth domain partition, change the JDBC connection to use a different pluggable database, namely PDB2.

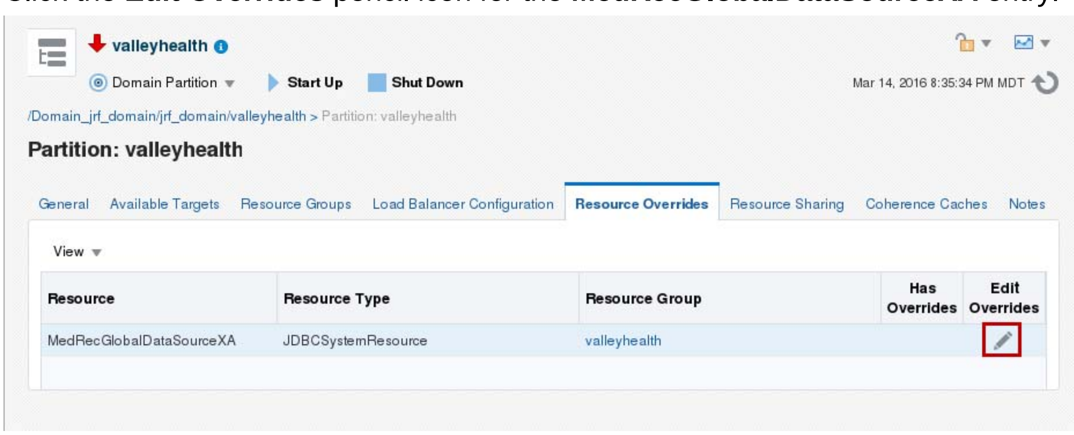
8. Override the JBDC connection settings for valleyhealth domain partition Resource Group.

   a. In the list of **Domain Partitions,** click the valleyhealth domain partition.

b. From the **Domain Partition** menu, select **Administration** and then select **Resource Overrides.**



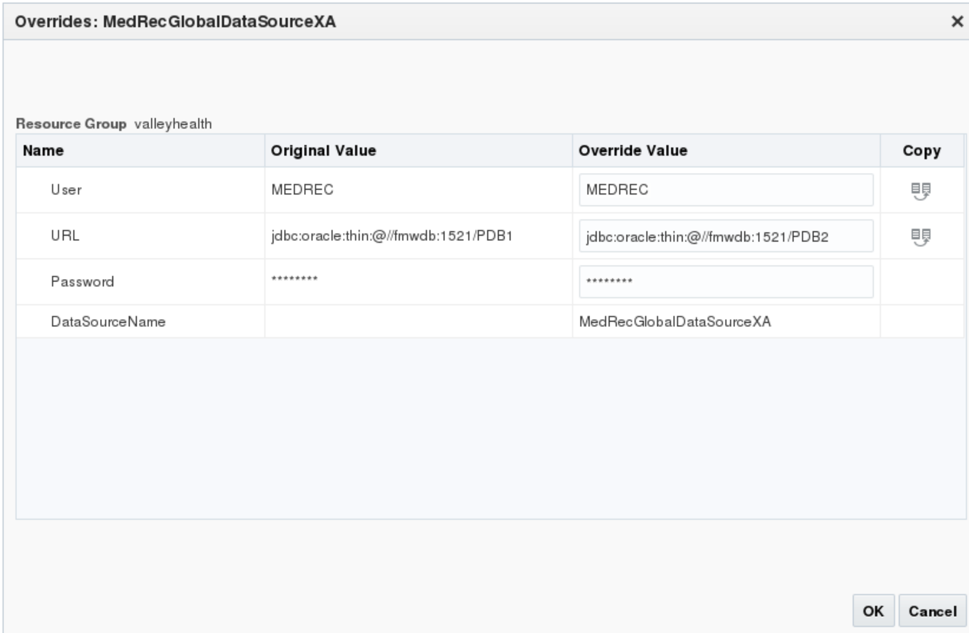
c. Click the **Edit Overrides** pencil icon for the **MedRecGlobalDataSourceXA** entry.



d. Override the **User** property with the value `MEDREC`.

e. Override the **URL** property with the value `jdbc:oracle:thin:@//fmwdb:1521/PDB2`.

f. Override the **Password** property with value for the password from the Course Practice Environment: Security Credentials document for the MEDREC user for the Oracle Database product.



g. Click **OK.**

9. Start the **valleyhealth** domain partition.
   a. Next to the **Domain Partition** menu, click **Start Up.**
   b. After the **Domain Partition** starts, click **Close.**
10. Test the **valleyhealth** domain partition.
    a. In the Browser window, navigate to **http://fmw1:7101/valley/medrec.**

Practice Solution: Creating a Domain Partition from a Resource Group Template

Perform the following tasks if you did not complete this practice and want to use the finished solution.

Note: If you started creating domain partitions, this script will not work. To run the script in that case, delete the partially created domain partitions before running this script.

Assumptions

- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf_domain admin server is running.
- The nodemanager for host fmw1, nodemanager-fmw1 is running.
- The nodemanager for host fmw2, nodemanager-fmw2 is running.
- The machines fm1 and fmw2 have been created.
- The dynamic cluster appcluster has been created.
- The resource group template MedRecResourceGroupTemplate has been created.

Solution Tasks

1. Connect to fmw1 as the `oracle` user.
2. Run the solution script.
   a. Open a Terminal window as the `oracle` user and navigate to the practice directory.
   ```
   $ cd /u01/practices/part1/practice04-02
   ```
   b. Run the solution script.
   ```
   $ ./solution.sh
   ```
   Note: This script starts the WebLogic Scripting Tool (WLST) passing it the WLST script `create_domain_partitions_from_rgt.py`, which creates the domain partitions for you. The WLST script prints out messages. Be patient and do not close the Terminal window until you see these messages:
   ```
   ...
   >>> Solution for Practice 04-02 applied successfully!

   Exiting WebLogic Scripting Tool.
   ```
3. Close the Terminal window.

Practice 4-3: Creating a Resource Group from a Resource Group Template

Overview

In this practice, you create a resource group at the WebLogic Server domain level using a resource group template. Resource groups can be used within domain partitions or standalone, within a WebLogic Server domain, to contain a related set of resources and access them from a virtual target.

Assumptions

- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf_domain admin server is running.
- The nodemanager for host fmw1, nodemanager-fmw1 is running.
- The nodemanager for host fmw2, nodemanager-fmw2 is running.
- The machines fm1 and fmw2 have been created.
- The dynamic cluster appcluster has been created.
- The resource group template MedRecResourceGroupTemplate has been created.

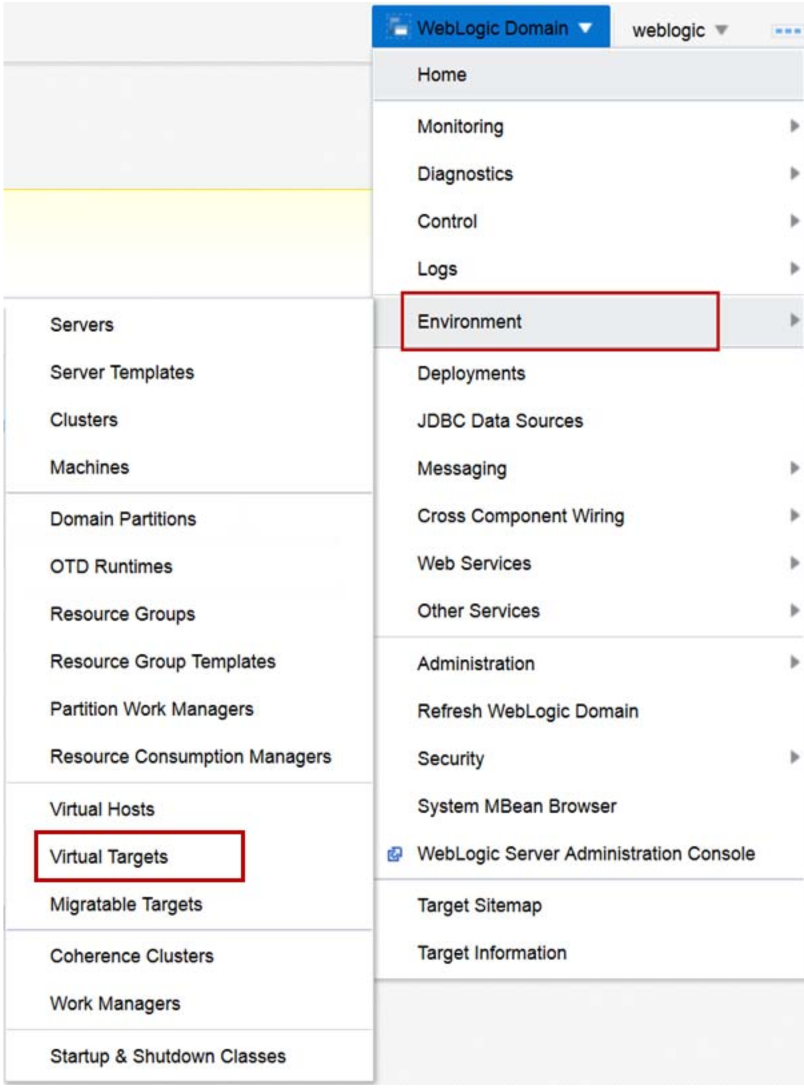
Tasks

1. Log in to the jrf_domain WebLogic Server domain using Fusion Middleware Control.
   a. Open a Browser window and navigate to **http://fmw1:7001/em.**
   b. For **User Name,** enter `weblogic` and for **Password,** enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click **Login.**

      The Fusion Middleware Control home page for the **jrf_domain** is displayed.

2. Create a Virtual Target for avitekclinic.

   a. In Fusion Middleware Control, from the **WebLogic Domain** menu, select **Environment** and click **Virtual Targets.**



   b. Click **Create.**

   c. For **Name,** enter `avitek.`

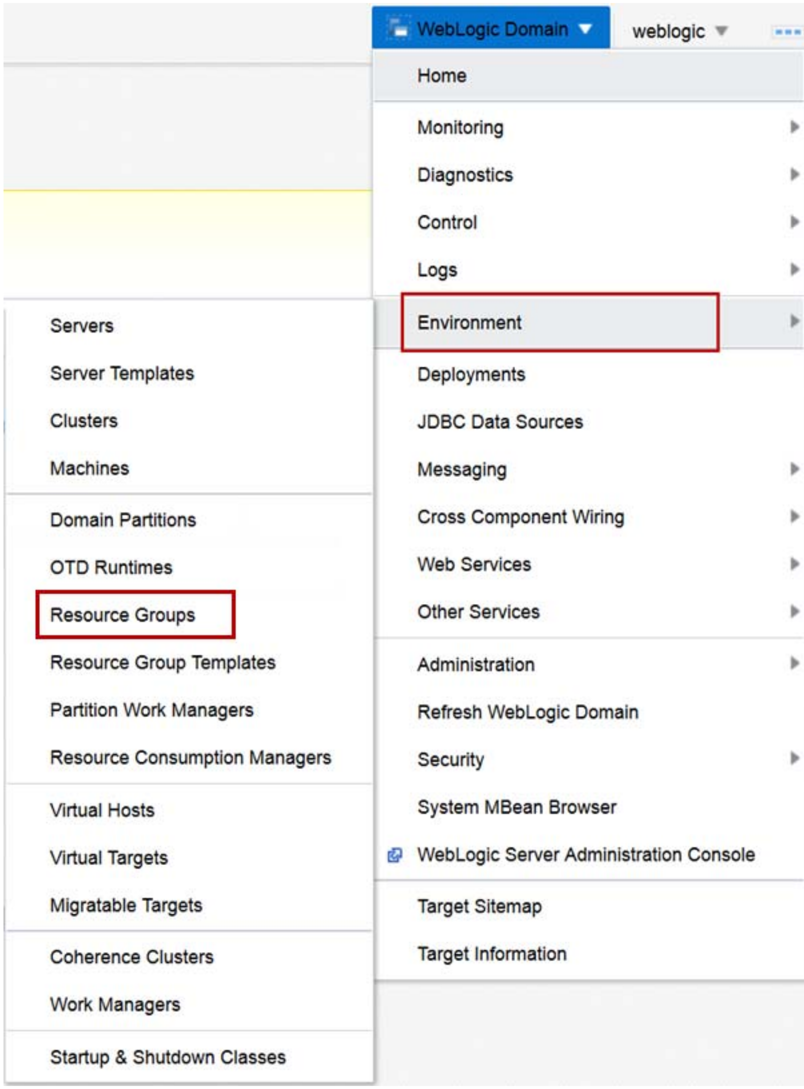
   d. For **Jri Prefix,** enter `/avitek.`

   e. Click **Next.**

   f. From the **Cluster** menu, select **appcluster.**

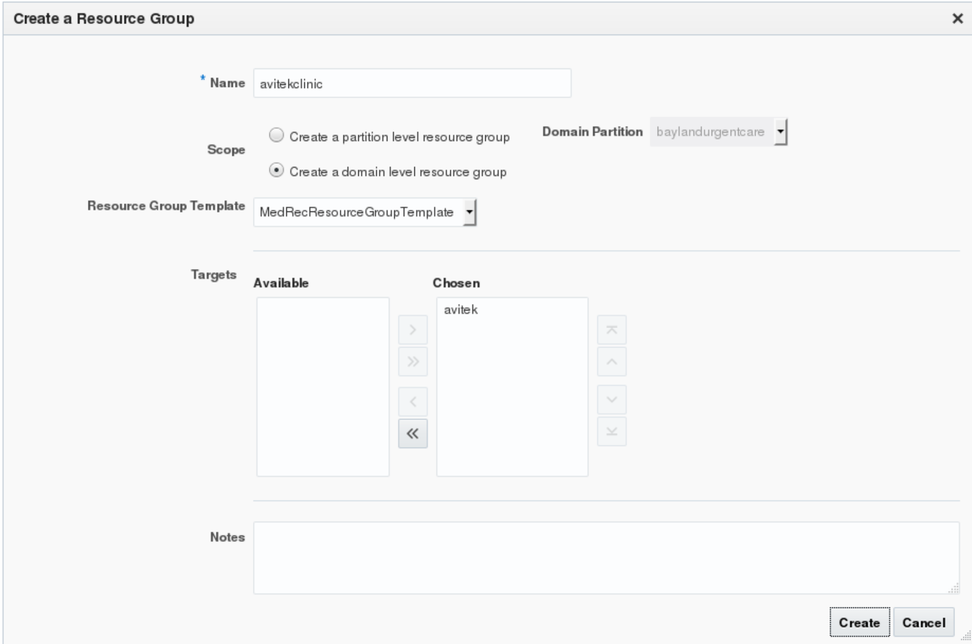
   g. Click **Create.**

3. Create a Resource Group for avitekclinic.
   a. In Fusion Middleware Control, from the **WebLogic Domain** menu, select **Environment** and click **Resource Groups.**



   b. Click **Create.**
   c. For **Name,** enter `avitekclinic.`
   d. For **Scope,** click **Create a domain level resource group.**
   e. For **Resource Group Template,** select **MedRecResourceGroupTemplate.**

f. In **Targets,** select **avitekclinic** and move it to the **Chosen** Targets.



g. Click **Create.**

The Resource Group **avitekclinic** is created successfully and is started.

4. Test the avitekclinic Resource Group.

a. In the Browser window, navigate to **http://fmw1:7101/avitek/medrec.**

Practice Solution: Creating a Resource Group from a Resource Group Template

Perform the following tasks if you did not complete this practice and want to use the finished solution.

Note: If you started creating a resource group, this script will not work. To run the script in that case, delete the partially created resource group prior to running this script.

Assumptions

- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf_domain admin server is running.
- The nodemanager for host fmw1, nodemanager-fmw1 is running.
- The nodemanager for host fmw2, nodemanager-fmw2 is running.
- The machines fm1 and fmw2 have been created.
- The dynamic cluster appcluster has been created.
- The resource group template MedRecResourceGroupTemplate has been created.

Solution Tasks

1. Connect to fmw1 as the `oracle` user.
2. Run the solution script.
   a. Open a Terminal window as the `oracle` user and navigate to the practice directory.
   ```
   $ cd /u01/practices/part1/practice04-03
   ```
   b. Run the solution script.
   ```
   $ ./solution.sh
   ```
   Note: This script starts the WebLogic Scripting Tool (WLST) passing it the WLST script `create_rg_from_rgt.py`, which creates the new generic data source for you. The WLST script prints out messages. Be patient and do not close the Terminal window until you see these messages:
   ```
   ...
   >>> Solution for Practice 04-03 applied successfully!

   Exiting WebLogic Scripting Tool.
   ```
3. Close the Terminal window.

Practices for Lesson 5 - Oracle Traffic Director

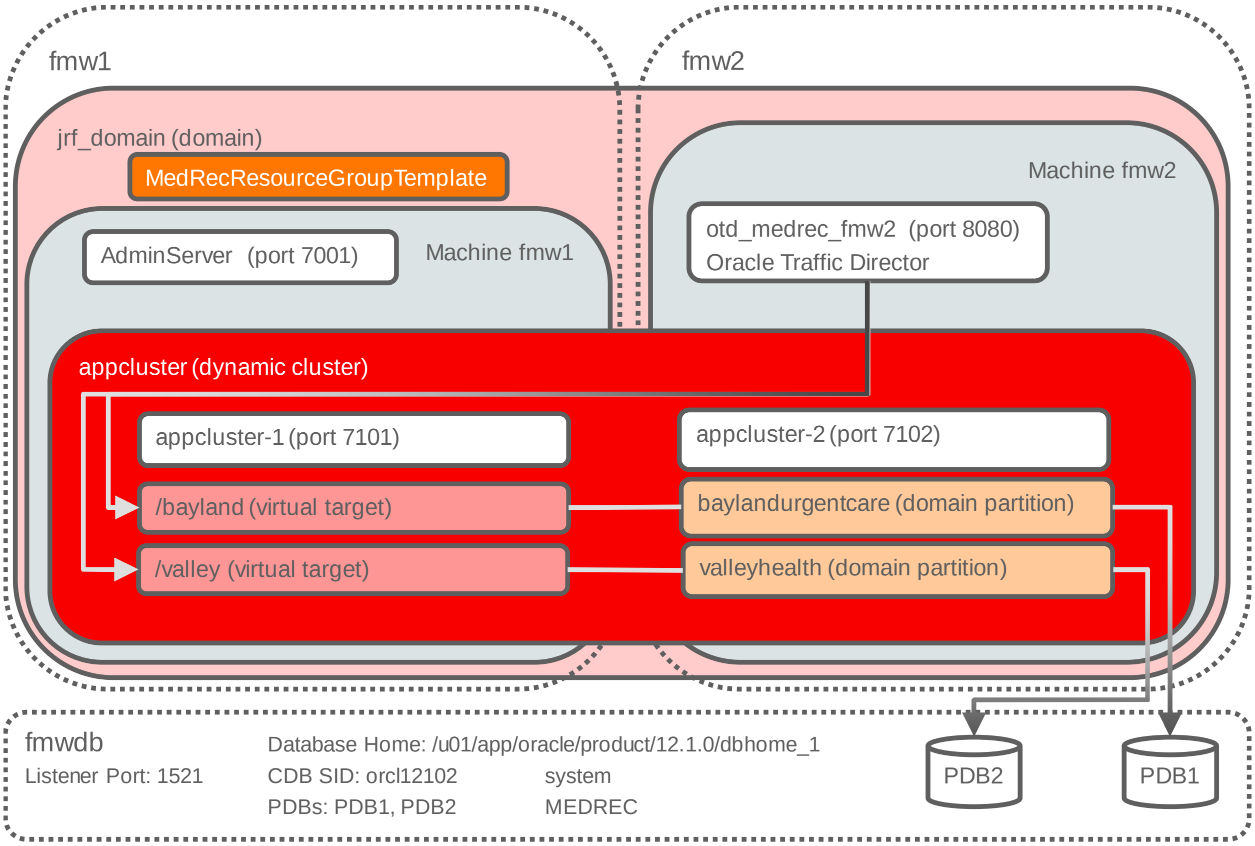
Chapter 5

Practices for Lesson 5: Overview

Overview

In these practices, you create an Oracle Traffic Director configuration and register the Oracle Traffic Director Runtime with WebLogic Server. Then, you create a domain partition that uses the Resource Group Template and Oracle Traffic Director configuration. Finally, you set the Oracle Traffic Director logging levels.

The following diagram illustrates the configuration:



Practice 5-1: Creating an Oracle Traffic Director Configuration

Overview

In this practice, using Fusion Middleware Control, you create an Oracle Traffic Director configuration that you use in the MedRec applications.

Assumptions

- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf_domain admin server is running.
- The nodemanager for host fmw1, nodemanager-fmw1 is running.
- The nodemanager for host fmw2, nodemanager-fmw2 is running.
- The machines fm1 and fmw2 have been created.
- The dynamic cluster appcluster has been created.
- The resource group template MedRecResourceGroupTemplate has been created.

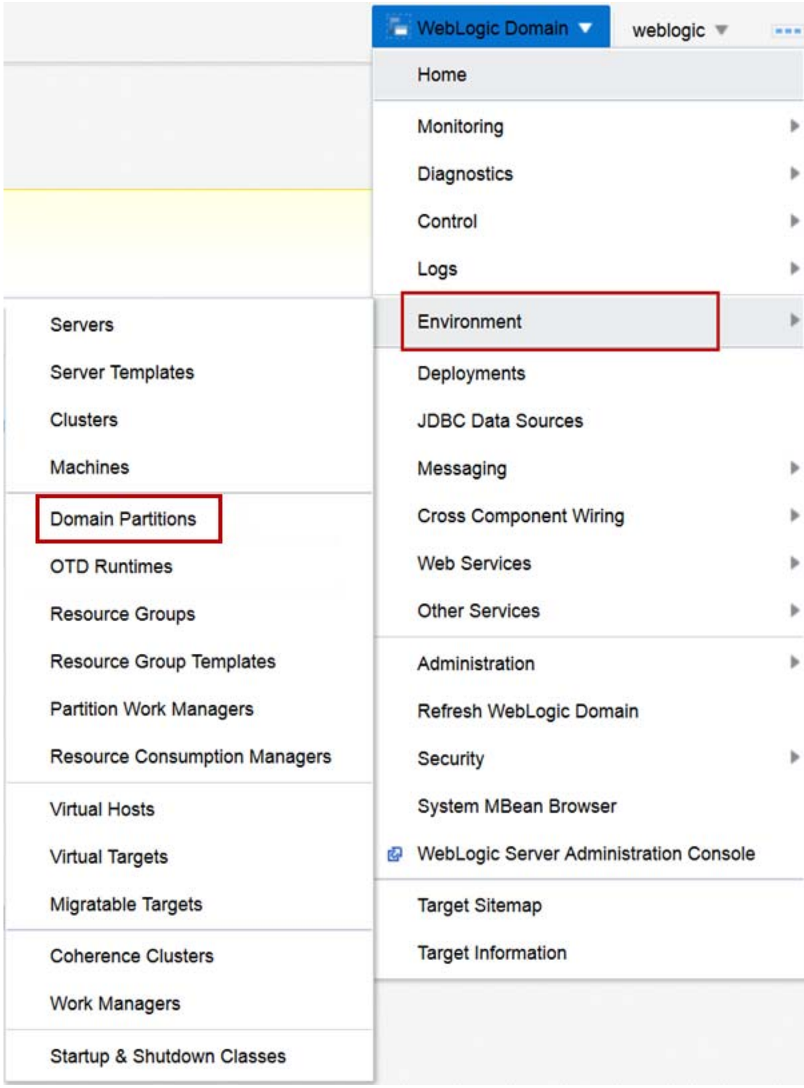
Tasks

1. Log in to the jrf_domain WebLogic Server Domain using Fusion Middleware Control.
   a. Open a browser window and navigate to **http://fmw1:7001/em.**
   b. For **User Name,** enter `weblogic` and for **Password,** enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click **Login.**

      The Fusion Middleware Control home page for the **jrf_domain** is displayed.

2. If the domain partitions baylandurgentcare and valleyhealth are still configured and running, stop and delete them.

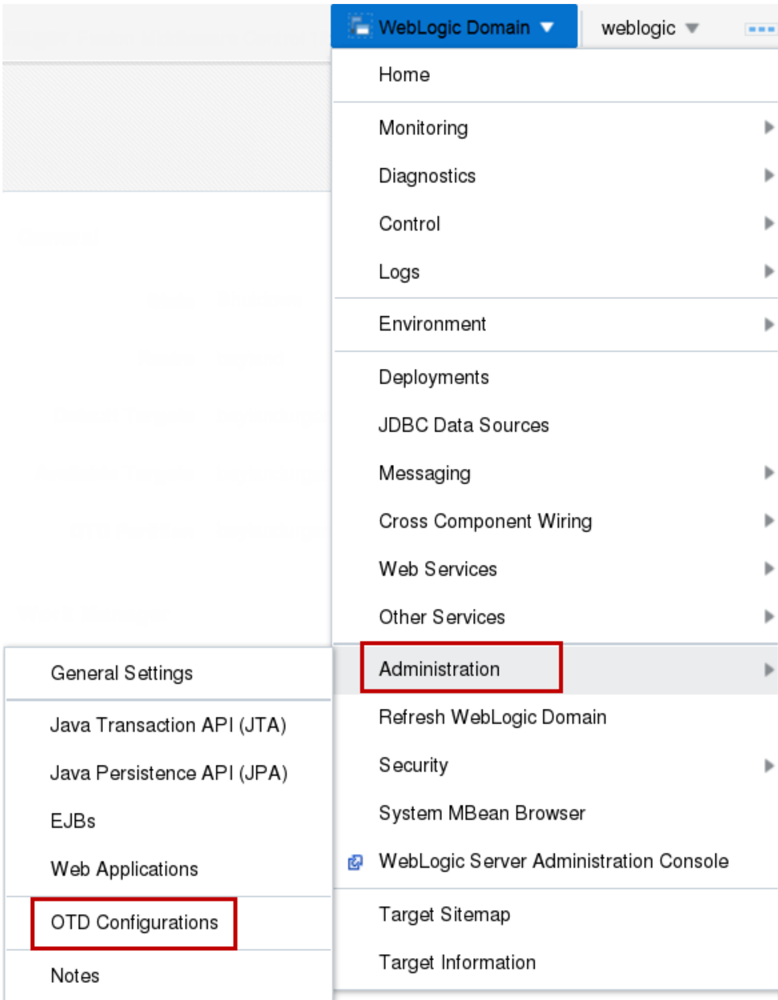
   a. From the **WebLogic Domain** menu, select **Environment** and then click **Domain Partitions.**



b. Click the table entry for the **baylandurgentcare** domain partition.
c. Click **Control,** select **Stop,** and select **Force stop now.**
d. Click **Close.**
e. Click the table entry for the **baylandurgentcare** domain partition.
f. Click **Delete.**
g. In the **Delete Domain Partition** dialog box, click **OK.**
h. Click the table entry for the **valleyhealth** domain partition.
i. Click **Control,** select **Stop,** and select **Force stop now.**
j. Click **Close.**
k. Click the table entry for the **valleyhealth** domain partition.
l. Click **Delete.**
m. In the **Delete Domain Partition** dialog box, click **OK.**

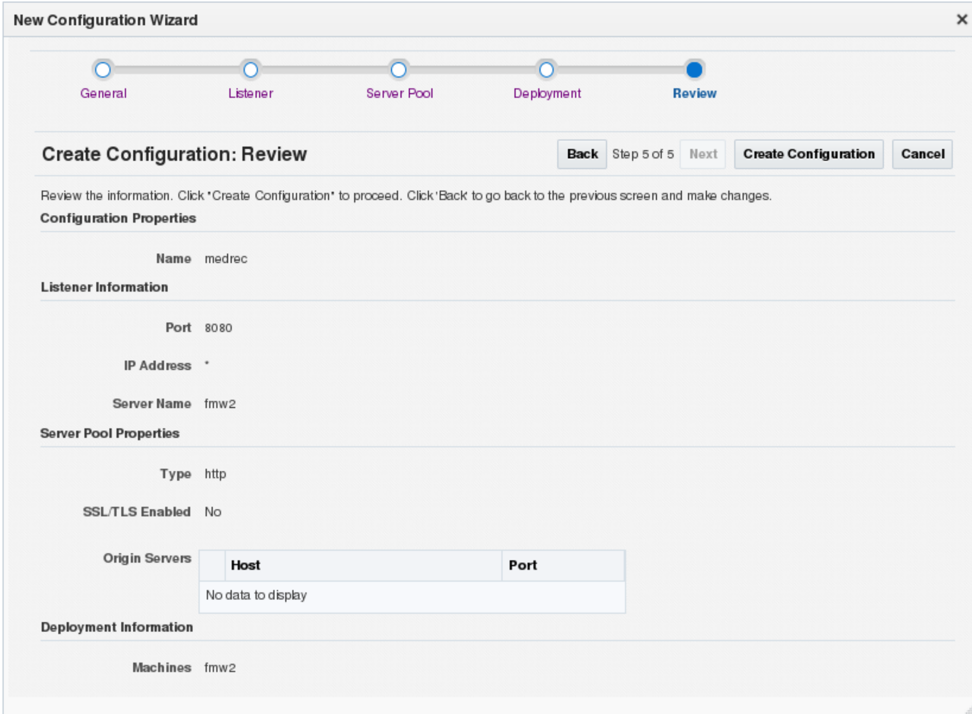
   The baylandurgentcare and valleyhealth domain partitions are deleted—note that you will create these partitions again in these practices—but with an Oracle Traffic Director configuration and a Resource Group Template. Also, note that the Virtual Targets for both domain partitions will be used again in these practices.

3. Create an Oracle Traffic Director Configuration.
   a. From the **WebLogic Domain** menu, select **Administration** and then click **OTD Configurations.**



   b. Click **Create.**
   c. For **Name,** enter `medrec` and click **Next.**
   d. For **Port,** enter `8080`.
   e. For **IP Address**, enter `*`.
   f. For **Server Name,** enter `fmw2` and click **Next.**
   g. In the **Server Pool** configuration, click **Next.**

      Note that the Origin Server pool will be created with the domain partitions by the Lifecycle Manager.
   h. For **Deployment**, select the **fmw2** machine and click **Next.**



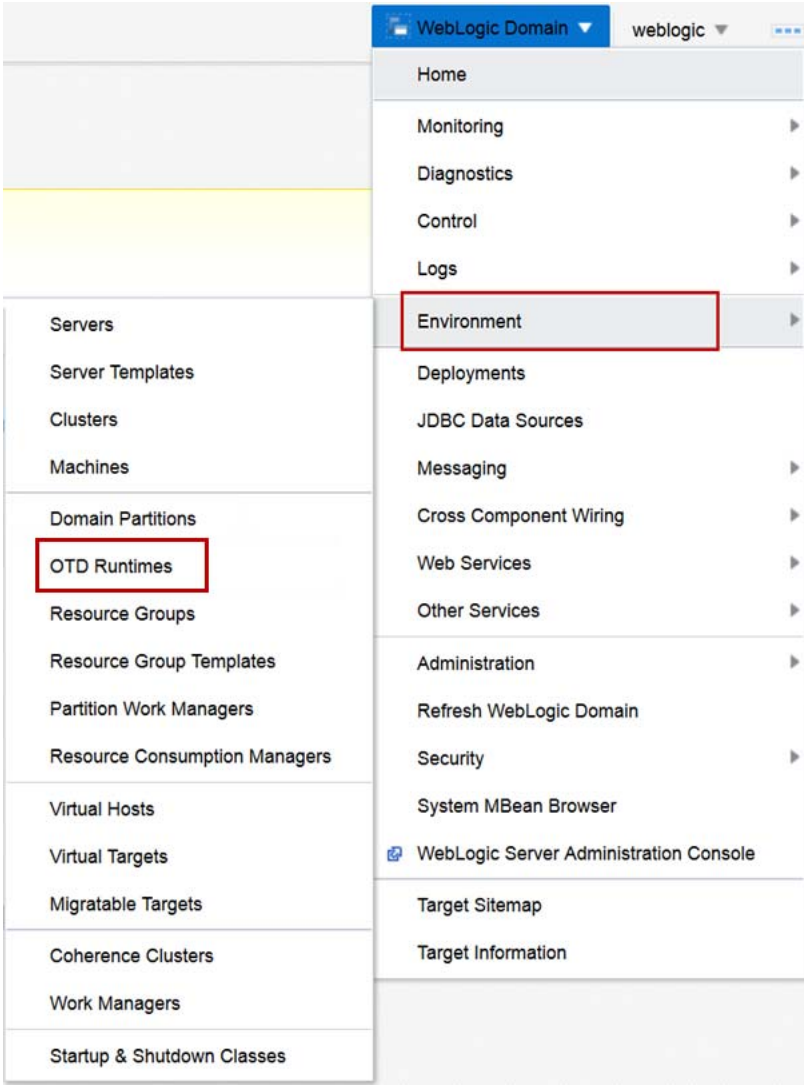
i. Review the configuration and click **Create Configuration.**

The Oracle Traffic Director configuration **medrec** creates successfully.

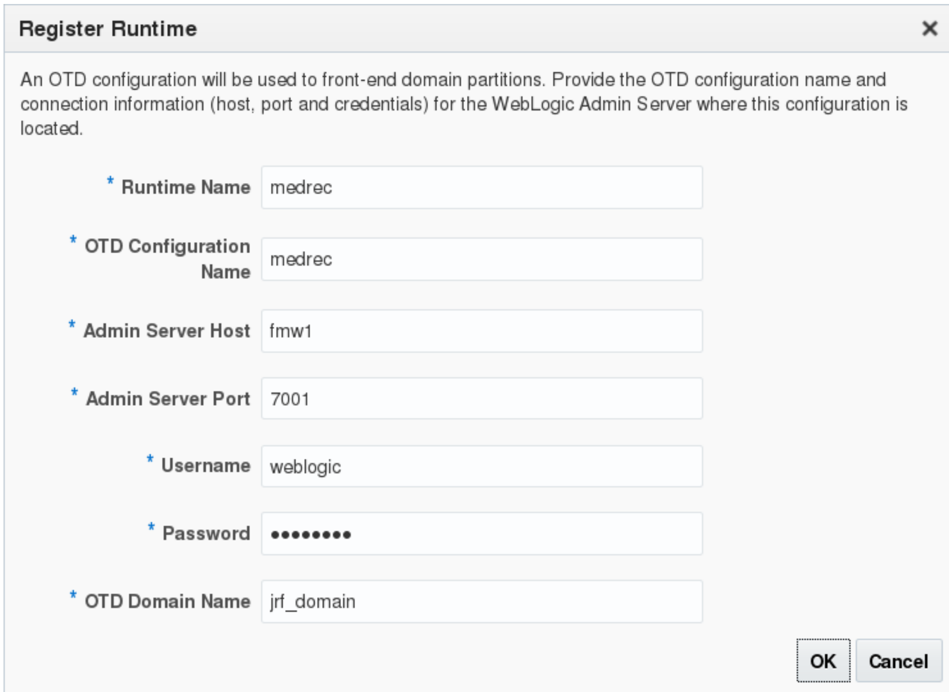
4. Start the medrec Oracle Traffic Director configuration.
   a. In the Fusion Middleware Control window, click the m**edrec** configuration entry.
   b. Click **Start Instances.**
   c. In the Confirmation dialog box, click **Close.**

   The Oracle Traffic Director instances for the **medrec** configuration start successfully.

5. Register the OTD Runtime.
   a. From the **WebLogic Domain** menu, select **Environment** and then click **OTD Runtimes.**



   b. Click **Register Runtime.**
   c. For **Runtime Name,** enter `medrec`.
   d. For **OTD Configuration Name,** enter `medrec`.
   e. For **Admin Server Host,** enter `fmw1`.
   f. For **Admin Server Port,** enter `7001`.
   g. For **Username,** enter `weblogic`.
   h. For **Password,** enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product.
   i. For **OTD Domain Name,** enter `jrf_domain` and click **OK.**



The Oracle Traffic Director Runtime **medrec** registers successfully.

Practice Solution: Creating an Oracle Traffic Director Configuration

Perform the following tasks if you did not complete this practice and want to use the finished solution.

> **Note:** If you started creating Oracle Traffic Director configurations, this script will not work. To run the script in that case, delete the partially created Oracle Traffic Director configurations before running this script.

Assumptions

- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf_domain admin server is running.
- The nodemanager for host fmw1, nodemanager-fmw1 is running.
- The nodemanager for host fmw2, nodemanager-fmw2 is running.
- The machines fm1 and fmw2 have been created.
- The dynamic cluster appcluster has been created.
- The resource group template MedRecResourceGroupTemplate has been created.

Solution Tasks

1. Connect to fmw1 as the `oracle` user.
2. Run the solution script.
   a. Open a Terminal window as the `oracle` user and navigate to the practice directory.
      $ **cd /u01/practices/part1/practice05-01**
   b. Run the solution script.
      $ **./solution.sh**
   c. Enter any passwords as you are prompted.

   Note: This script starts the WebLogic Scripting Tool (WLST) passing it the WLST script `create_otd_configuration.py`, which creates the new generic data source for you. The WLST script prints out messages. Be patient and do not close the Terminal window until you see these messages:

   ```
   ...
   >>> Solution for Practice 05-01 applied successfully!

   Exiting WebLogic Scripting Tool.
   ```
3. Close the Terminal window.

Practice 5-2: Creating Domain Partitions That Use Oracle Traffic Director

Overview

In this practice, you create domain partitions that use the Oracle Traffic Director for load balancing.

Assumptions

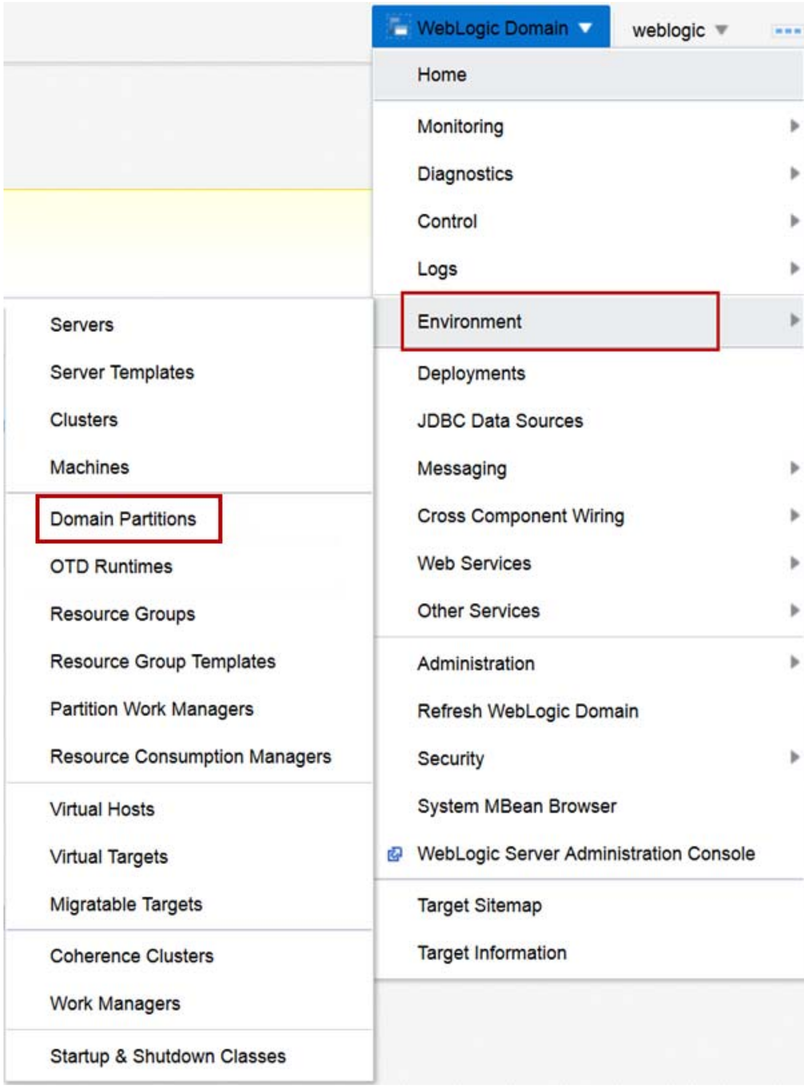
- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf_domain admin server is running.
- The nodemanager for host fmw1, nodemanager-fmw1 is running.
- The nodemanager for host fmw2, nodemanager-fmw2 is running.
- The machines fm1 and fmw2 have been created.
- The dynamic cluster appcluster has been created.
- The resource group template MedRecResourceGroupTemplate has been created.
- The Oracle Traffic Director configuration medrec has been created.
- The Oracle Traffic Director server otd_medrec_fmw2 is running.

Tasks

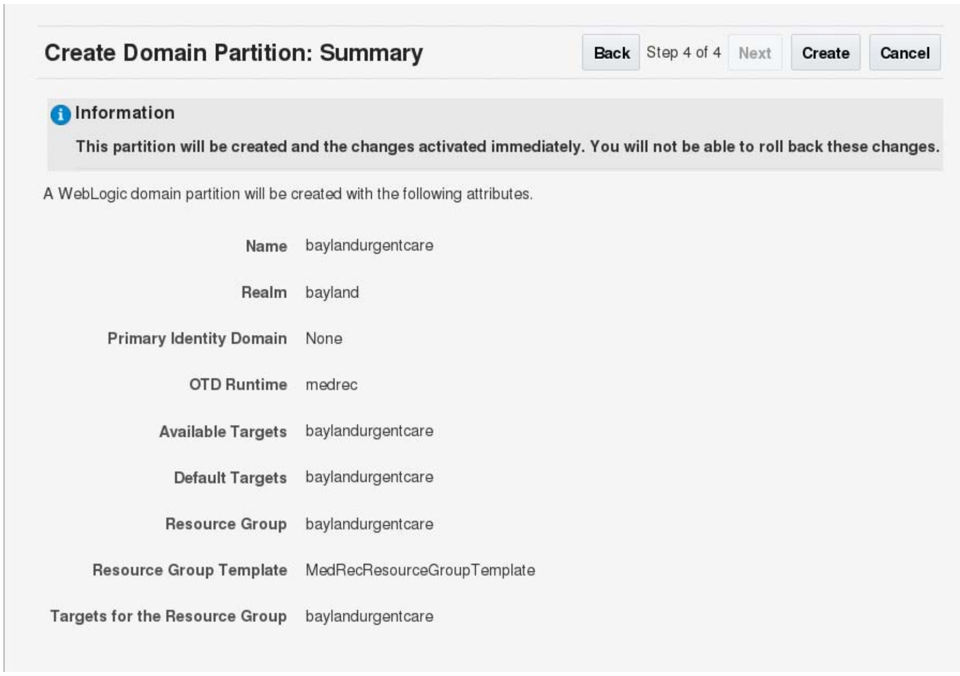
1. Log in to the jrf_domain WebLogic Server Domain using Fusion Middleware Control.
   a. Open a browser window and navigate to **http://fmw1:7001/em.**
   b. For **User Name,** enter `weblogic` and for **Password,** enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click **Login.**

      The Fusion Middleware Control home page for the **jrf_domain** is displayed.

2. Create the baylandurgentcare domain partition.
   a. From the **WebLogic Domain** menu, select **Environment** and then click **Domain Partitions.**



   b. Click **Create.**
   c. For **Name,** enter `baylandurgentcare`.
   d. For **Security Realm,** select bayland.
   e. In the **Load Balancer Configuration** section of the page, select **Use OTD for load balancing.**
   f. For **OTD Runtime,** select **medrec.**
   g. Click **Next.**
   h. For the Virtual Target baylandurgentcare, choose both **Select** and **Set as Default** and click **Next.**
   i. For **Resource Group Name,** enter `baylandurgentcare`.
   j. For **Resource Group Template,** select **MecRedResourceGroupTemplate.**
   k. Move the **baylandurgentcare** target to **Selected Targets.**
   l. Click **Next.**



m. Review the domain partition summary and click **Create.**

   The baylandurgentcare domain partition is created.

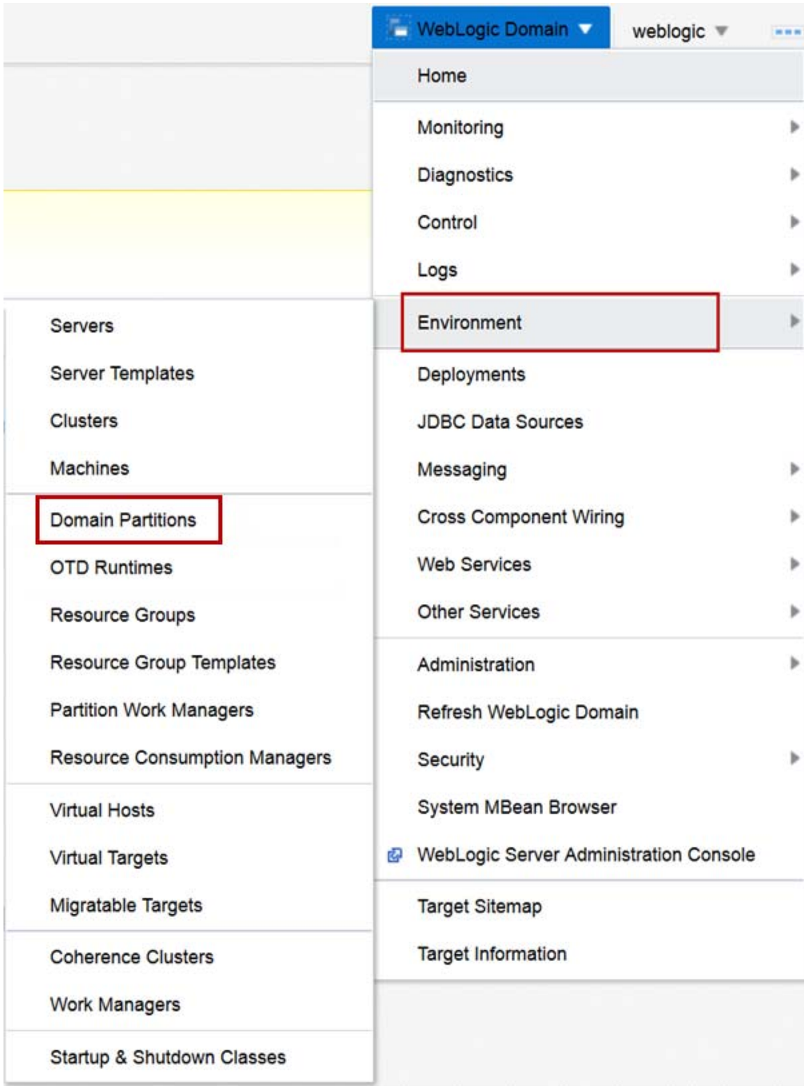
3. Start the baylandurgentcare domain partition.
   a. In the list of Domain Partitions, click the **baylandurgentcare** entry.
   b. Click **Control** and click **Start**.
   c. Once the Confirmation dialog box indicates the Start Operation is successful, click **Close.**

      The baylandurgentcare domain partition successfully starts.

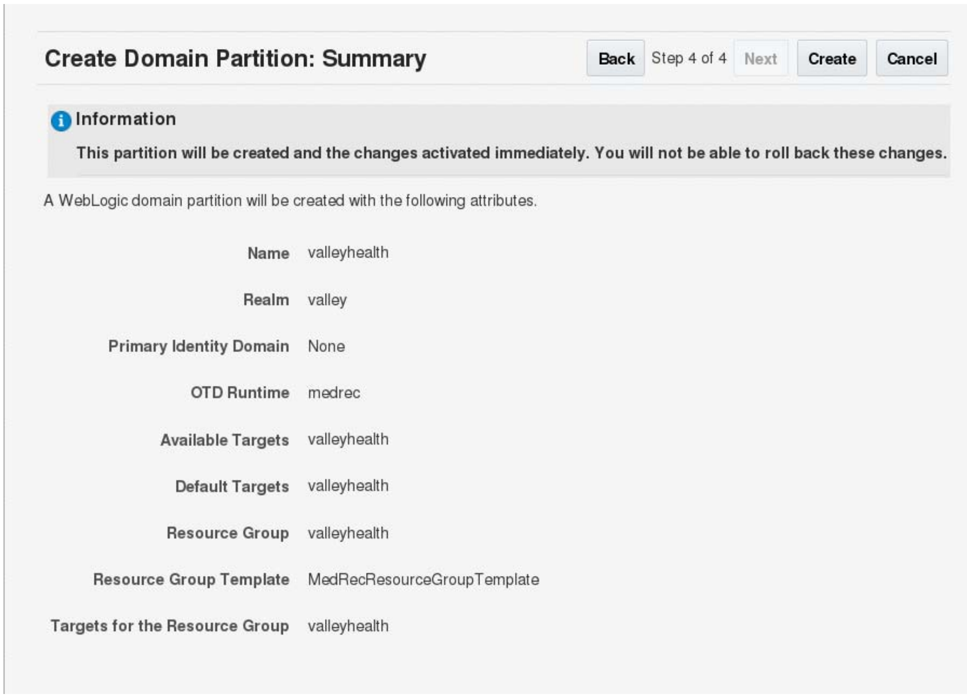
4. Test the baylandurgentcare domain partition.
   a. In the browser window, navigate to **http://fmw2:8080/bayland/medrec.**

      Note that when you created the domain partition and started it, the Lifecycle Manager performed the necessary configuration for OTD for the baylandurgentcare deployment.

5. Create the valleyhealth domain partition.
   a. From the **WebLogic Domain** menu, select **Environment** and then click **Domain Partitions.**



   b. Click **Create.**
   c. For **Name,** enter `valleyhealth`.
   d. For **Security Realm,** select **valley.**
   e. In the **Load Balancer Configuration** section of the page, select **Use OTD for load balancing.**
   f. For **OTD Runtime,** select **medrec.**
   g. Click **Next.**
   h. For the Virtual Target **valleyhealth**, choose both **Select** and **Set as Default** and click **Next.**
   i. For **Resource Group Name,** enter `valleyhealth`.
   j. For **Resource Group Template,** select **MecRedResourceGroupTemplate.**
   k. Move the **valleyhealth** target to **Selected Targets.**
   l. Click **Next.**



m. Review the domain partition summary and click **Create.**

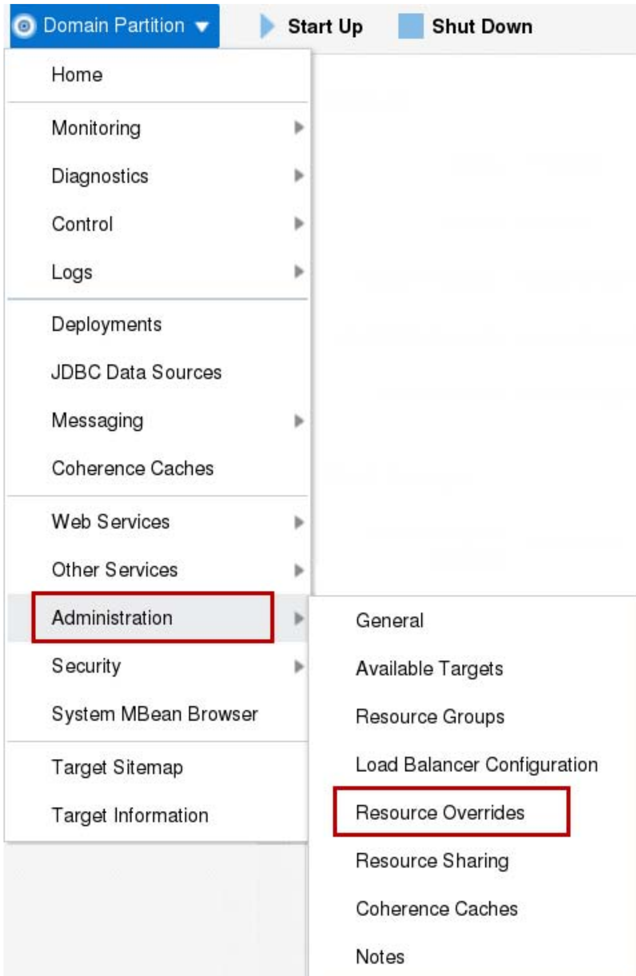
   The valleyhealth domain partition is created.

   However, before starting the valleyhealth domain partition, change the JDBC connection to use a different pluggable database, namely PDB2.

6. Override the JBDC connection settings for the valleyhealth domain partition Resource Group.

   a. In the list of **Domain Partitions,** click the valleyhealth domain partition.

b. From the **Domain Partition** menu, select **Administration** and then select **Resource Overrides.**



c. Click the **Edit Overrides** pencil icon for the **MedRecGlobalDataSourceXA** entry.

d. Override the **User** property with the value `MEDREC`.

e. Override the **URL** property with the value `jdbc:oracle:thin:@//fmwdb:1521/PDB2`.

f. Override the **Password** property with value for the password from the Course Practice Environment: Security Credentials document for the MEDREC user for the Oracle Database product.

g. Click **OK.**

7. Start the **valleyhealth** domain partition.

   a. Next to the **Domain Partition** menu, click **Start Up.**

   b. After the **Domain Partition** starts, click **Close.**

8. Test the **valleyhealth** domain partition.

   a. In the browser window, navigate to **http://fmw2:8080/valley/medrec**.

Practice 5-3: Configuring Oracle Traffic Director Logging Level

Overview

In this practice, you work with the Oracle Traffic Director logs to locate and configure log files. First, you use Fusion Middleware Control to view log files, specifically the access log and server log. Next, you configure the logs to provide finer tracing and manually roll the log files. Finally, you configure and view the Traffic Director performance pages.

Assumptions

- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf_domain admin server is running.
- The nodemanager for host fmw1, nodemanager-fmw1 is running.
- The nodemanager for host fmw2, nodemanager-fmw2 is running.
- The machines fm1 and fmw2 have been created.
- The dynamic cluster appcluster has been created.
- The resource group template MedRecResourceGroupTemplate has been created.
- The Oracle Traffic Director configuration medrec has been created and registered.
- The Oracle Traffic Director server otd_medrec_fmw2 is running.
- Domain partitions baylandurgentcare and valleyhealth are running.

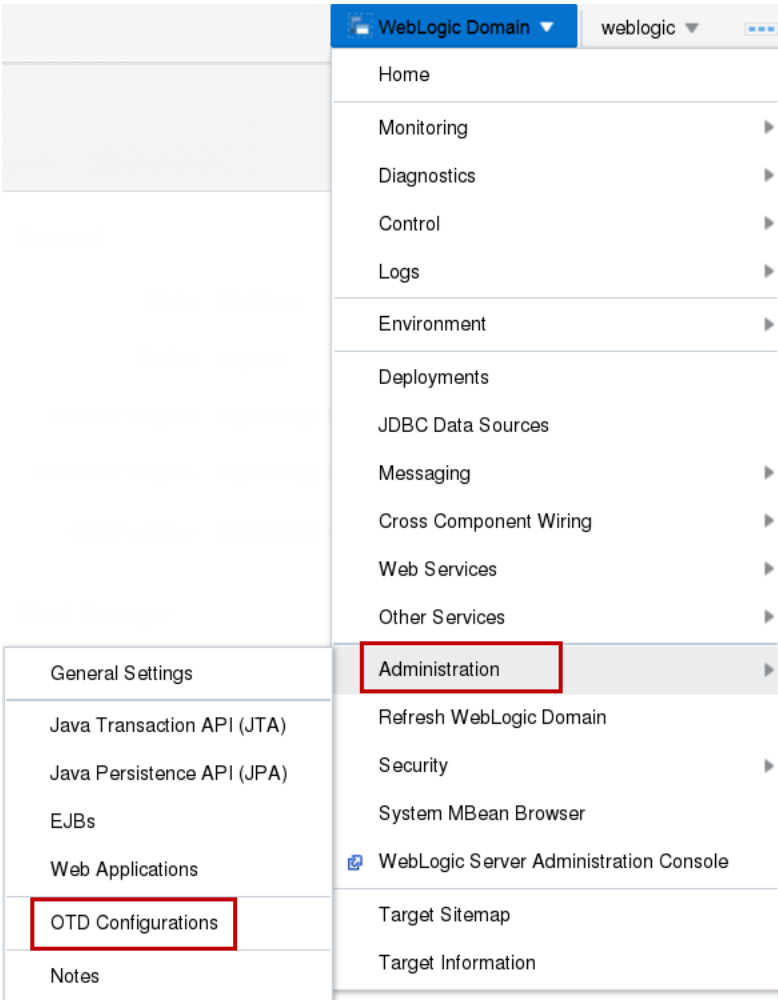
Tasks

1. Log in to the jrf_domain WebLogic Server Domain using Fusion Middleware Control.
    a. Open a Browser window and navigate to **http://fmw1:7001/em.**
    b. For **User Name,** enter `weblogic` and for **Password,** enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click **Login.**

       The Fusion Middleware Control home page for the **jrf_domain** is displayed.

2. View the Oracle Traffic Director log files.

   a. In Fusion Middleware Control, from the **WebLogic Domain** menu, select **Administration** and click **OTD Configurations.**



   b. Click the **medrec** configuration.

   c. From the **Traffic Director Configuration** menu, select **Logs** and select **View Log Messages.**

   d. If there are no log messages to display, click the **Search** icon and expand the date range – for example, from the most recent 10 minutes to the most recent 10 hours.

   e. In the Search dialog box, also include all the Message Types – check **Incident Error, Error, Warning, Notification, Trace** and **Unknown.**

   f. Click **Search.**

   g. In the rightmost column, click the **baylandurgentcare.log** link.

      The log entries for baylandurgentcare are displayed. Note that you can download the `baylandurgentcare.log` file.

   h. Click the **Log Files** link in the breadcrumbs display, just beneath the **Traffic Director Configuration** menu.

   i. Click the **`access.log`** file entry and click the **View Log File** button.

   j. Browse the `access.log`.

k. Click the **Log Files** link in the breadcrumbs display, just beneath the **Traffic Director Configuration** menu.

l. Click the **`server.log`** file entry and click the **View Log File** button.

m. Browse the `server.log`.

3. Configure the Oracle Traffic Director logging level.

   a. From the **Traffic Director Configuration** menu, select **Administration** and select **Logging.**

   b. For **Log Level,** select **TRACE: 1 (fine)** and click **Apply.**

      View the `server.log`. Now messages should be displayed that include Warning, Notification, and Error.

4. Manually roll the Oracle Traffic Director log files.

   a. From the **Traffic Director Configuration** menu, select **Administration** and select **Log Rotation.**

   b. Click **Rotate Logs Now.**

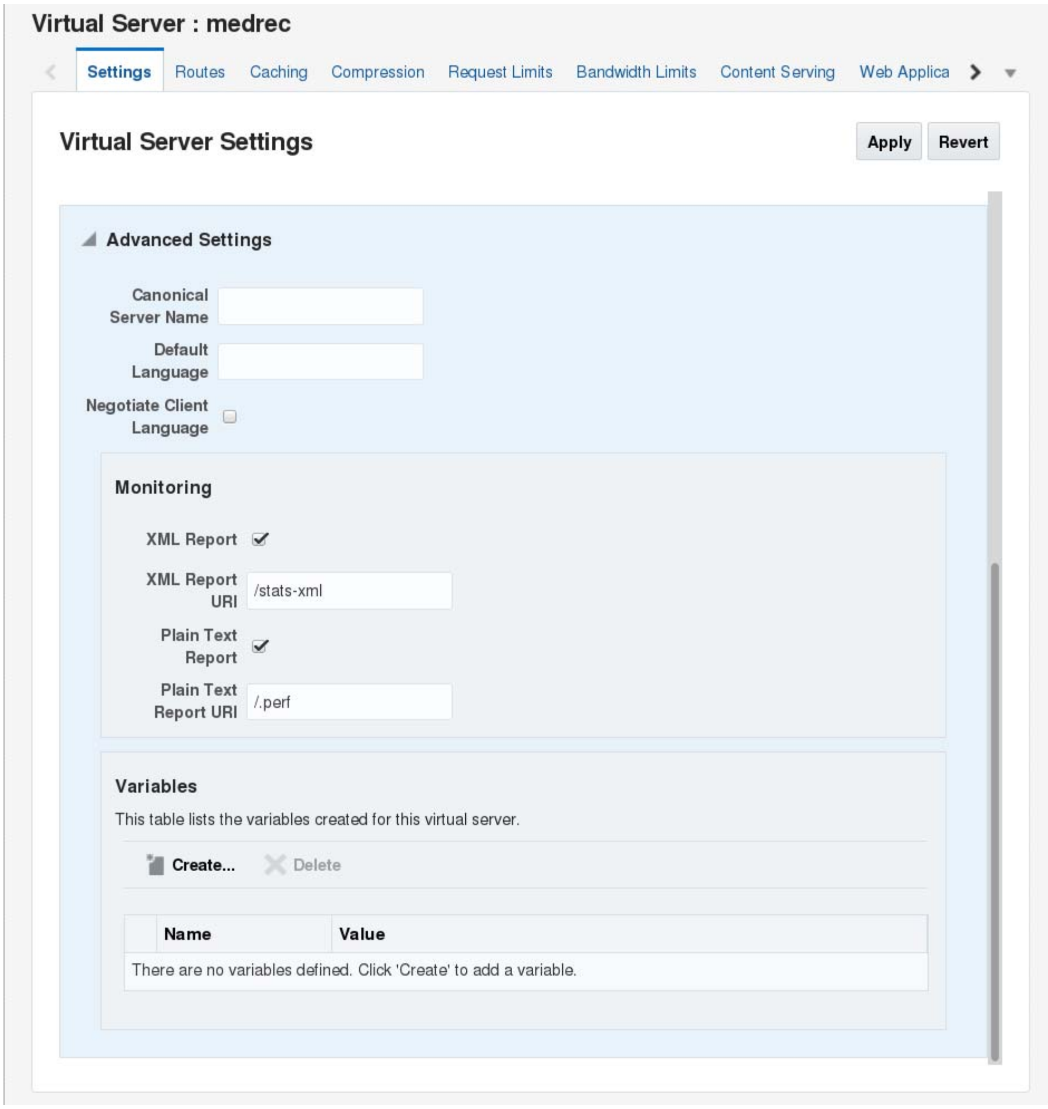
5. Configure the Oracle Traffic Director monitoring and performance pages.

   a. From the **Traffic Director Configuration** menu, select **Administration** and select **Virtual Servers.**

   b. Click the **medrec** Virtual Server.

   c. In the **Virtual Server Settings** configuration page, scroll down and select **Advanced Settings.**

   d. Select the check box for **XML Report** and **Plain Text Report.**



e. Click **Apply.**

6. View the Oracle Traffic Director monitoring and performance pages.
   a. Open a browser window and navigate to **http://fmw2:8080/.perf.**
   b. The performance-monitoring page for Oracle Traffic Director is displayed. Note that it displays information for queuing, keepalive, sessions, cache, pools, and origin servers.
   c. Open a browser window and navigate to **http://fmw2:8080/stats-xml.**
   d. The statistics page for Oracle Traffic Director is displayed. Note that it displays information for the server threads, CPU utilization, and server pool statistics.

Practices for Lesson 6 - Database Multitenancy

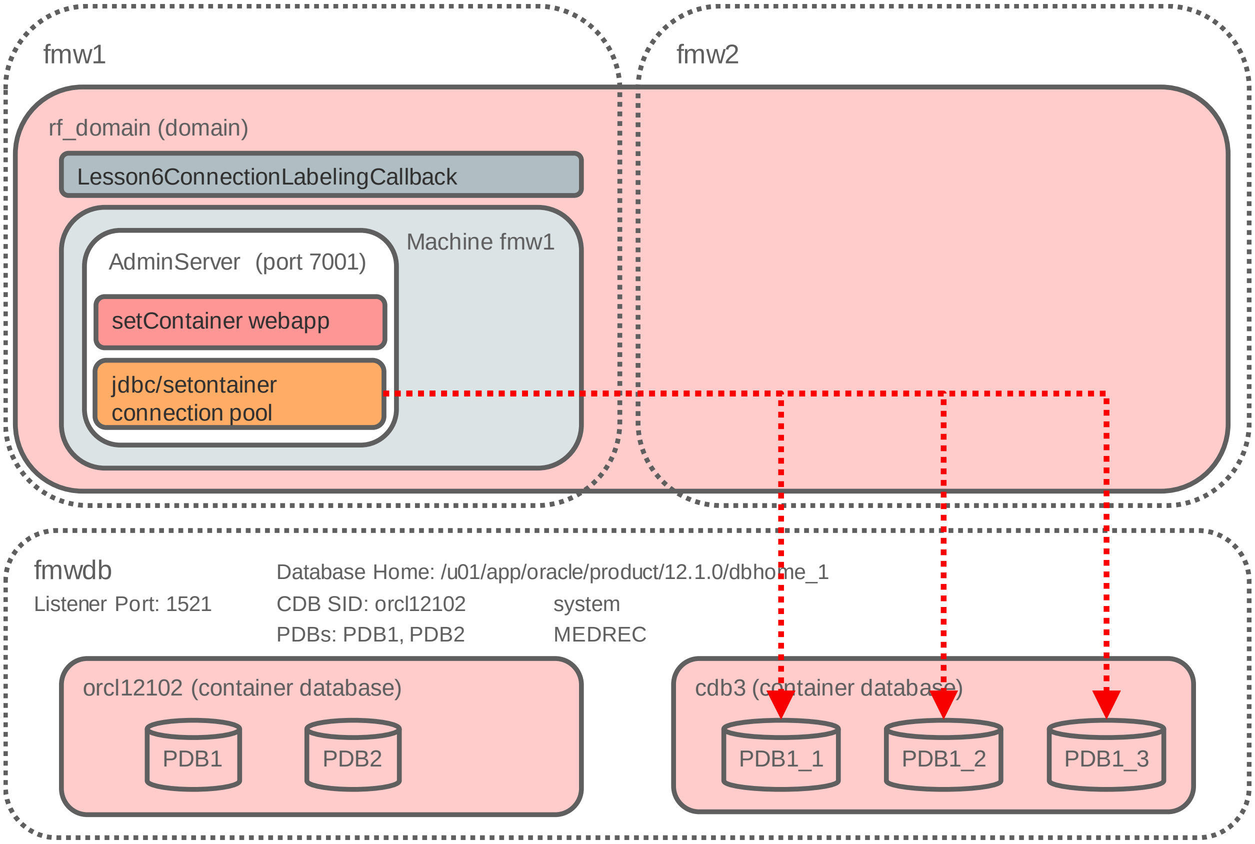
Chapter 6

Practices for Lesson 6: Overview

Overview

In these practices, you create a new container database and then add pluggable databases to the new container database. Finally, you use a test WebLogic Server application to exercise the SET CONTAINER capability.

The following diagram describes the database and application configuration:



Practice 6-1: Creating a Container Database

Overview

In this practice, using the Database Configuration Assistant you create a new Container Database (CDB) to host new pluggable databases.

Assumptions

- Oracle Database 12.1.0.2 is installed.

Tasks

1. Connect to the machine fmwdb as the `oracle` user, in a VNC session. Open a terminal session as the `oracle` user.
2. Set the environment.
   a. For the currently configured database `orcl12102`, on host fmwdb:

```
$ . oraenv
ORACLE_SID = [] ? orcl12102
The Oracle base has been set to /u01/app/oracle
$
```

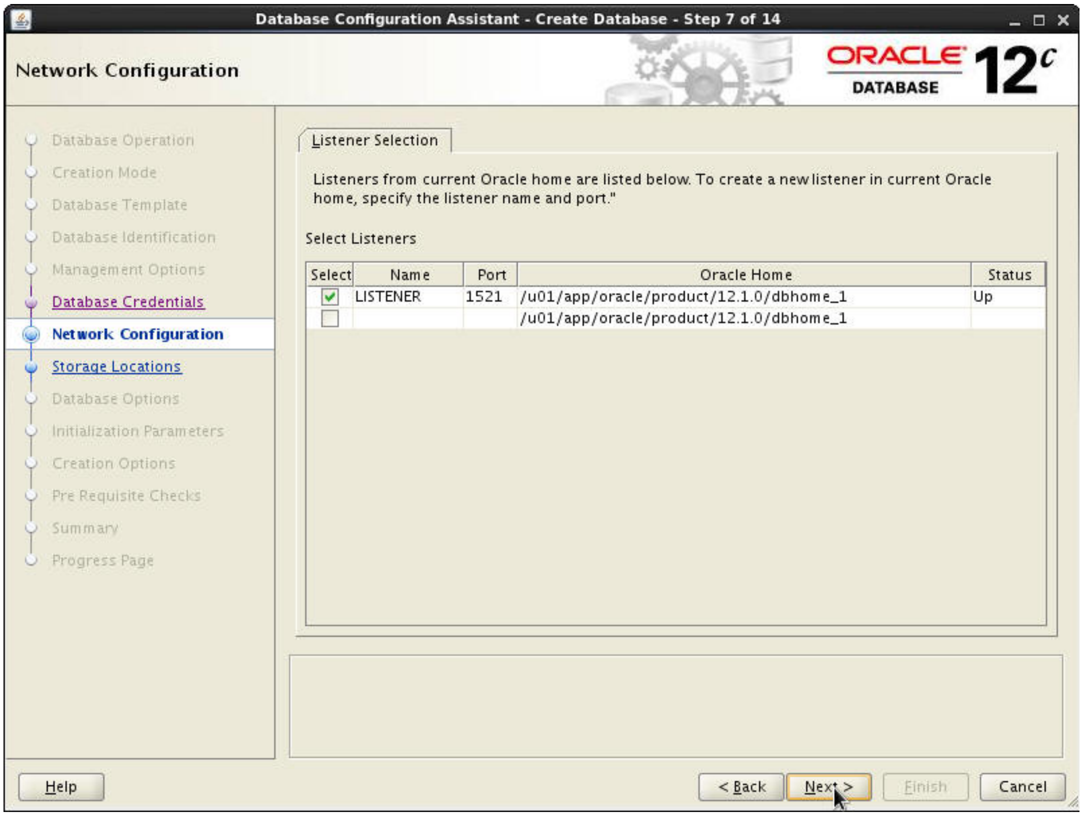
3. Create a Container Database (CDB) named `cdb2` using Database Configuration Assistant by starting `dbca` and performing the following steps in the graphical wizard.

```
$ dbca
```

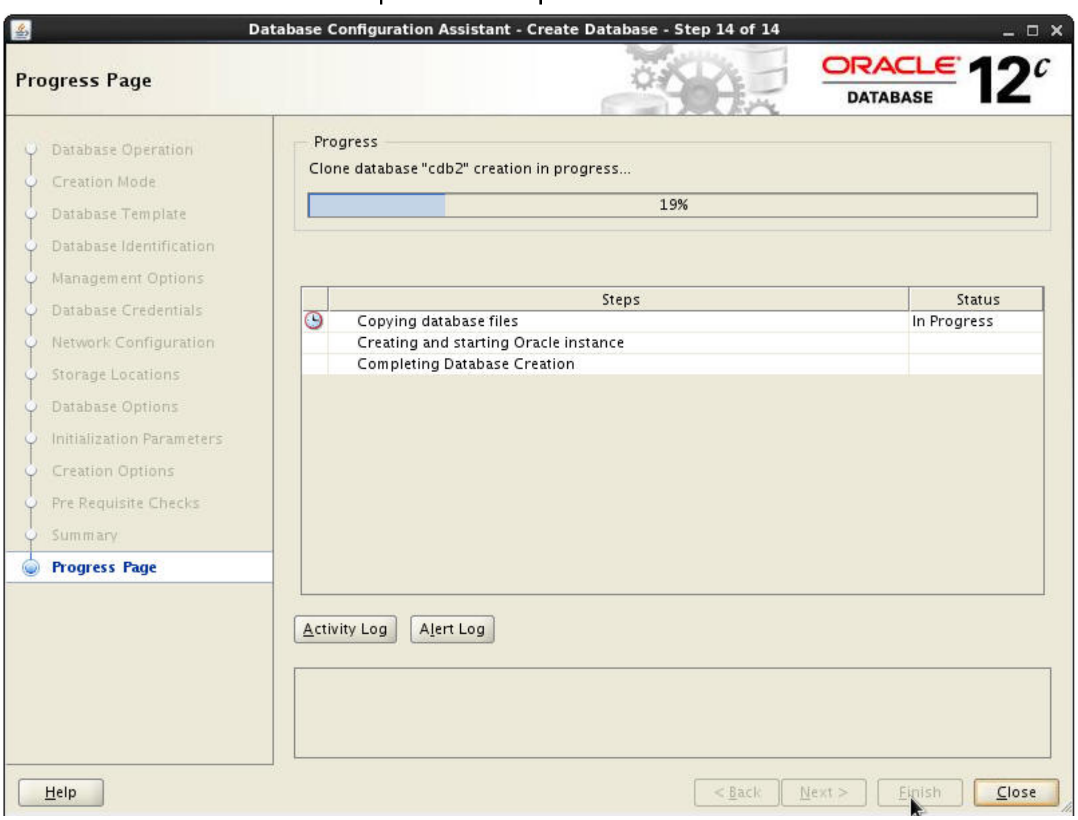
Step	Window/Page Description	Choices or Values
a.	Step 1: Database Operation	Select "Create Database." Click Next.
b.	Step 2: Creation Mode	Select "Advanced Mode." Click Next.
c.	Step 3: Database Template	Select "General Purpose or Transaction Processing." Click Next.
d.	Step 4: Database Identification	Enter Global Database Name: `cdb2` SID: `cdb2` Select **"Create As Container Database."** Select **"Create An Empty Container Database."** Click Next.
e.	Step 5: Management Options	Deselect "Configure Enterprise Manager (EM) Database Express." Click Next.
f.	Step 6: Database Credentials	Select "Use same Administrative password…"

Step	Window/Page Description	Choices or Values
		Enter: Password: enter he password from the Course Practice Environment: Security Credentials document for the Administrative user for the Oracle Database product. Confirm password: enter the password from the Course Practice Environment: Security Credentials document for the Administrative user for the Oracle Database product. Click Next.
g.	Step 7: Network Configuration	Click Next.
h.	Step 8: Storage Locations	Confirm Storage type is “File System.” Select “Use Common Location for All Database Files.” Click Next.
i.	Step 9: Database Options	Click Next.
j.	Step 10: Initialization Parameters	Select “Character Sets.” Select “Use Unicode (**AL32UTF8**).” Click Next.
k.	Step 11: Creation Option	Select “Create Database.” Click Next.
l.	Step 12: Pre Requisite Checks	Click Next.
m.	Step 13: Summary	Click Finish.
n.	Step 14: Progress Page	On the Database Configuration Assistant page (for password management), click Finish.

The screenshot below corresponds to step g.



The screenshot below corresponds to step n.



Practice 6-2: Creating Pluggable Databases

Overview

In this practice, using SQLPlus, you unplug a current pluggable database named `PDB1` (from database `orcl12102`) and use it to create new pluggable databases in database `cdb2`.

Assumptions

- Oracle Database 12.1.0.2 is installed.
- The Oracle Database orcl12102 is running.
- The Oracle Database cdb2 is running.

Tasks

1. Connect to the machine fmwdb as the `oracle` user, in a VNC session. Open a terminal session as the `oracle` user.
2. Set the environment for database `orcl12102.`
   a. For the currently configured database `orcl12102`, on host fmwdb:

```
$ . oraenv
ORACLE_SID = [] ? orcl12102
The Oracle base has been set to /u01/app/oracle
$
```

3. Unplug the `PDB1` database from `orcl12102.`
   a. Execute `sqlplus:`

```
$ sqlplus / as sysdba

SQL*Plus: Release 12.1.0.2.0 Production on Mon Jan 25 15:12:47
2016

Copyright (c) 1982, 2014, Oracle.  All rights reserved.

Connected to:
Oracle Database 12c Enterprise Edition Release 12.1.0.2.0 -
64bit Production
With the Partitioning, OLAP, Advanced Analytics and Real
Application Testing options

SQL>
```

   b. Close the PDB1 pluggable database.

```
SQL> ALTER PLUGGABLE DATABASE PDB1 CLOSE IMMEDIATE;

Pluggable database altered.

SQL>
```

c. Unplug the PDB1 pluggable database in to the file PDB1.XML.

```
SQL> ALTER PLUGGABLE DATABASE PDB1 UNPLUG INTO 'PDB1.XML';

Pluggable database altered.

SQL>
```

d. Drop the PDB1 pluggable database.

```
SQL> DROP PLUGGABLE DATABASE PDB1;

Pluggable database dropped.

SQL>
```

e. Exit SQLPlus.

```
SQL> EXIT;
Disconnected from Oracle Database 12c Enterprise Edition Release
12.1.0.2.0 - 64bit Production
With the Partitioning, OLAP, Advanced Analytics and Real
Application Testing options
$
```

4. Set the environment for database cdb2.

a. For the currently configured database cdb2, on host fmwdb:

```
$ . oraenv
ORACLE_SID = [] ? cdb2
The Oracle base has been set to /u01/app/oracle
$
```

5. Create the pluggable databases in the container database cdb2.

a. Execute sqlplus:

```
$ sqlplus / as sysdba

SQL*Plus: Release 12.1.0.2.0 Production on Mon Jan 25 15:12:47
2016

Copyright (c) 1982, 2014, Oracle.  All rights reserved.

Connected to:
Oracle Database 12c Enterprise Edition Release 12.1.0.2.0 -
64bit Production
With the Partitioning, OLAP, Advanced Analytics and Real
Application Testing options

SQL>
```

b. Ensure that you are connected to the correct database by viewing the data files – the set of data files for the cdb2 database are displayed.

```
SQL> SELECT FILE_NAME FROM DBA_DATA_FILES;

FILE_NAME
-----------------------------------------------------------------
----------------
/u01/app/oracle/oradata/cdb2/users01.dbf
/u01/app/oracle/oradata/cdb2/undotbs01.dbf
/u01/app/oracle/oradata/cdb2/system01.dbf
/u01/app/oracle/oradata/cdb2/sysaux01.dbf

SQL>
```

c. Create the new PDB1_1 pluggable database.

```
SQL> CREATE PLUGGABLE DATABASE PDB1_1 AS CLONE USING 'PDB1.XML'
COPY STORAGE UNLIMITED TEMPFILE REUSE
FILE_NAME_CONVERT=('PDB1','PDB1_1','DEVCIS','DEVCISPDB1_1','FULL
','FULLPDB1_1');

Pluggable database created.

SQL>
```

d. Open the new PDB1_1 pluggable database for read write.

```
SQL> ALTER PLUGGABLE DATABASE PDB1_1 OPEN READ WRITE;

Pluggable database altered.

SQL>
```

e. Create the new PDB1_2 pluggable database.

```
SQL> CREATE PLUGGABLE DATABASE PDB1_2 AS CLONE USING 'PDB1.XML'
COPY STORAGE UNLIMITED TEMPFILE REUSE
FILE_NAME_CONVERT=('PDB1','PDB1_2','DEVCIS','DEVCISPDB1_2','FULL
','FULLPDB1_2');

Pluggable database created.

SQL>
```

f. Open the new PDB1_1 pluggable database for read write.

```
SQL> ALTER PLUGGABLE DATABASE PDB1_2 OPEN READ WRITE;

Pluggable database altered.

SQL>
```

g. Create the new PDB1_3 pluggable database.

```
SQL> CREATE PLUGGABLE DATABASE PDB1_3 AS CLONE USING 'PDB1.XML'
COPY STORAGE UNLIMITED TEMPFILE REUSE
FILE_NAME_CONVERT=('PDB1','PDB1_3','DEVCIS','DEVCISPDB1_3','FULL
','FULLPDB1_3');


Pluggable database created.


SQL>
```

h. Open the new PDB1_3 pluggable database for read write.

```
SQL> ALTER PLUGGABLE DATABASE PDB1_3 OPEN READ WRITE;


Pluggable database altered.


SQL>
```

i. Ensure the pluggable databases are open for read write, by selecting from the `CDB_PDBS` table and viewing the pluggable database names and status.

```
SQL> SELECT PDB_NAME, STATUS FROM CDB_PDBS;


PDB_NAME
-----------------------------------------------------------------
----------------
STATUS
---------
PDB1_1
NORMAL

PDB$SEED
NORMAL

PDB1_2
NORMAL

PDB1_3
NORMAL

SQL>
```

j. Exit SQLPlus.

```
SQL> EXIT;
Disconnected from Oracle Database 12c Enterprise Edition Release
12.1.0.2.0 - 64bit Production
With the Partitioning, OLAP, Advanced Analytics and Real
Application Testing options
$
```

6. Set the environment for database `orcl12102`.
   a. For the currently configured database `orcl12102`, on host fmwdb:

```
$ . oraenv
ORACLE_SID = [] ? orcl12102
The Oracle base has been set to /u01/app/oracle
$
```

7. Create the pluggable database PDB1 in container database `orcl12102`.
   a. Execute `sqlplus`:

```
$ sqlplus / as sysdba

SQL*Plus: Release 12.1.0.2.0 Production on Mon Jan 25 15:12:47
2016

Copyright (c) 1982, 2014, Oracle.  All rights reserved.

Connected to:
Oracle Database 12c Enterprise Edition Release 12.1.0.2.0 -
64bit Production
With the Partitioning, OLAP, Advanced Analytics and Real
Application Testing options

SQL>
```

   b. Ensure that you are connected to the correct database by viewing the data files – the set of data files for the orcl12102 database are displayed.

```
SQL> SELECT FILE_NAME FROM DBA_DATA_FILES;

FILE_NAME
-----------------------------------------------------------------
----------------
/u01/app/oracle/oradata/orcl12102/users01.dbf
/u01/app/oracle/oradata/orcl12102/undotbs01.dbf
/u01/app/oracle/oradata/orcl12102/system01.dbf
/u01/app/oracle/oradata/orcl12102/sysaux01.dbf

SQL>
```

c. Create the PDB1 pluggable database.

```
SQL> CREATE PLUGGABLE DATABASE "PDB1" USING 'PDB1.XML'
SOURCE_FILE_NAME_CONVERT=NONE  NOCOPY  STORAGE UNLIMITED
TEMPFILE REUSE;
Pluggable database created.


SQL>
```

d. Open the PDB1 pluggable database for read write.

```
SQL> ALTER PLUGGABLE DATABASE PDB1 OPEN READ WRITE;


Pluggable database altered.


SQL>
```

e. Ensure the PDB1 pluggable database is open for read write, by selecting from the `CDB_PDBS` table and viewing the pluggable database names and status.

```
SQL> SELECT PDB_NAME, STATUS FROM CDB_PDBS;


PDB_NAME
----------------------------------------------------------------
----------------
STATUS
---------
PDB1
NORMAL


PDB$SEED
NORMAL


PDB2
NORMAL


SQL>
```

f. Exit SQLPlus.

```
SQL> EXIT;
Disconnected from Oracle Database 12c Enterprise Edition Release
12.1.0.2.0 - 64bit Production
With the Partitioning, OLAP, Advanced Analytics and Real
Application Testing options
$
```

Practice 6-3: Using Pluggable Databases in a Java Application

Overview

In this practice, you use a Java application that can utilize connection labeling and the SET CONTAINER directive to change the applications' JDBC connection pool to different pluggable databases.

Assumptions

- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf_domain admin server is running.

Tasks

1. Connect to fmw1 as the `oracle` user.
   a. Refer to the instructions provided by your instructor to connect to your machine.
   b. Unless stated otherwise, you will be working within fmw1 for the remainder of this practice.
2. Copy the connection labeling class JAR file to the jrf_domain.
   a. Open a Terminal window as the `oracle` user, and navigate to `/u01/fmw/wls1221_infra/user_projects/domains/jrf_domain/lib`.

   Tip: There is a launcher for a Terminal window in the panel at the top of the desktop.

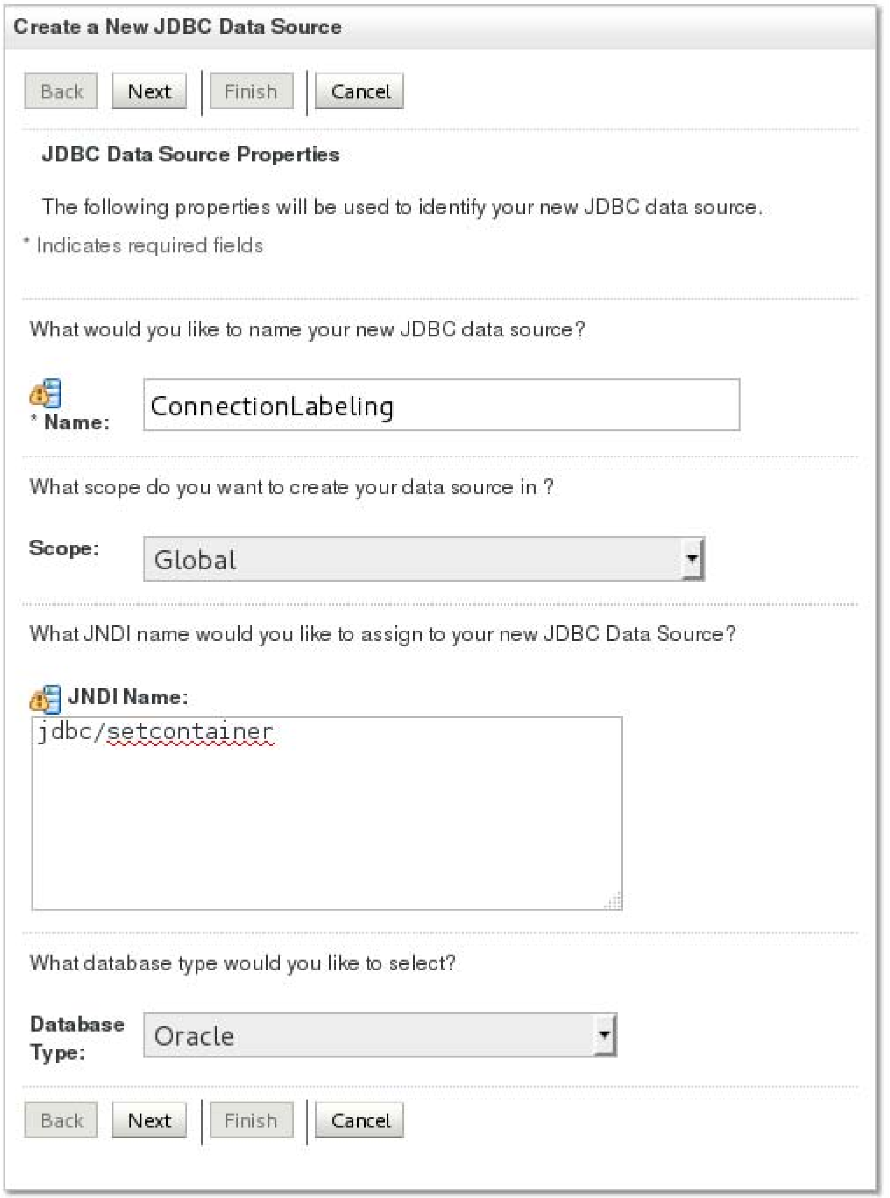
   ```
   $ cd
   /u01/fmw/wls1221_infra/user_projects/domains/jrf_domain/lib/
   ```

   b. Copy the `Lesson6ConnectionLabelingCallback.jar` file into the domain `lib` directory.

   ```
   $ cp /u01/practices/part1/practice06-
   03/Lesson6ConnectionLabelingCallback.jar .
   ```

3. Log in to the jrf_domain WebLogic Server Domain using the WebLogic Server Administration Console.
   a. Open a Browser window and navigate to **http://fmw1:7001/console.**
   b. For **User Name,** enter `weblogic` and for **Password,** enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click **Login.**

   The WebLogic Server Administration Console home page for the **jrf_domain** is displayed.
4. Create a Data Source that uses the connection labeling callback class.
   a. In the WebLogic Server Administration Console, from the **Domain Structure** panel, click **Services** and click **Data Sources.**
   b. Click **New,** and click **Generic Data Source.**
   c. For **Name,** enter `ConnectionLabeling.`
   d. For **JNDI Name,** enter `jdbc/setcontainer.`



e. Click **Next.**

f. In **JDBC Data Source Properties,** click **Next.**

g. In **Transaction Options,** click **Next.**

h. For **Database Name,** enter `PDB1_1`.

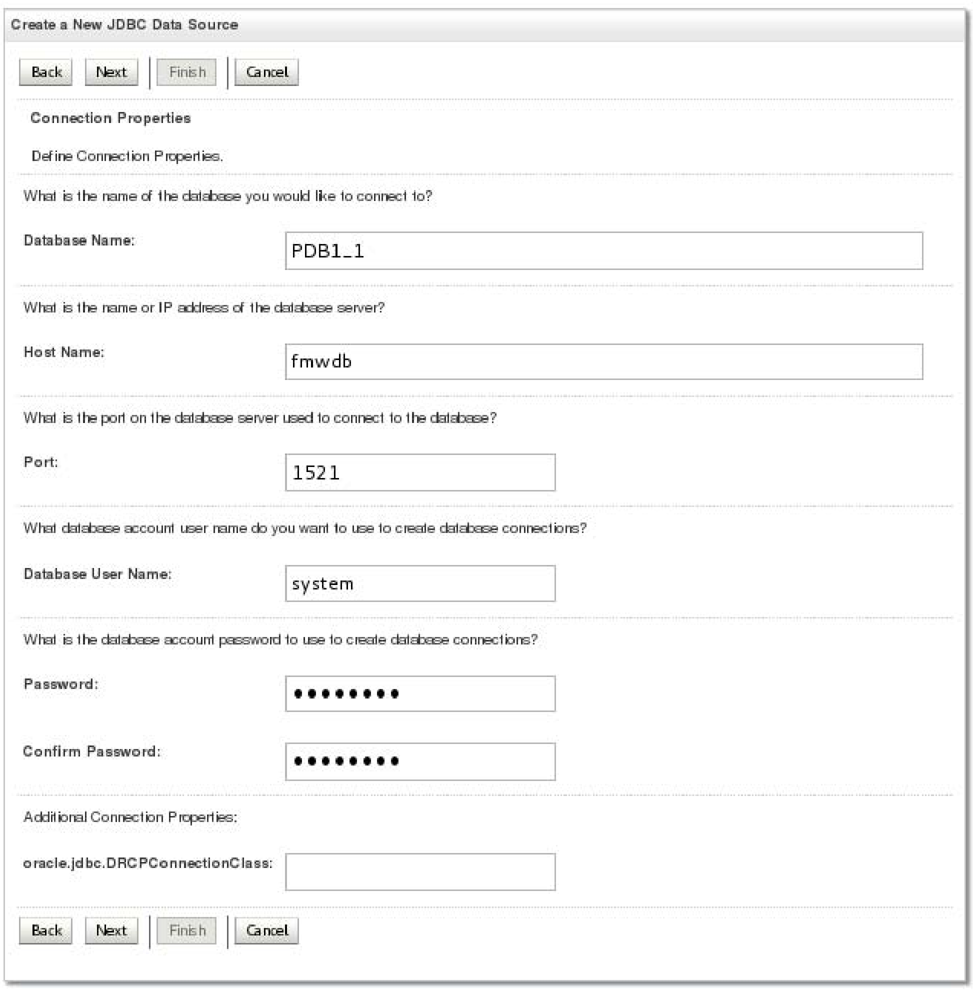
i. For **Host Name,** enter `fmwdb`.

j. For **Port,** enter `1521`.

k. For **Database User Name,** enter `system`.

l. For **Password,** enter the password from the Course Practice Environment: Security Credentials document for the system user for the Oracle Database product. .

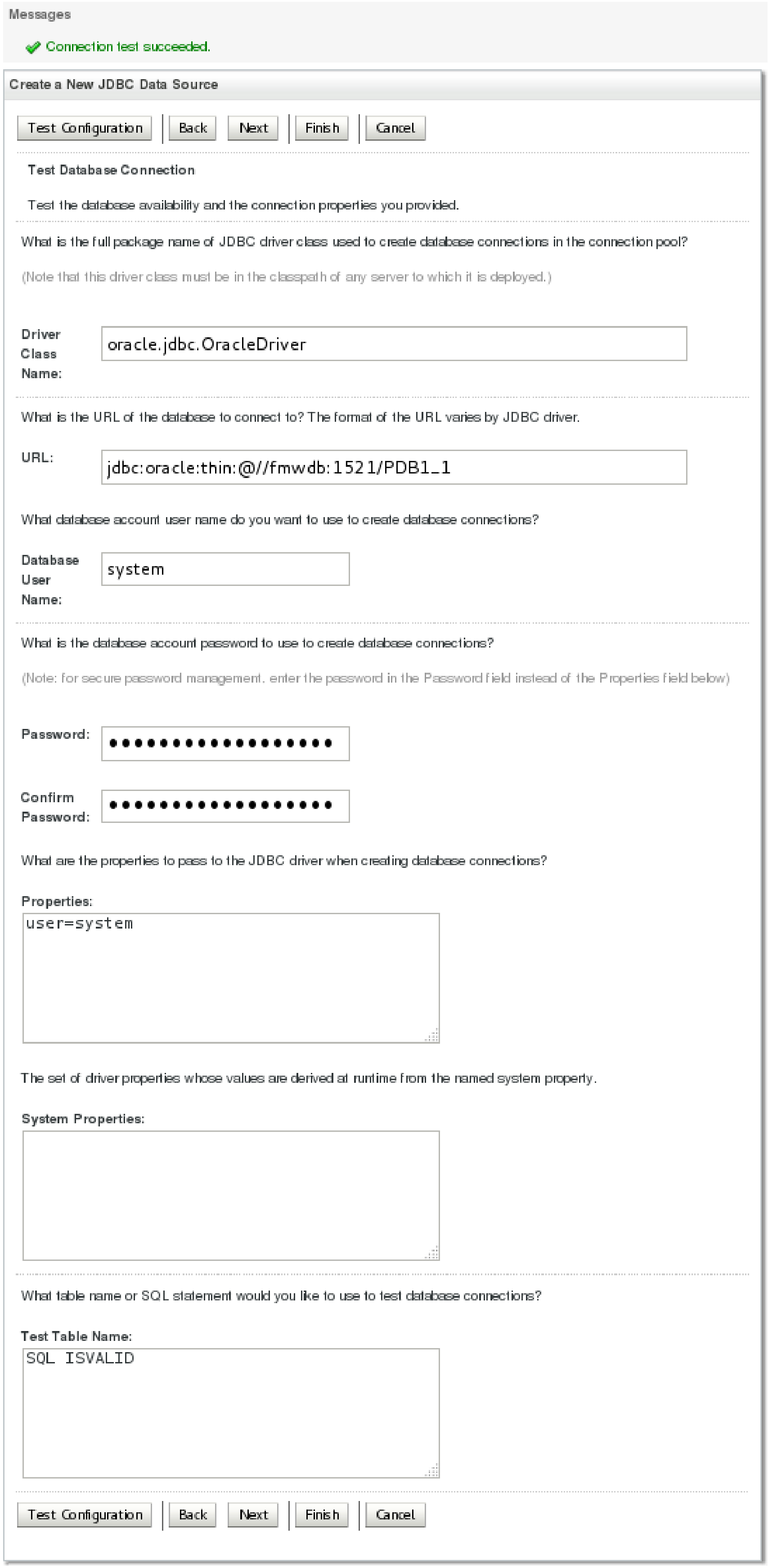
m. For **Confirm Password,** enter the password from the Course Practice Environment: Security Credentials document for the system user for the Oracle Database product.



n. Click **Next.**

o. Click **Test Configuration.**

The connection test succeeds.


Messages
Connection test succeeded.
Create a New JDBC Data Source
Test Configuration
Back
Next
Finish
Cancel
Test Database Connection
Test the database availability and the connection properties you provided.
What is the full package name of JDBC driver class used to create database connections in the connection pool?
(Note that this driver class must be in the classpath of any server to which it is deployed.)
Driver Class Name:
oracle.jdbc.OracleDriver
What is the URL of the database to connect to? The format of the URL varies by JDBC driver.
URL:
jdbc:oracle:thin:@//fmwdb:1521/PDB1_1
What database account user name do you want to use to create database connections?
Database User Name:
system
What is the database account password to use to create database connections?
(Note: for secure password management, enter the password in the Password field instead of the Properties field below)
Password:
Confirm Password:
What are the properties to pass to the JDBC driver when creating database connections?
Properties:
user=system
The set of driver properties whose values are derived at runtime from the named system property.
System Properties:
What table name or SQL statement would you like to use to test database connections?
Test Table Name:
SQL ISVALID
Test Configuration
Back
Next
Finish
Cancel

p. Click **Next.**

q. In **Select Targets,** select the check box for **AdminServer.**

r. Click **Finish.**

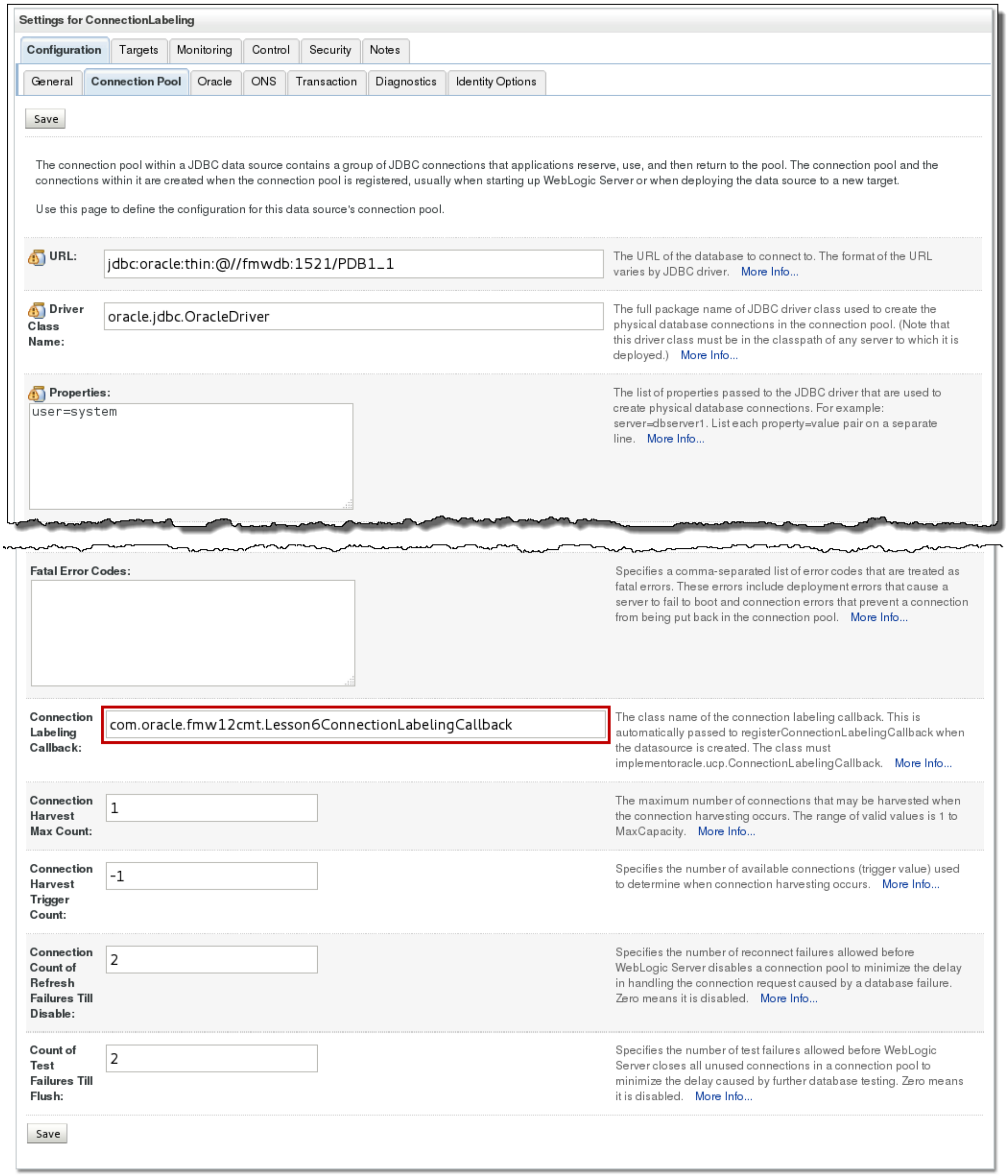
5. Configure the Connection Labeling Callback.

   a. In the Summary of **JDBC Data Sources** table, click the **ConnectionLabeling** Data Source.

   b. Click the **Connection Pool** tab.

   c. Scroll down and click **Advanced.**

   d. For **Connection Labeling Callback,** enter `com.oracle.fmw12cmt.Lesson6ConnectionLabelingCallback.`



e. Click **Save.**

The connection pool changes are activated; however, the AdminServer must be restarted for the changes to take effect.

6. Restart the AdminServer.

a. In the WebLogic Server Administration Console, from the **WebLogic Domain Structure** panel, click **Environment** and click **Servers.**

b. Click the **Control** tab.

c. Select the **AdminServer.**

d. Click **Shutdown** and select **Force Shutdown Now.**

e. On the fmw1 machine, open a Terminal window as the `oracle` user and navigate to `/u01/fmw/wls1221_infra/user_projects/domains/jrf_domain`

$ **cd /u01/fmw/wls1221_infra/user_projects/domains/jrf_domain/**

f. Execute `startWebLogic.sh` to start the WebLogic Administration Server.

$ **./startWebLogic.sh**

g. Keep this terminal window open to view the output from the connection labeling callback class.

7. Log in to the jrf_domain WebLogic Server Domain using the WebLogic Server Administration Console.

   a. Open a Browser window and navigate to **http://fmw1:7001/console.**

   b. For **User Name,** enter `weblogic` and for **Password,** enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click **Login.**

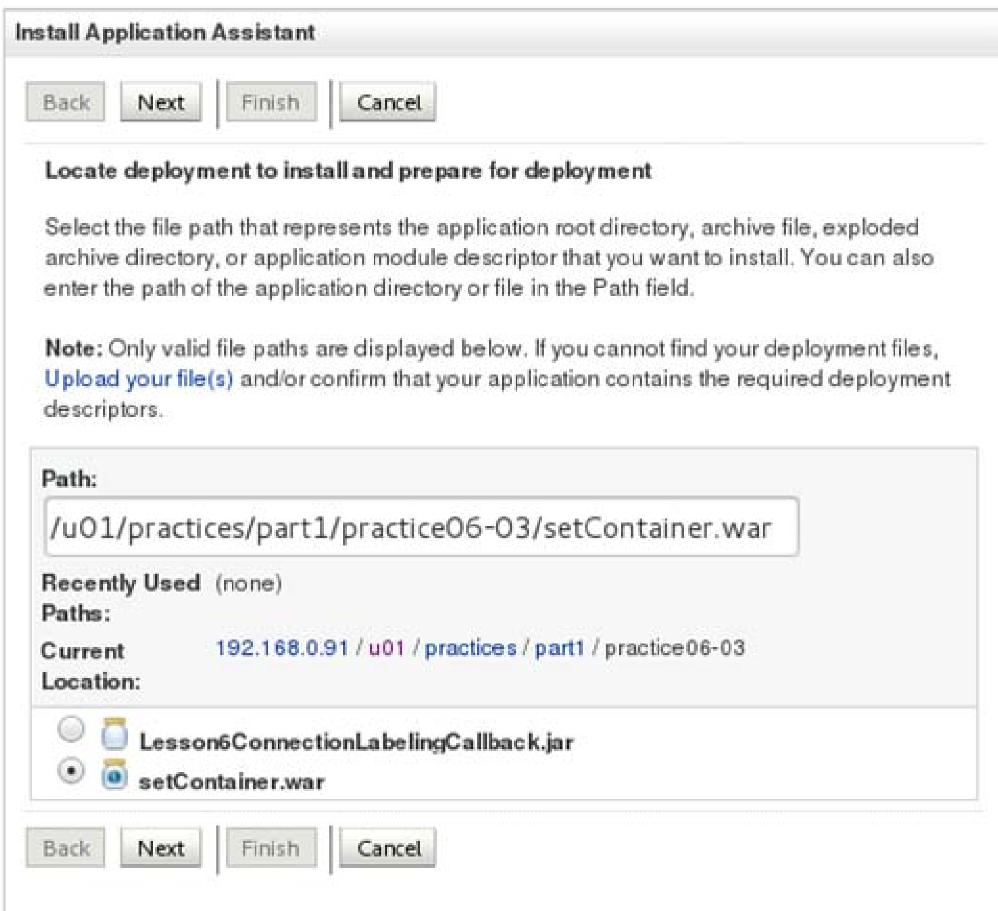
   The WebLogic Server Administration Console home page for the **jrf_domain** is displayed.

8. Deploy the setContainer web application.

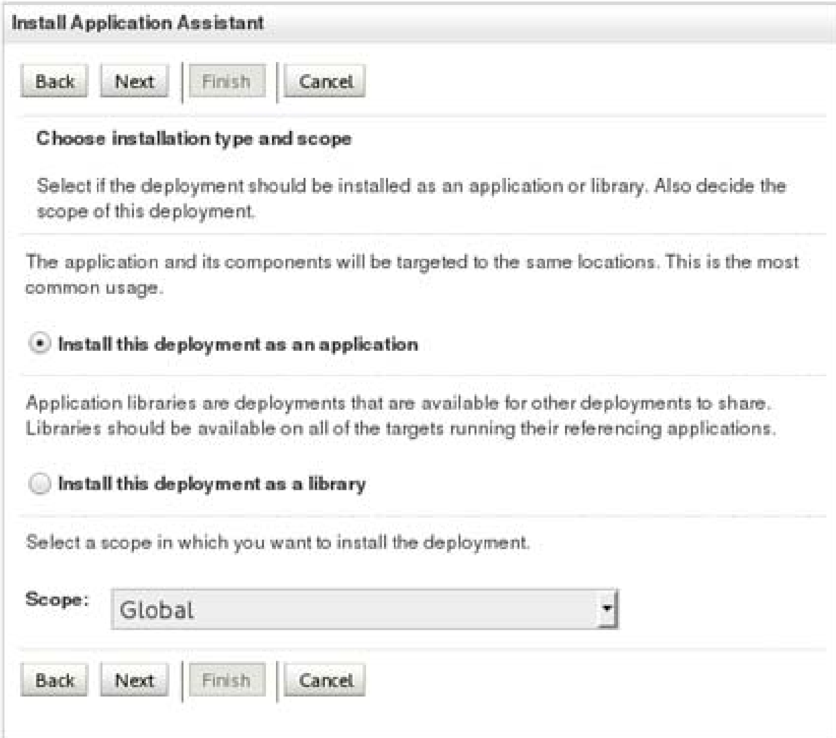
   a. In the WebLogic Server Administration Console, from the **Domain Structure** panel, click **Deployments.**

   b. Click **Install.**

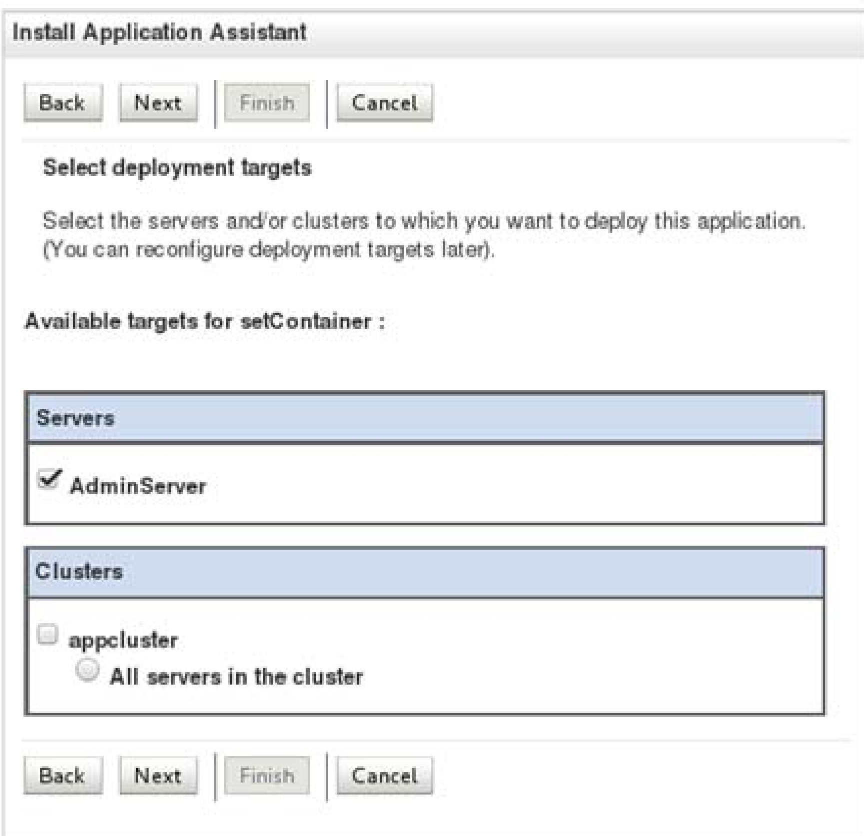
   c. In the **Install Application Assistant,** navigate to `/u01/practices/part1/practice06-03/setContainer.war`, select **setContainer.war** and click **Next.**



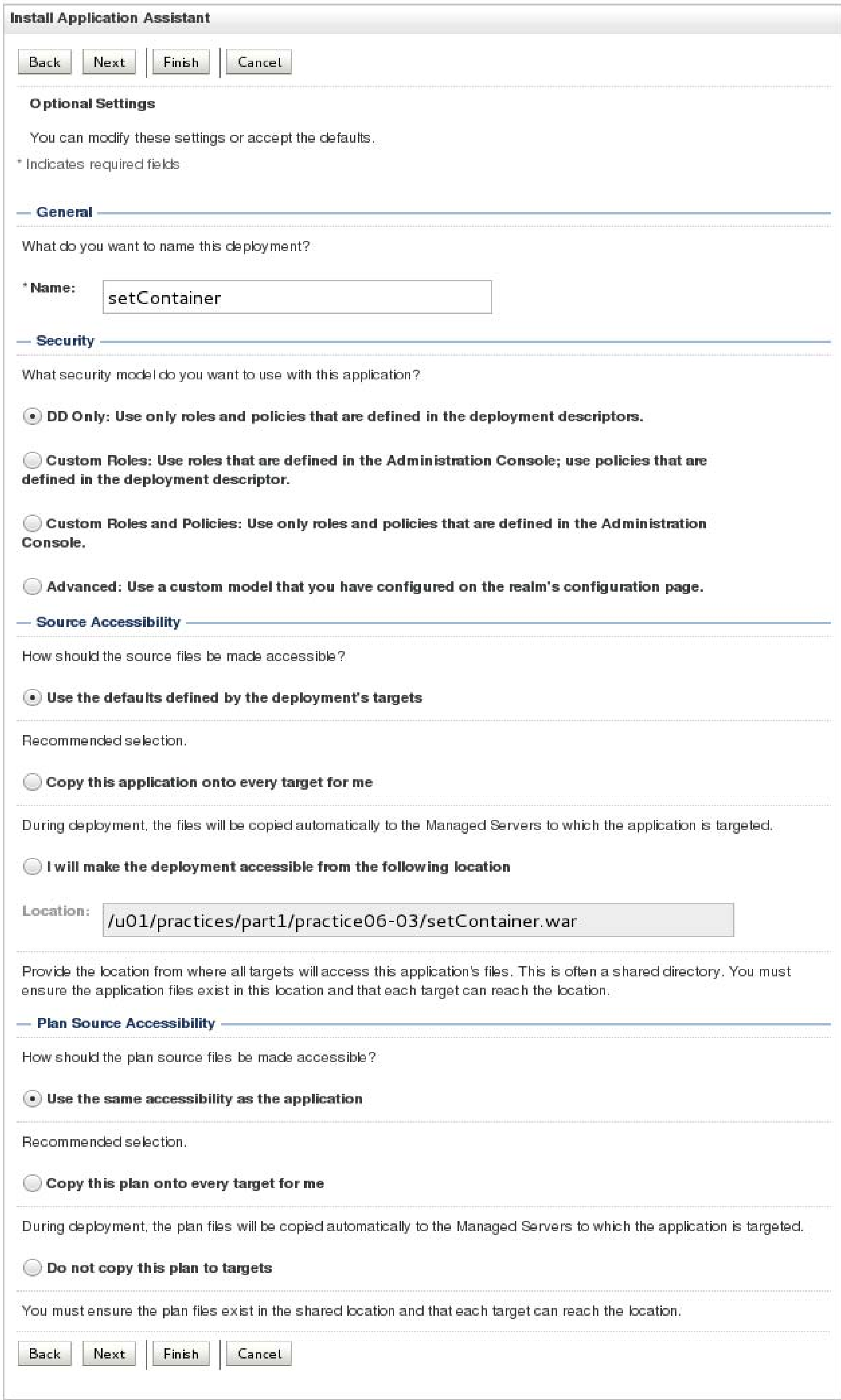
d. Select **Install the deployment as an application** and click **Next.**



e. For deployment targets, select **AdminServer** and click **Next.**



f. In the **Optional Settings** page, click **Next.**



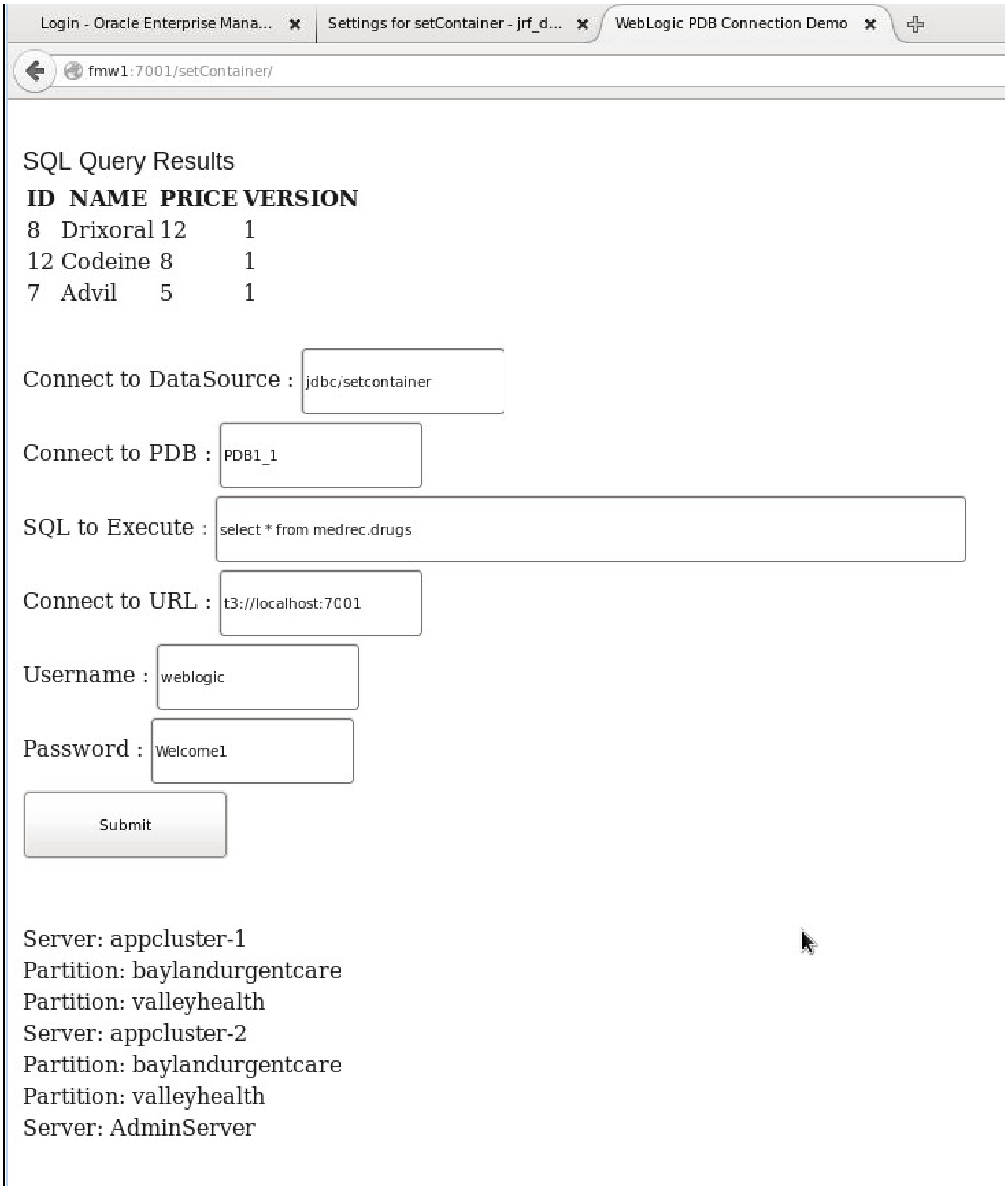
g. Click **Finish.**

The web application is deployed. It uses the connection labeling callback class to switch database connections to other pluggable databases.

9. Test the setContainer web application.

   a. Open a Browser window and navigate to **http://fmw1:7001/setContainer**.

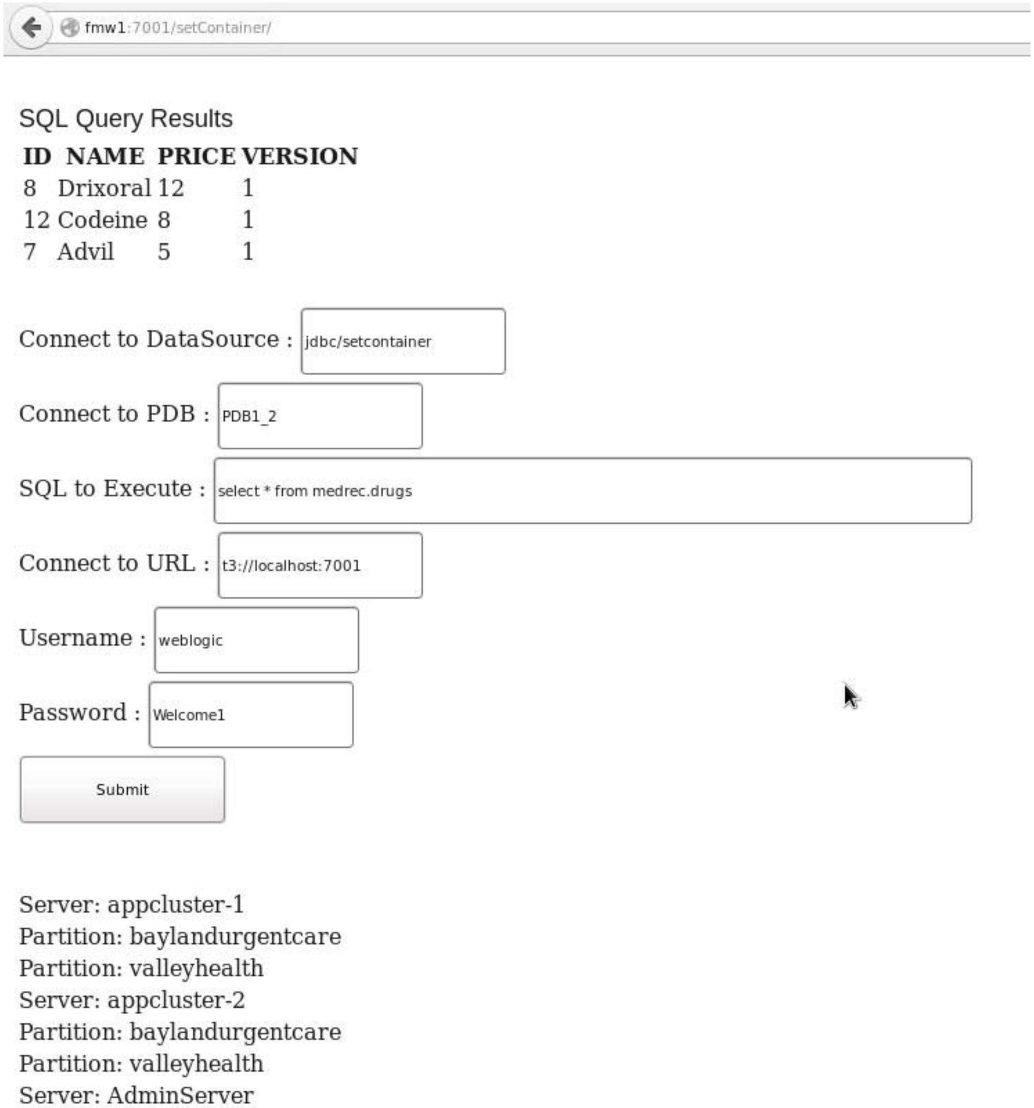
      The page displays the SQL results and fields to enter connection parameters.



   b. For **Connect to PDB,** enter `PDB1_2`.

   c. Click **Submit.**

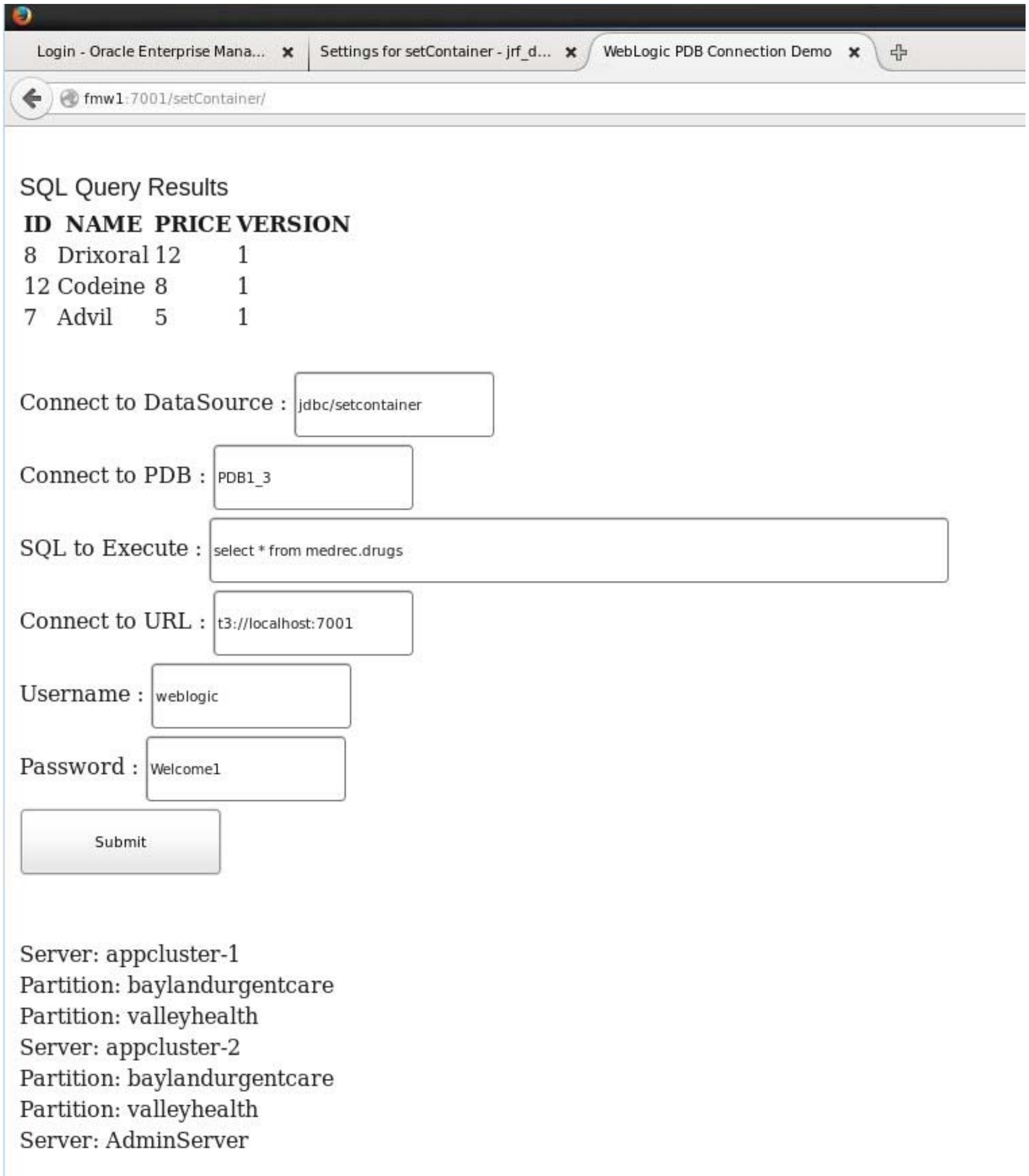
      The connection labeling callback class uses the SET CONTAINER capability to use the PDB1_2 pluggable database and the results from executing the SQL query are displayed.



d. For **Connect to PDB,** enter `PDB1_3`.

e. Click **Submit.**

   The connection labeling callback class uses the SET CONTAINER capability to use the PDB1_2 pluggable database and the results from executing the SQL query are displayed.



f. View the terminal window running startWebLogic.sh.
The connection labeling callback class messages to the system display.


Admin Server(JRF Domain)

File Edit View Search Terminal Help

```
]" is now listening on 192.168.0.160:7001 for protocols iiop, t3, ldap, snmp, ht
tp.>
<Feb 18, 2016 3:39:44 PM UTC> <Notice> <Server> <BEA-002613> <Channel "Default[2
]" is now listening on 0:0:0:0:0:0:0:1%lo:7001 for protocols iiop, t3, ldap, snm
p, http.>
<Feb 18, 2016 3:39:44 PM UTC> <Notice> <WebLogicServer> <BEA-000360> <The server
 started in RUNNING mode.>
<Feb 18, 2016 3:39:44 PM UTC> <Notice> <WebLogicServer> <BEA-000365> <Server sta
te changed to RUNNING.>
setContainer -->
###--> Lesson6ConnectionLabelingCallback: No match 'PDB1_1' != 'null' ##
###--> Lesson6ConnectionLabelingCallback: In configure for value PDB1_1
getting server runtimes
printing servers
server: appcluster-1
2
partition: baylandurgentcare
partition: valleyhealth
server: appcluster-2
2
partition: baylandurgentcare
partition: valleyhealth
server: AdminServer
0
setContainer -->
###--> Lesson6ConnectionLabelingCallback: No match 'PDB1_2' != 'PDB1_1' ##
###--> Lesson6ConnectionLabelingCallback: No match 'PDB1_2' != 'null' ##
###--> Lesson6ConnectionLabelingCallback: In configure for value PDB1_2
getting server runtimes
printing servers
server: appcluster-1
2
partition: baylandurgentcare
partition: valleyhealth
server: appcluster-2
2
partition: baylandurgentcare
partition: valleyhealth
server: AdminServer
0
setContainer -->
###--> Lesson6ConnectionLabelingCallback: No match 'PDB1_1' != 'PDB1_2' ##
###--> Lesson6ConnectionLabelingCallback: Exact match found = PDB1_1
getting server runtimes
printing servers
server: appcluster-1
2
partition: baylandurgentcare
partition: valleyhealth
server: appcluster-2
2
partition: baylandurgentcare
partition: valleyhealth
server: AdminServer
0
setContainer -->
###--> Lesson6ConnectionLabelingCallback: No match 'PDB1_3' != 'PDB1_1' ##
###--> Lesson6ConnectionLabelingCallback: No match 'PDB1_3' != 'PDB1_2' ##
```

Practices for Lesson 7 - Coherence with Multitenancy

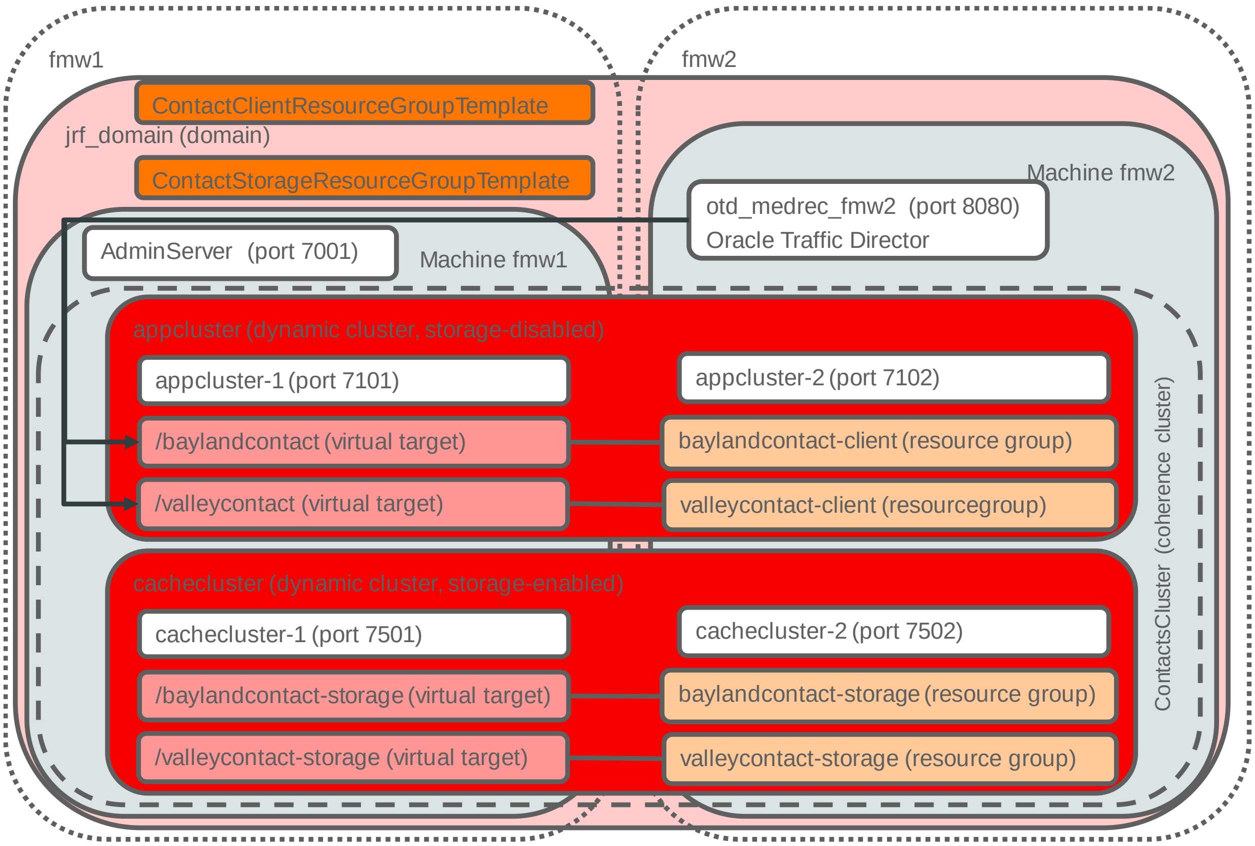
Chapter 7

Practices for Lesson 7: Overview

Overview

In these practices, you create Coherence Clusters managed by WebLogic Server, Resource Group Templates for the coherence application and storage, and create domain partitions for the coherence applications. You also configure one cache as a shared cache for both applications and then test the applications to demonstrate the shared cache.

The diagram below illustrates the configuration:



Practice 7-1: Creating Coherence Clusters Managed by WebLogic Server

Overview

In this practice, using Fusion Middleware Control, you create a new dynamic cluster and coherence cluster for the contacts' sample application. Finally, you restart both dynamic clusters to ensure they are running for subsequent practices.

Assumptions

- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf_domain admin server is running.
- The nodemanager for host fmw1, nodemanager-fmw1 is running.
- The nodemanager for host fmw2, nodemanager-fmw2 is running.
- The machines fm1 and fmw2 have been created.
- The dynamic cluster appcluster has been created.

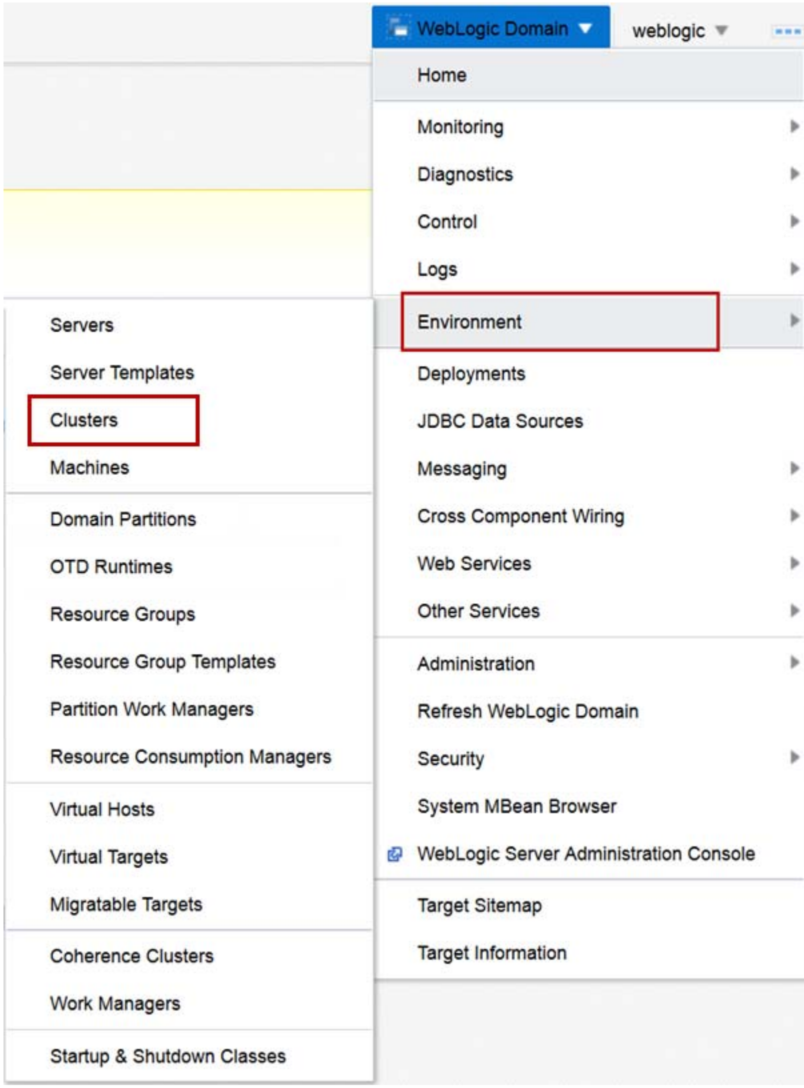
Tasks

1. Log in to the jrf_domain WebLogic Server Domain using Fusion Middleware Control.
    a. Open a browser window and navigate to **http://fmw1:7001/em.**
    b. For **User Name,** enter `weblogic` and for **Password,** enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click **Login.**

       The Fusion Middleware Control home page for the **jrf_domain** is displayed.

2. Create the Dynamic Cluster cachecluster.

   a. From the **WebLogic Domain** menu, select **Environment** and then click **Clusters.**



   b. Click **Create** and select **Dynamic Cluster.**

   c. For **Name,** enter `cachecluster` and click **Next.**

   d. For **Dynamic Cluster Size,** enter **2.**

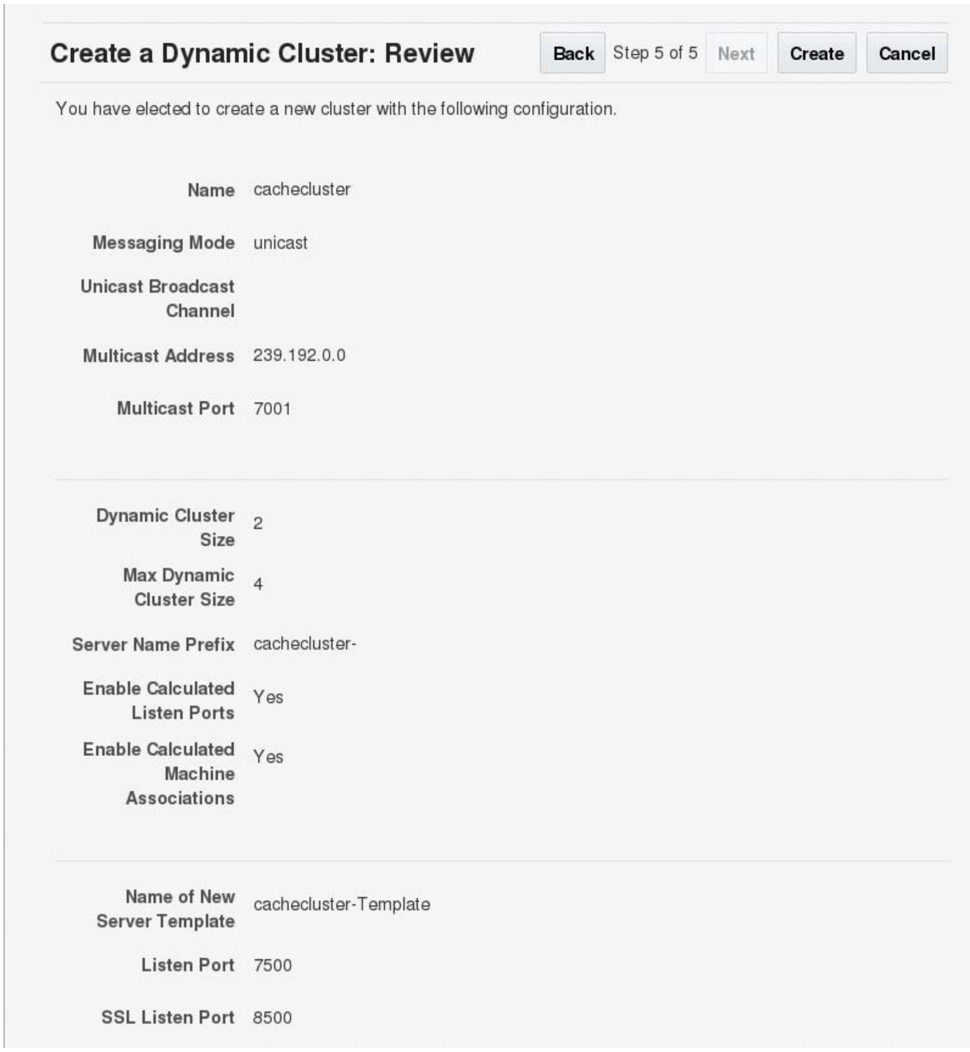
   e. For **Max Dynamic Cluster Size,** enter **4,** and click **Next.**

   f. Select **Use any machine configured in this domain** and click **Next.**

   g. Select **Assign each dynamic server unique listen ports.**

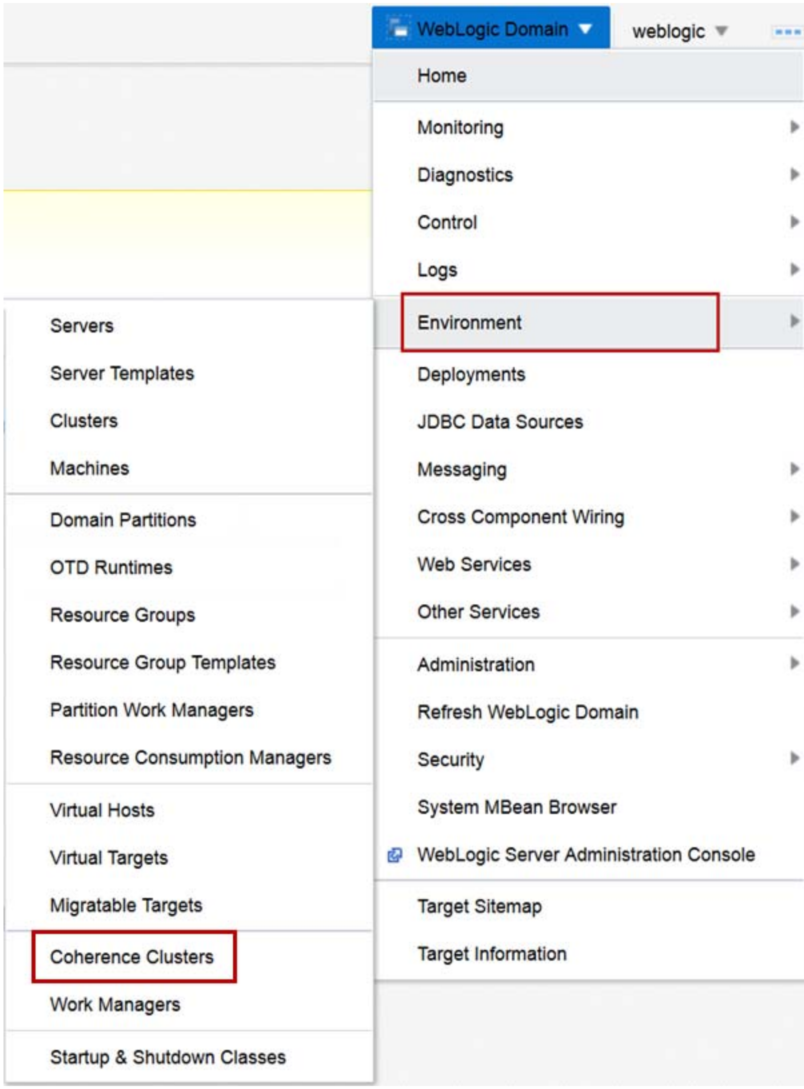
   h. For **Base Listen Port,** enter `7500`.

   i. For **SSL Base Listen Port,** enter `8500` and click **Next.**



j. Click **Create.**

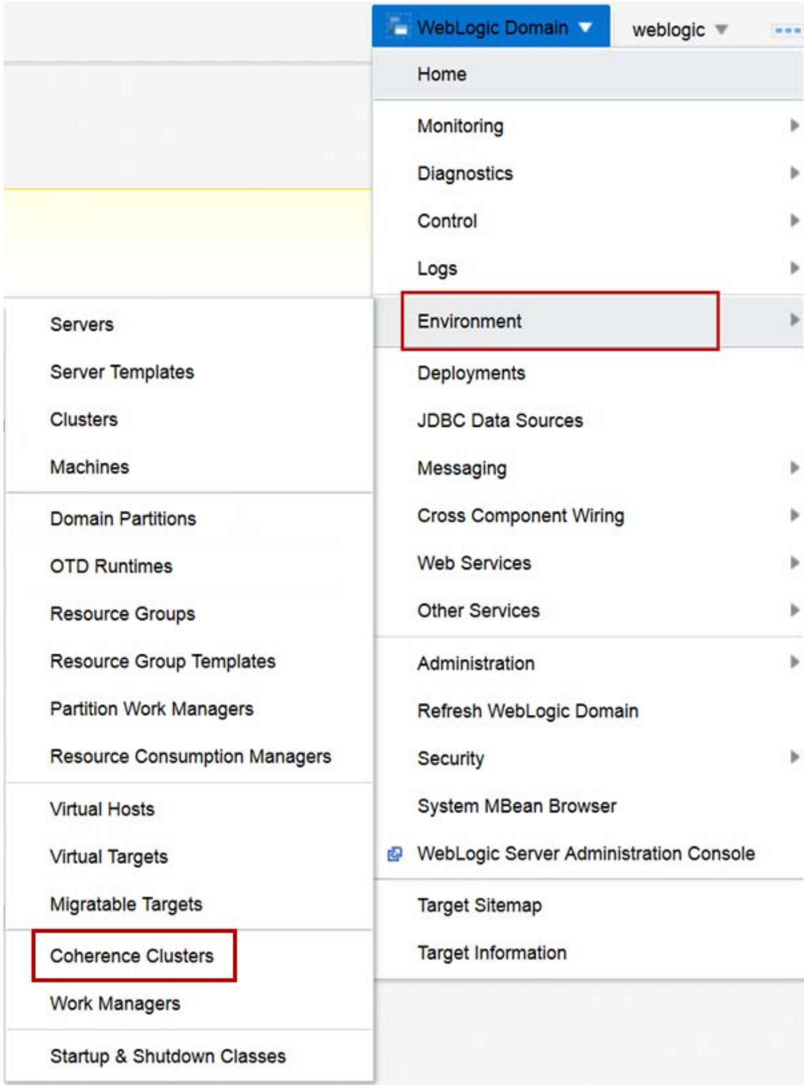
3. Create the Coherence Cluster ContactsCluster.
   a. From the **WebLogic Domain** menu, select **Environment** and then click **Coherence Clusters.**



   b. Click **Create.**
   c. For **Name,** enter `ContactsCluster` and click **Next.**
   d. Select **appcluster** and **cachecluster** as members of this Coherence Cluster and click **Next.**

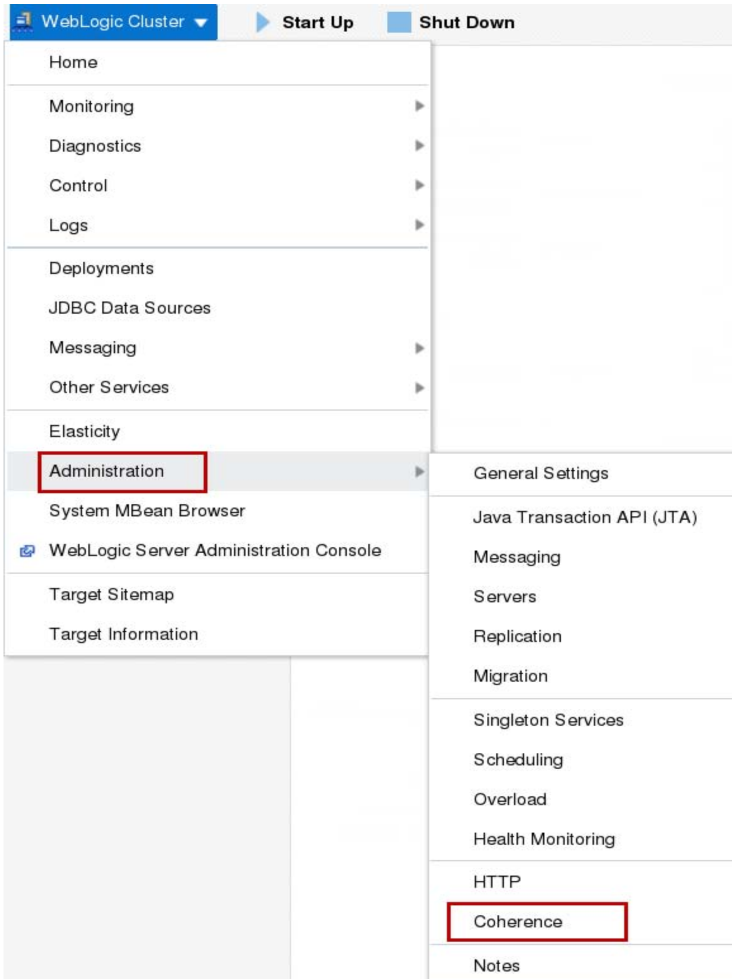
e. Click **Create.**

4. Ensure the appcluster is not storage enabled and the cachecluster is storage enabled.
   a. From the **WebLogic Domain** menu, select **Environment** and then click **Clusters.**



   b. Click the **appcluster** entry.

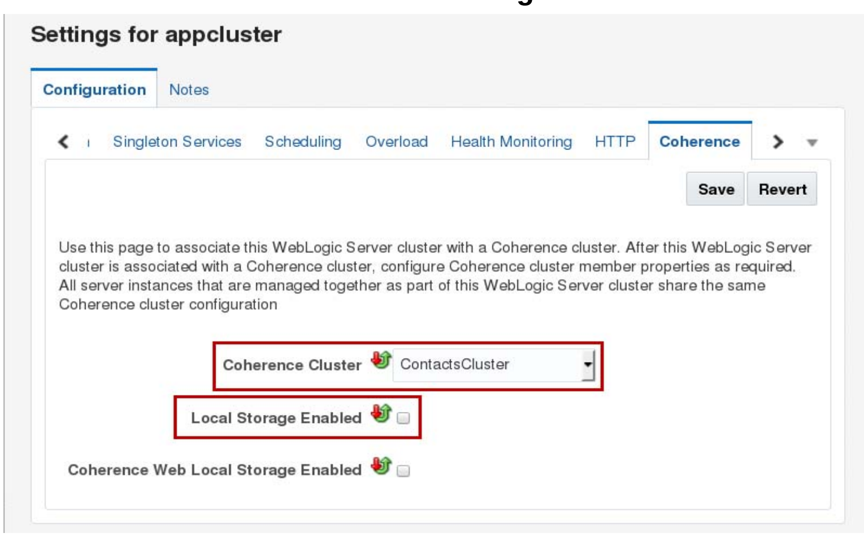
c. From the **WebLogic Cluster** menu, select **Administration** and then select **Coherence.**



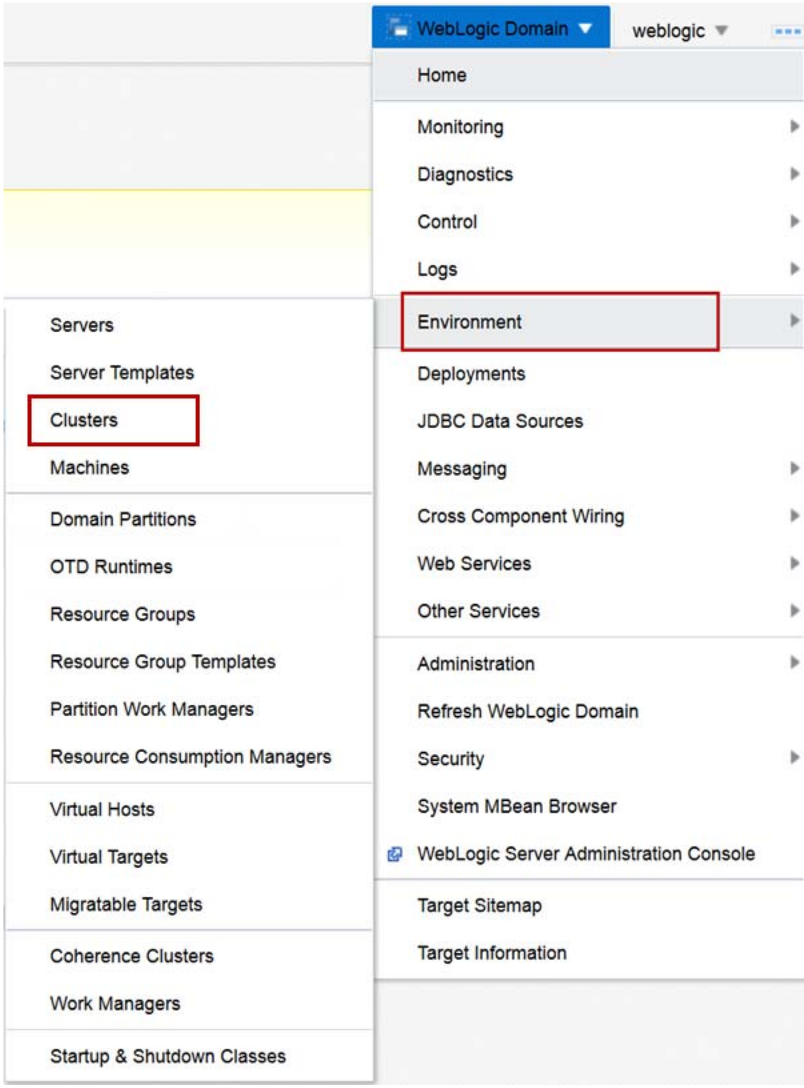
d. For **Coherence Cluster,** select **ContactsCluster.**

e. Deselect **Local Storage Enabled.**

f. Deselect **Coherence Web Local Storage Enabled** and click **Save.**

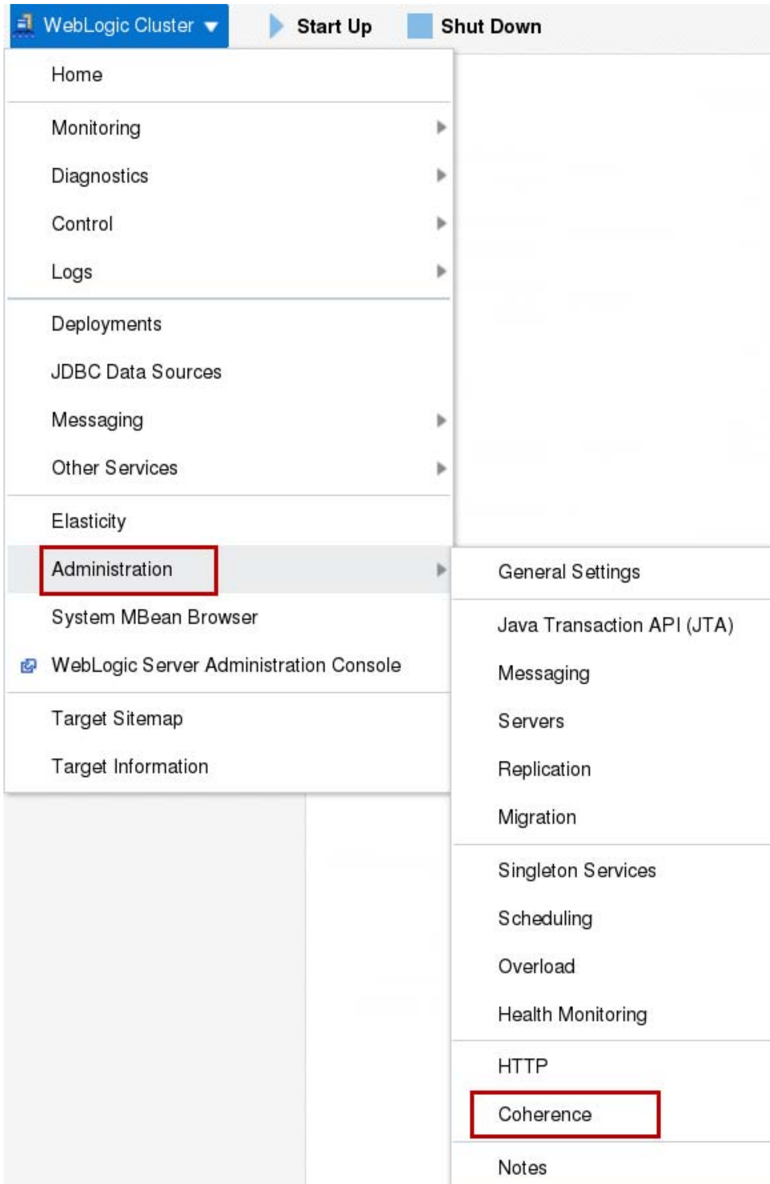


g. From the **WebLogic Domain** menu, select **Environment** and then click **Clusters.**



h. Click the **cachecluster** entry.

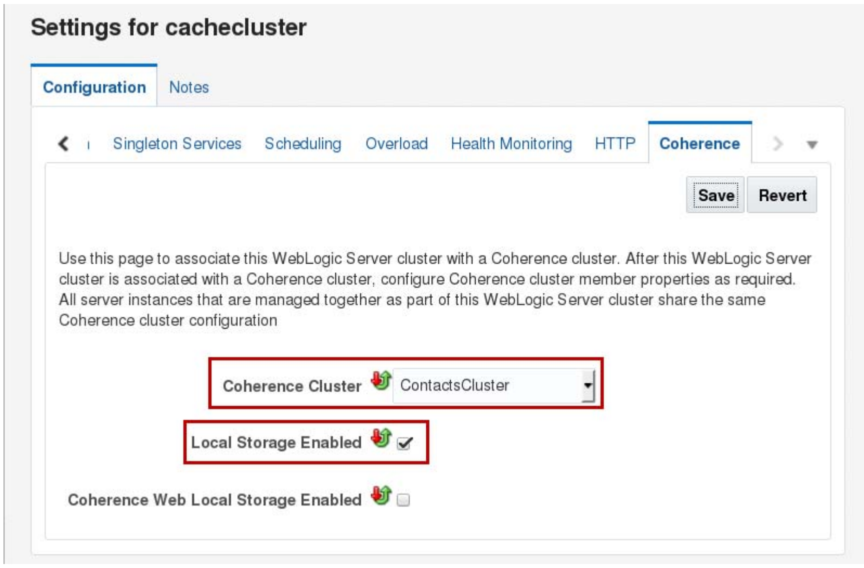
i. From the **WebLogic Cluster** menu, select **Administration** and then select **Coherence.**



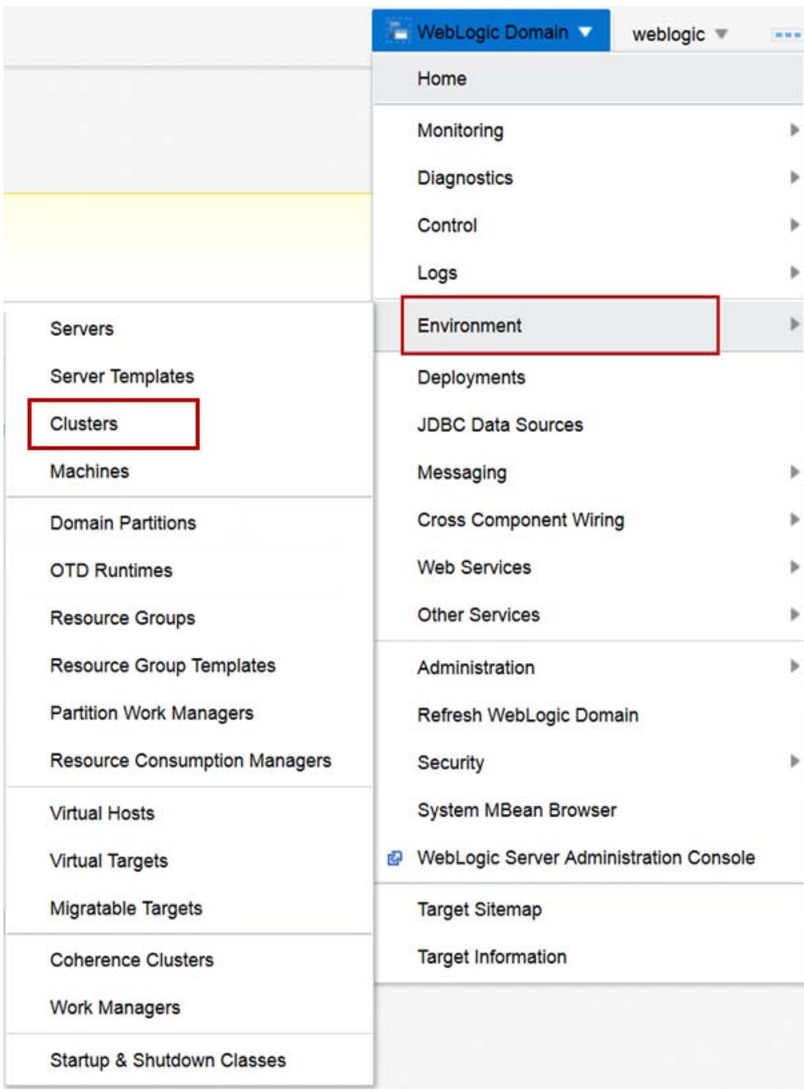
j. For **Coherence Cluster,** select **ContactsCluster.**

k. Select **Local Storage Enabled.**

l. Deselect **Coherence Web Local Storage Enabled** and click **Save.**



6. Restart the Dynamic Cluster appcluster.
   a. From the **WebLogic Domain** menu, select **Environment** and then click **Clusters.**



b. Click the table entry for **appcluster.**

c. Click **Control,** select **Shutdown,** and select **Force shutdown now.**

d. Click **Forcibly Shutdown Servers.**

e. Click the **Refresh** icon.

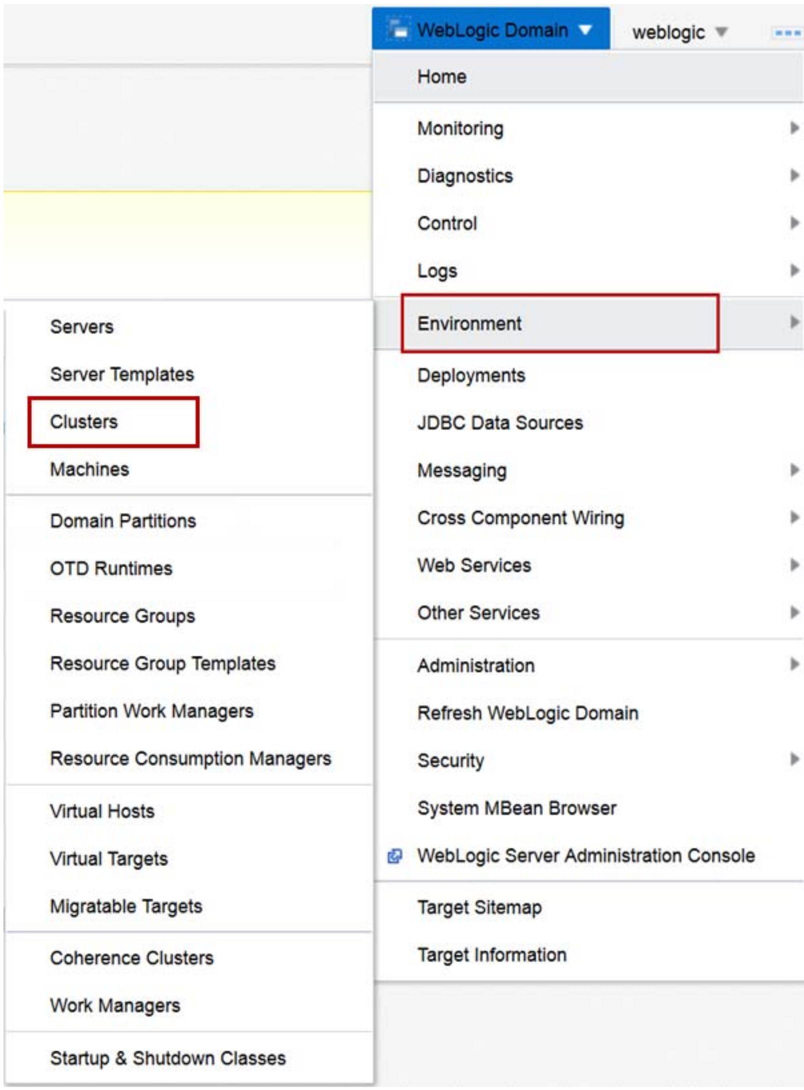
f. Click the table entry for **appcluster.**

g. Click **Control** and select **Start.**

h. Click **Start Servers.**

7. Start the Dynamic Cluster cachecluster.

a. From the **WebLogic Domain** menu, select **Environment** and then click **Clusters.**



b. Click the table entry for **cachecluster.**

c. Click **Control** and select **Start.**

d. Click **Start Servers.**

Practice Solution: Creating Coherence Clusters Managed by WebLogic Server

Perform the following tasks if you did not complete this practice and want to use the finished solution.

Note: If you started creating the dynamic cluster cachecluster or the coherence cluster, this script will not work. To run the script in that case, delete the partially created clusters before running this script.

Assumptions

- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf_domain admin server is running.
- The nodemanager for host fmw1, nodemanager-fmw1 is running.
- The nodemanager for host fmw2, nodemanager-fmw2 is running.
- The machines fm1 and fmw2 have been created.
- The dynamic cluster appcluster has been created.

Solution Tasks

1. Connect to fmw1 as the `oracle` user.
2. Run the solution script.
   a. Open a Terminal window as the `oracle` user and navigate to the practice directory.
      ```
      $ cd /u01/practices/part1/practice07-01
      ```
   b. Run the solution script.
      ```
      $ ./solution.sh
      ```
   c. Enter any passwords as you are prompted.

   Note: This script starts the WebLogic Scripting Tool (WLST) passing it the WLST script `create_coherence_clusters.py`, which creates the new generic data source for you. The WLST script prints out messages. Be patient and do not close the Terminal window until you see these messages:

   ```
   ...
   >>> Solution for Practice 07-01 applied successfully!

   Exiting WebLogic Scripting Tool.
   ```
3. Close the Terminal window.

Practice 7-2: Creating Coherence Resource Group Templates

Overview

In this practice, you create the resource group templates required for the contact sample application—one resource group template for the client application and cluster and a second resource group template for the storage application and cluster.

Assumptions

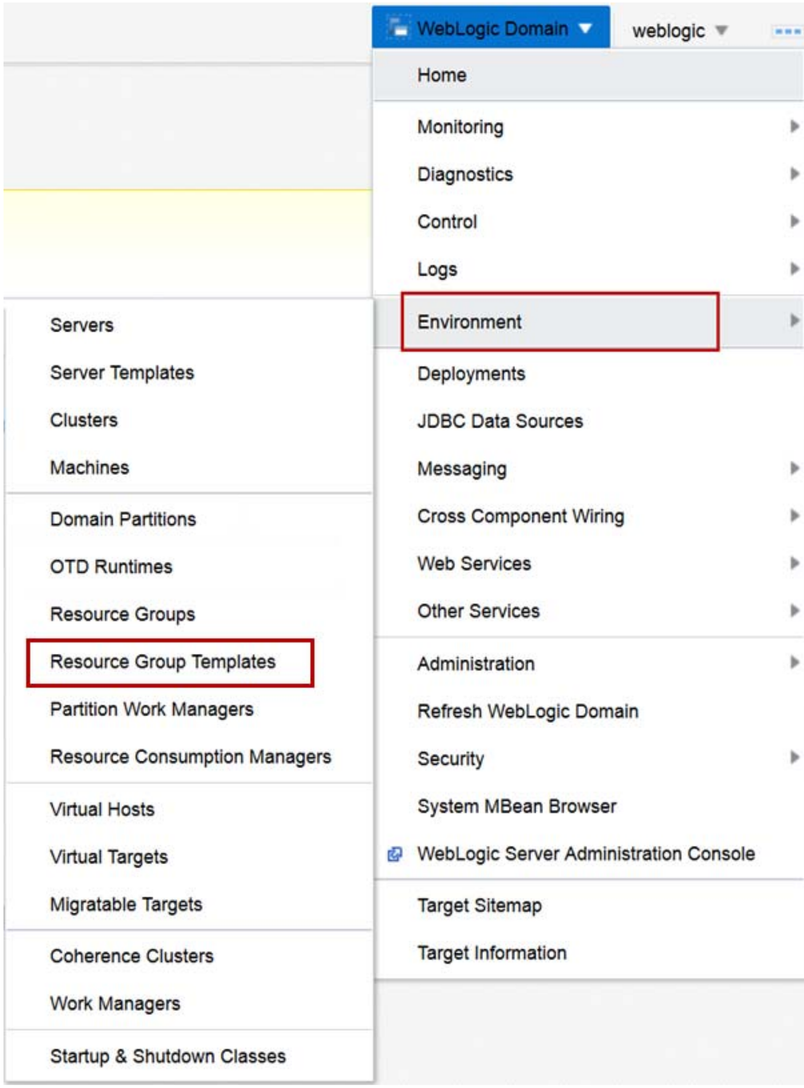
- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf_domain admin server is running.
- The nodemanager for host fmw1, nodemanager-fmw1 is running.
- The nodemanager for host fmw2, nodemanager-fmw2 is running.
- The machines fm1 and fmw2 have been created.
- The dynamic cluster appcluster has been created and is running.
- The dynamic cluster coherencecluster has been created and is running.

Tasks

1. Log in to the jrf_domain WebLogic Server Domain using Fusion Middleware Control.
   a. Open a Browser window and navigate to **http://fmw1:7001/em.**
   b. For **User Name,** enter `weblogic` and for **Password,** enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click **Login.**

      The Fusion Middleware Control home page for the **jrf_domain** is displayed.

2. Create the ContactClientResourceGroupTemplate.
   a. From the **WebLogic Domain** menu, select **Environment** and then click **Resource Group Templates.**



b. Click **Create.**

c. For **Name,** enter `ContactClientResourceGroupTemplate` and click **Create.**

d. Click the **ContactClientResourceGroupTemplate** table entry.

e. Click the **Deployments** tab.

f. Click **Deployment** and select **Deploy.**

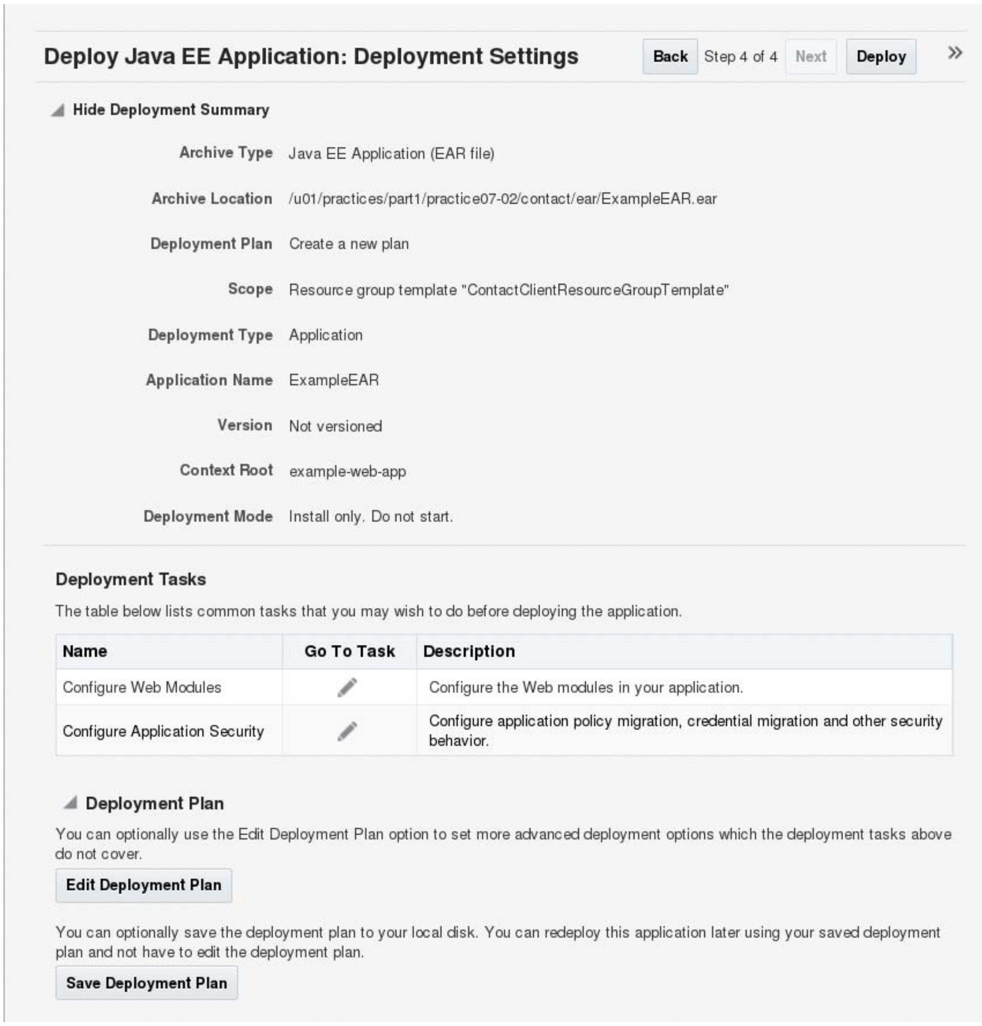
g. Select **Archive or exploded directory is on the server where Enterprise Manager is running.**

h. Click **Browse.**

i. Browse to `/u01/practices/part1/practice07-02/contact/ear/ExampleEAR.ear` and click **OK.**

j. Click **Next.**

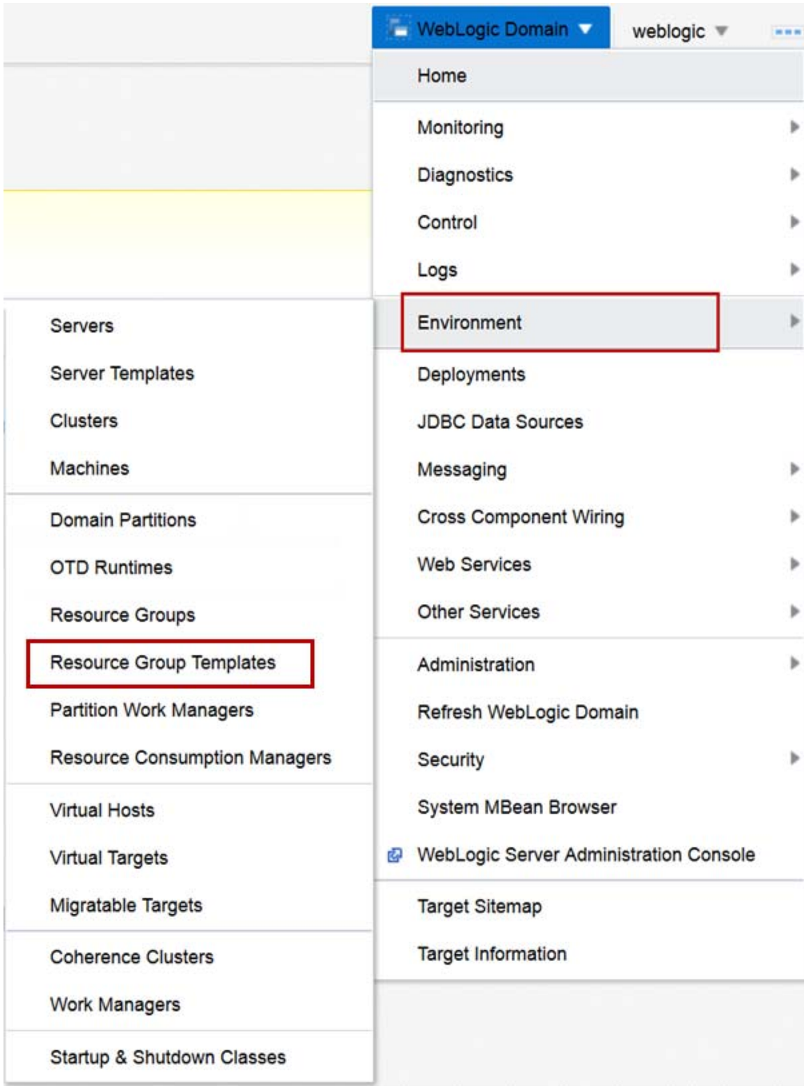
k. In the **Distribution** section, select **Install only. Do not start.** and click **Next.**



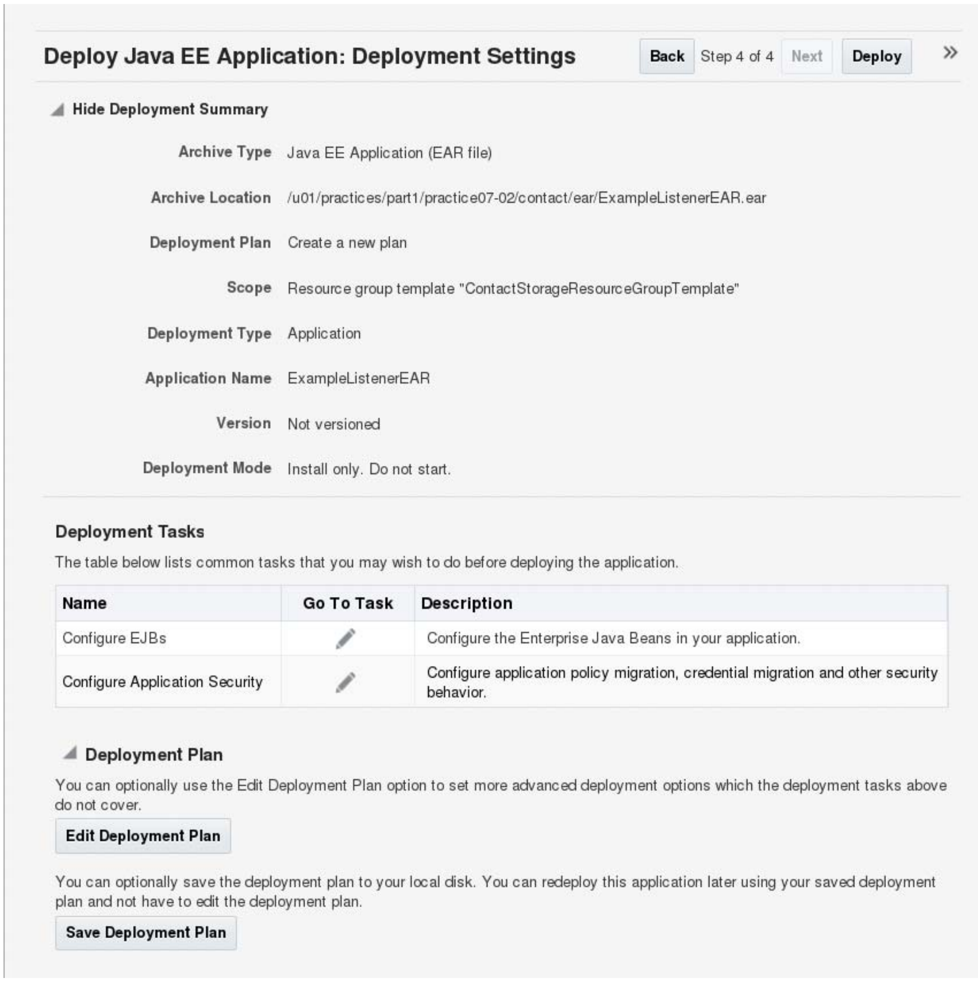
l. Review the Deployment Settings and click **Deploy.**

m. Click **Close.**

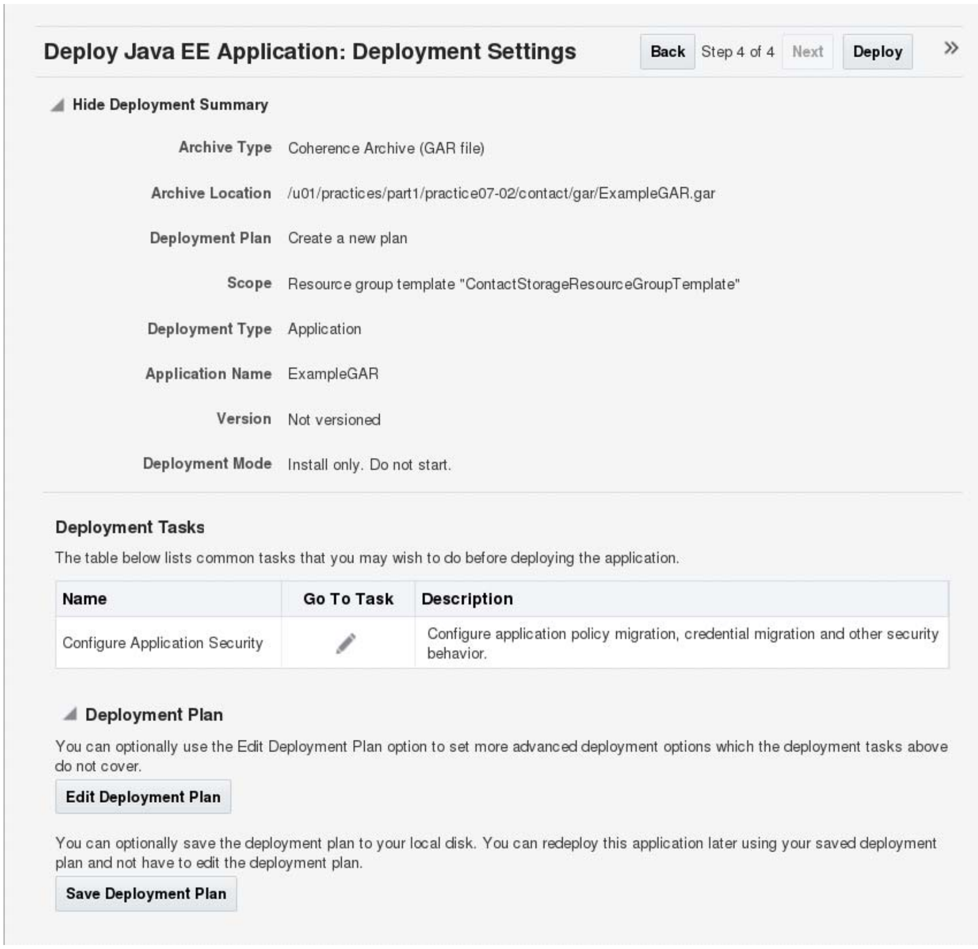
3. Create the ContactStorageResourceGroupTemplate.
   a. From the **WebLogic Domain** menu, select **Environment** and then click **Resource Group Templates.**



   b. Click **Create.**
   c. For **Name,** enter `ContactStorageResourceGroupTemplate` and click **Create.**
   d. Click the **Deployments** tab.
   e. Click **Deployment** and select **Deploy.**
   f. Select **Archive or exploded directory is on the server where Enterprise Manager is running.**
   g. Click **Browse.**
   h. Browse to `/u01/practices/part1/practice07-02/contact/ear/ExampleListenerEAR.ear` and click **OK.**
   i. Click **Next.**
   j. In the **Distribution** section, select **Install only. Do not start.** and click **Next.**



k. Review the Deployment Settings and click **Deploy.**
l. Click **Close.**
m. Click **Deployment** and select **Deploy.**
n. Select **Archive or exploded directory is on the server where Enterprise Manager is running.**
o. Click **Browse.**
p. Browse to `/u01/practices/part1/practice07-02/contact/gar/ExampleGAR.gar` and click **OK.**
q. Click **Next.**
r. In the **Distribution** section, select **Install only. Do not start.** and click **Next.**



s. Review the Deployment Settings and click **Deploy.**

t. Click **Close.**

Practice Solution: Creating Coherence Resource Group Templates

Perform the following tasks if you did not complete this practice and want to use the finished solution.

Note: If you started creating domain partitions, this script will not work. To run the script in that case, delete the partially created domain partitions before running this script.

Assumptions

- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf_domain admin server is running.
- The nodemanager for host fmw1, nodemanager-fmw1 is running.
- The nodemanager for host fmw2, nodemanager-fmw2 is running.
- The machines fm1 and fmw2 have been created.
- The dynamic cluster appcluster has been created and is running.
- The dynamic cluster coherencecluster has been created and is running.

Solution Tasks

1. Connect to fmw1 as the `oracle` user.
2. Run the solution script.
   a. Open a Terminal window as the `oracle` user and navigate to the practice directory.
      ```
      $ cd /u01/practices/part1/practice07-02
      ```
   b. Run the solution script.
      ```
      $ ./solution.sh
      ```
   c. Enter any passwords as you are prompted.

   Note: This script starts the WebLogic Scripting Tool (WLST) passing it the WLST script `create_coherence_rg.py`, which creates the domain partitions for you. The WLST script prints out messages. Be patient and do not close the Terminal window until you see these messages:

   ```
   ...
   >>> Solution for Practice 07-02 applied successfully!


   Exiting WebLogic Scripting Tool.
   ```
3. Close the Terminal window.

Practice 7-3: Creating Domain Partitions Using Coherence Resources

Overview

In this practice, you create the domain partition for the contact sample application—one domain partition for bayland and a second domain partition for valley. Finally, you start the domain partitions and test the contact application and shared coherence cache.

Assumptions

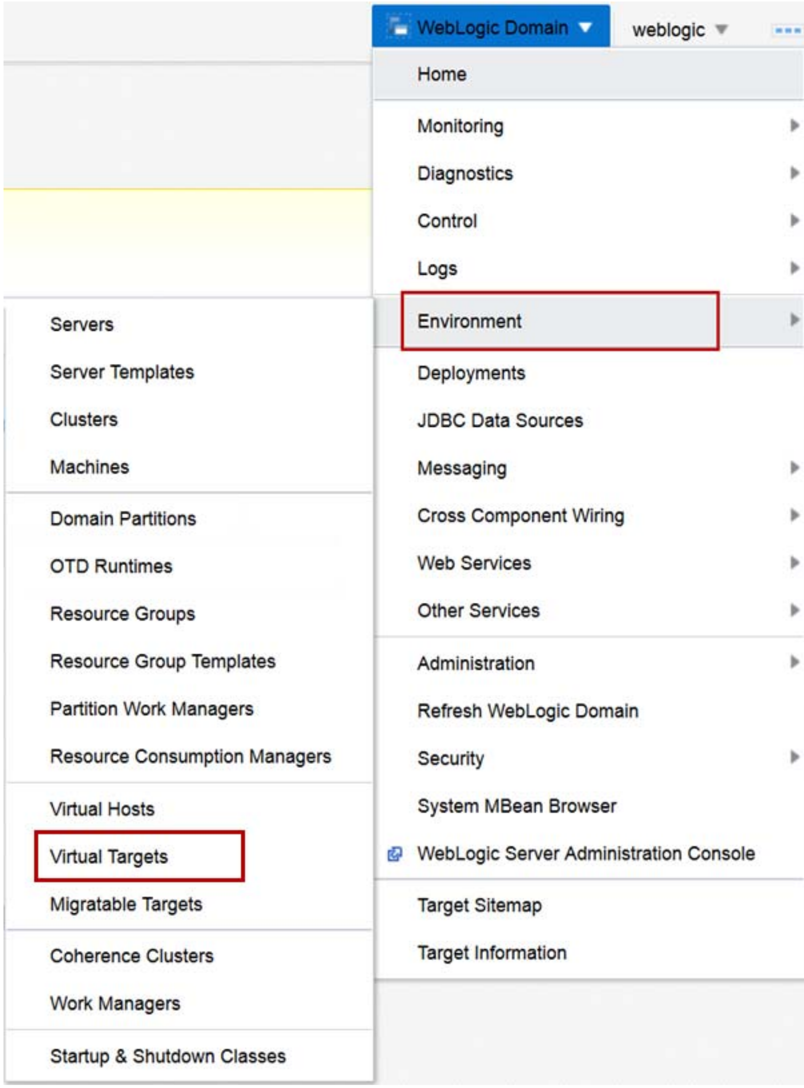
- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf_domain admin server is running.
- The nodemanager for host fmw1, nodemanager-fmw1 is running.
- The nodemanager for host fmw2, nodemanager-fmw2 is running.
- The machines fm1 and fmw2 have been created.
- The dynamic cluster appcluster has been created and is running.
- The dynamic cluster cachecluster has been created and is running.
- The resource group templates ContactClientResourceGroupTemplate and ContactStorageResourceGroupTemplate have been created.
- The Oracle Traffic Director configuration medrec has been created and registered.

Tasks

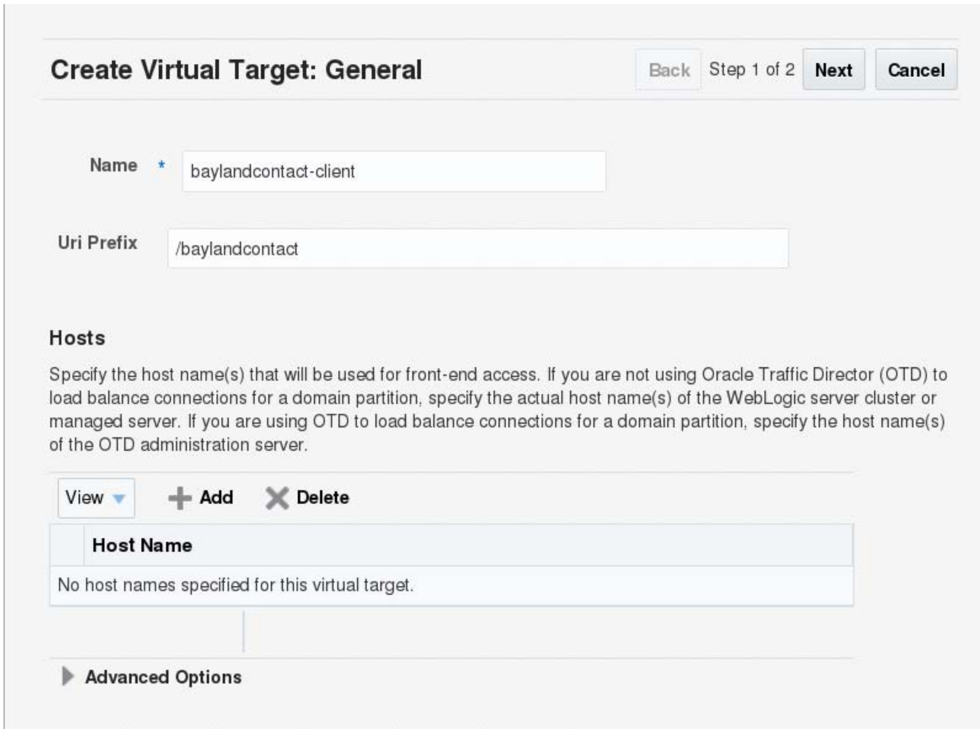
1. Log in to the jrf_domain WebLogic Server Domain using Fusion Middleware Control.
   a. Open a Browser window and navigate to **http://fmw1:7001/em.**
   b. For **User Name,** enter `weblogic` and for **Password,** enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click **Login.**

      The Fusion Middleware Control home page for the **jrf_domain** is displayed.

2. Create the baylandcontact-client virtual target.
   a. From the **WebLogic Domain** menu, select **Environment** and click **Virtual Targets.**



   b. Click **Create.**
   c. For **Name,** enter `baylandcontact-client`.
   d. For **Uri Prefix,** enter `/baylandcontact`.



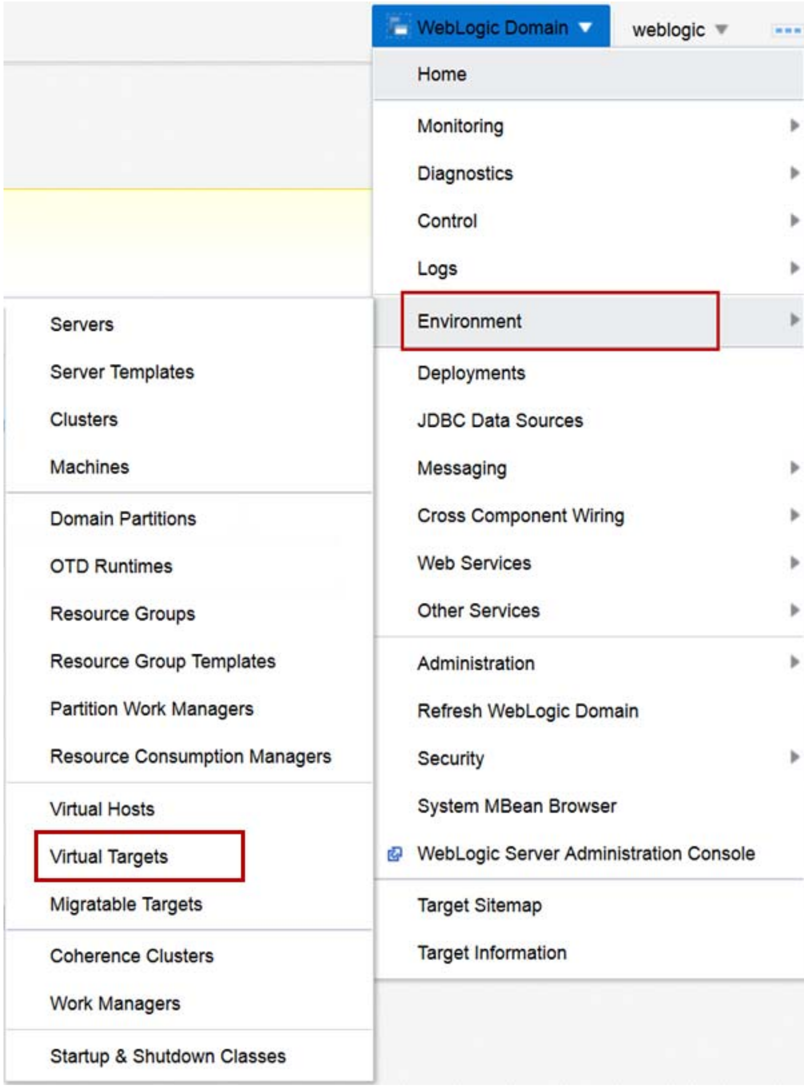
e. Click **Next.**

f. Select **appcluster** from the **Cluster** menu.

g. Click **Create.**

3. Create the baylandcontact-storage virtual target.

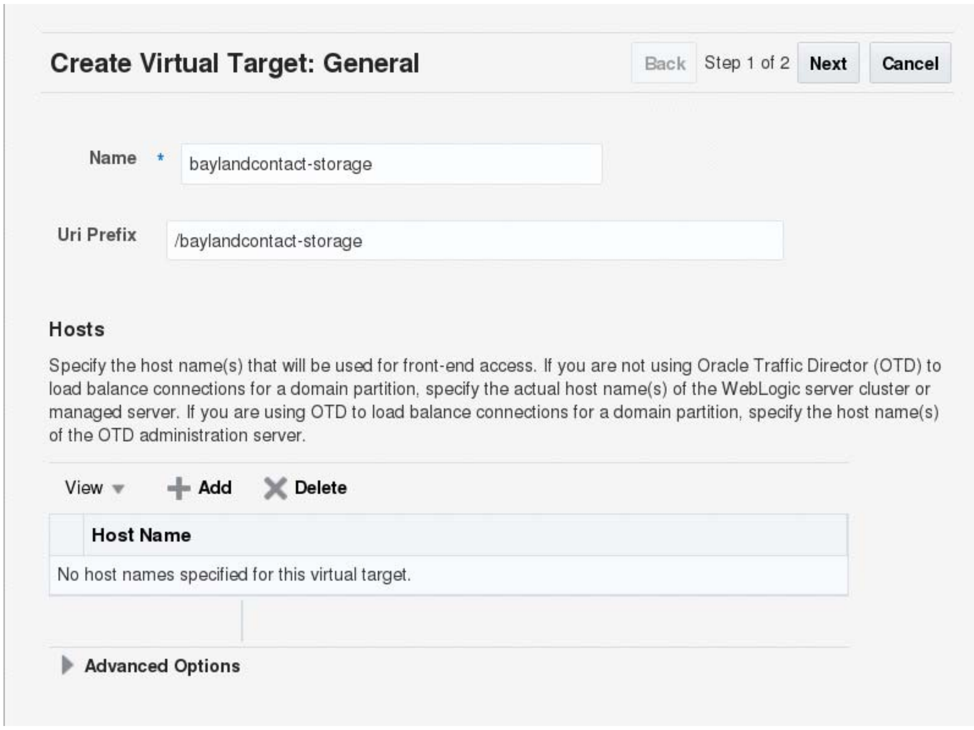
   a. From the **WebLogic Domain** menu, select **Environment** and click **Virtual Targets.**



   b. Click **Create.**

   c. For **Name,** enter `baylandcontact-storage`.

   d. For **Uri Prefix,** enter `/baylandcontact-storage.`

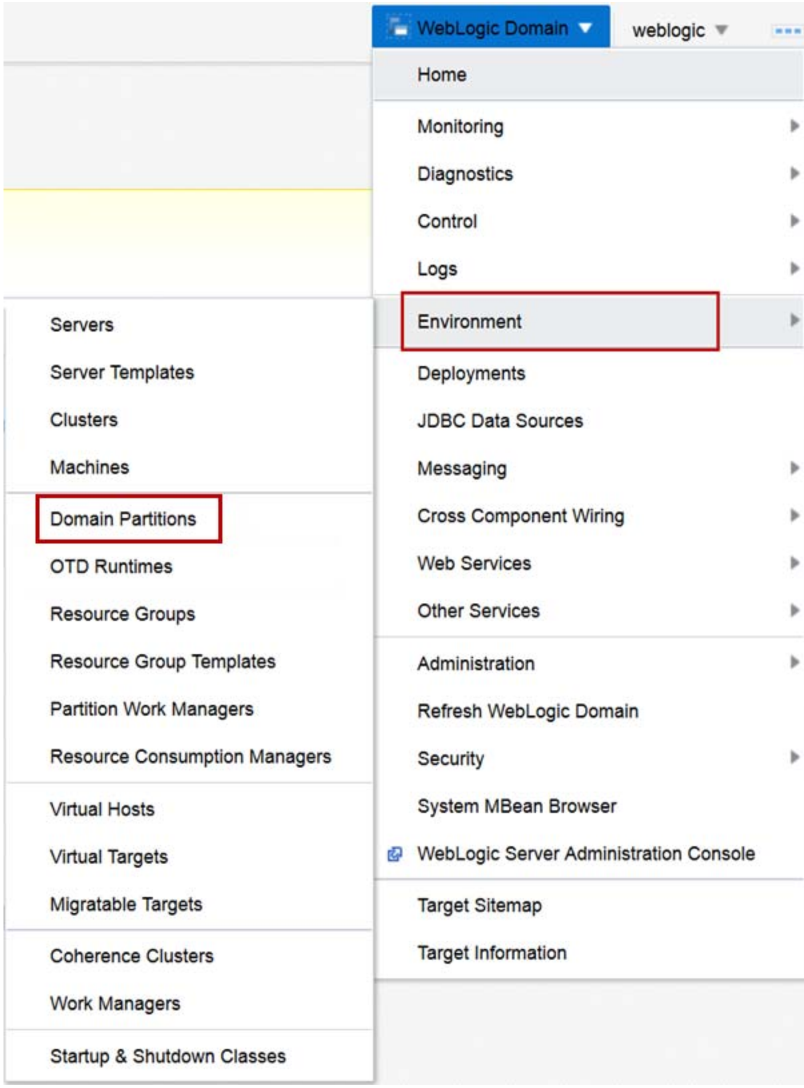


e. Click **Next.**

f. Select **cachecluster** from the **Cluster** menu.

g. Click **Create.**

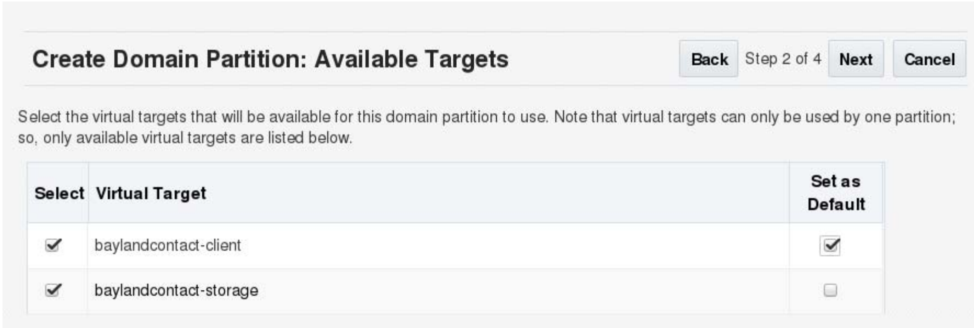
4. Create the baylandcontact domain partition.
   a. From the **WebLogic Domain** menu, select **Environment** and then click **Domain Partitions.**



   b. Click **Create.**
   c. For **Name,** enter `baylandcontact`.
   d. For **Security Realm,** select bayland.
   e. In the **Load Balancer Configuration** section of the page, select **Use OTD for load balancing.**
   f. For **OTD Runtime,** select **medrec.**
   g. Click **Next.**

h. For the Virtual Target **baylandcontact-client,** choose both **Select** and **Set as Default.**

i. For the Virtual Target **baylandcontact-storage,** choose **Select** and click **Next.**

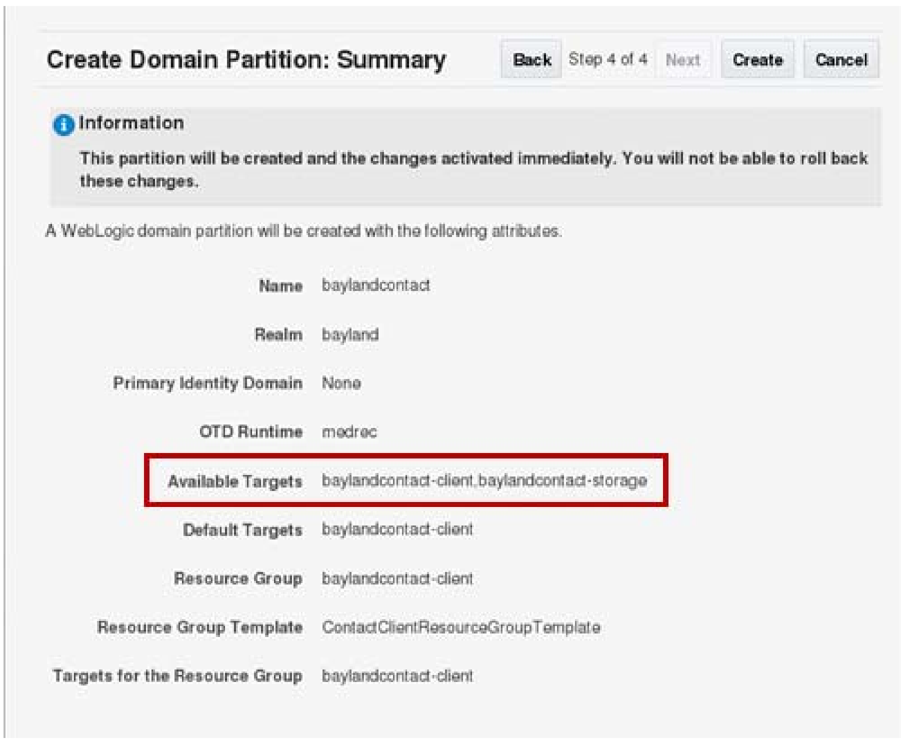


j. For **Resource Group Name,** enter `baylandcontact-client`.

k. For **Resource Group Template,** select **ContactClientResourceGroupTemplate.**

l. Move the **baylandcontact-client** target to **Selected Targets.**

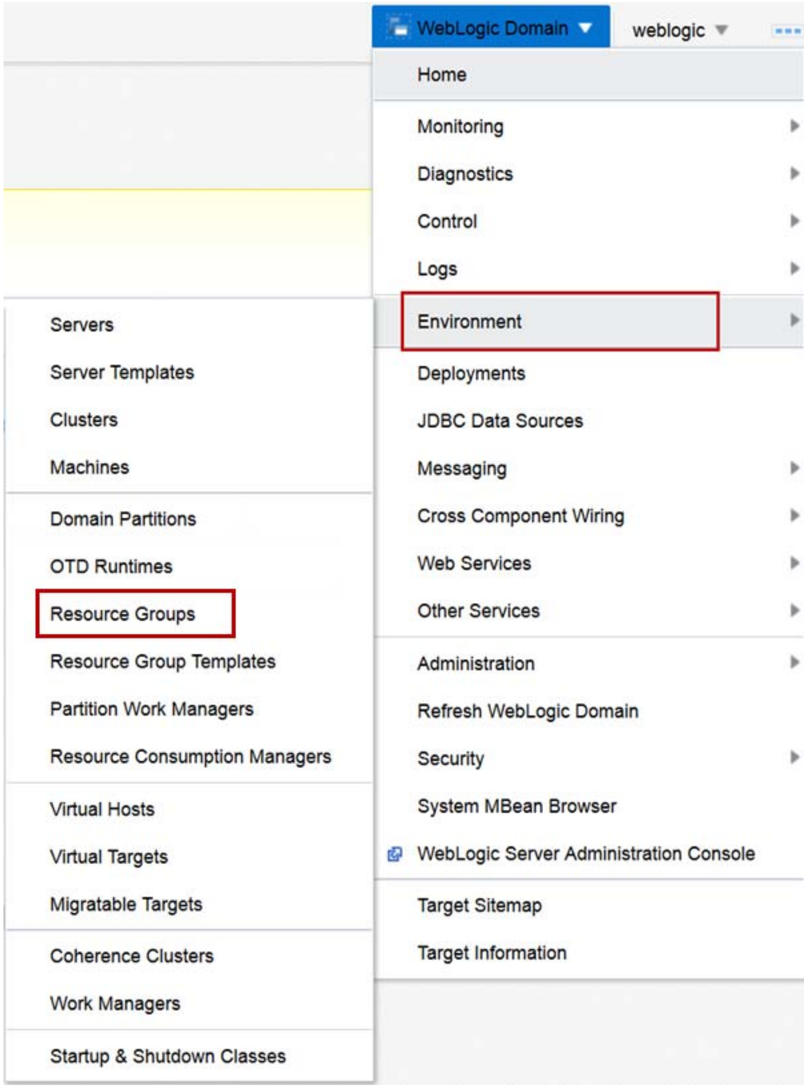
m. Click **Next.**



n. Click **Create.**

o. Click the **baylandcontact** domain partition.

p. From the **Domain Partition** menu, select **Administration** and select **Resource Groups.**

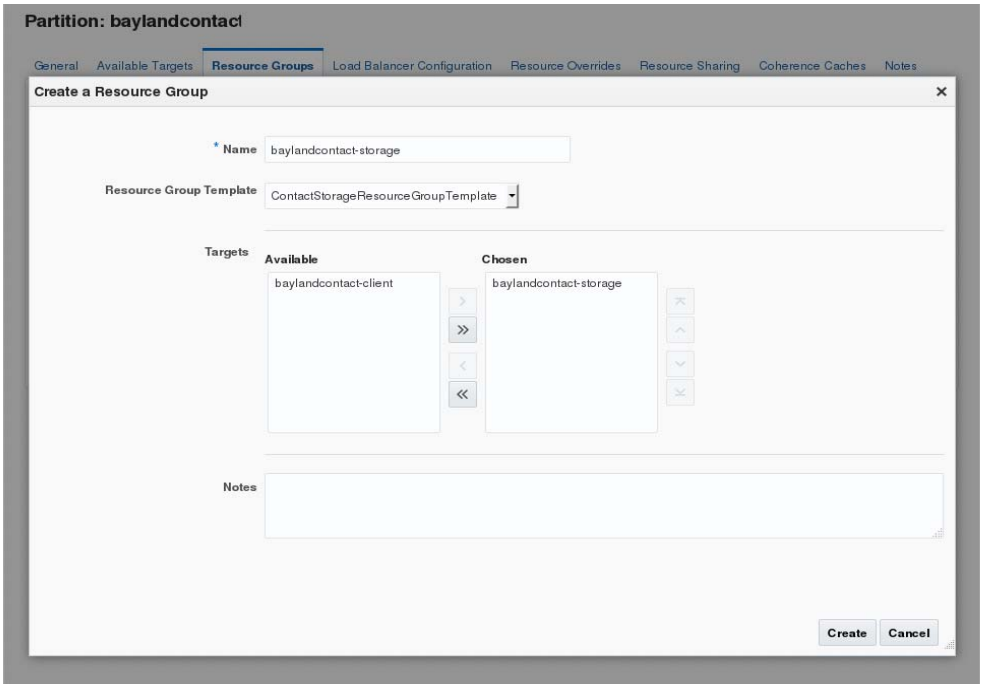


q. Click **Create.**

r. For **Name,** enter `baylandcontact-storage.`

s. For **Resource Group Template,** select **ContactStorageResourceGroupTemplate.**

t. For **Targets,** move **baylandcontact-storage** from **Available** to **Chosen** and click **Create.**



u. Click the **Coherence Caches** tab.

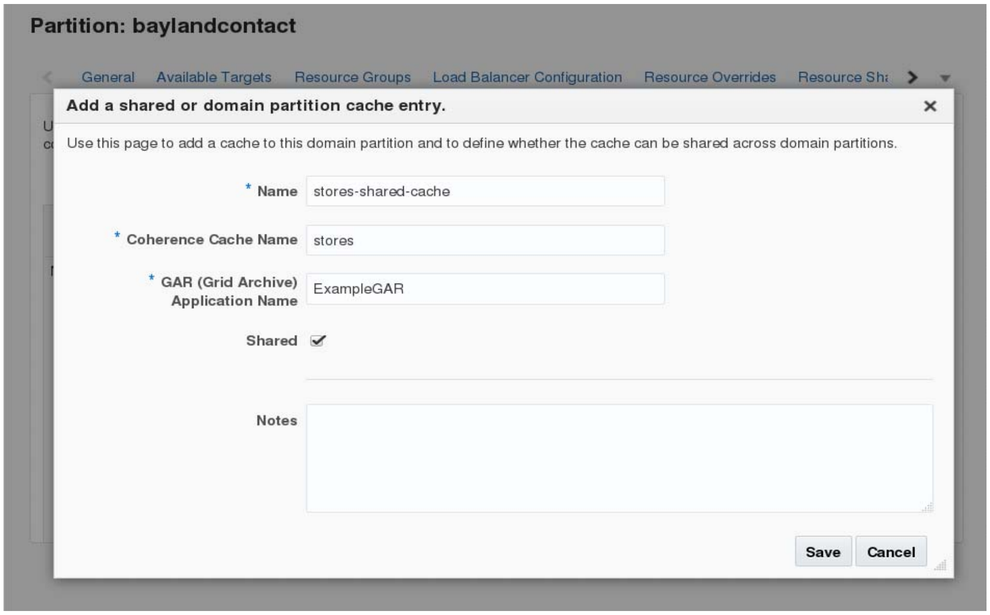
v. Click **Add.**

w. For **Name,** enter `stores-shared-cache.`

x. For **Coherence Cache Name,** enter `stores.`

y. For **GAR (Grid Archive) Application Name,** enter `ExampleGAR.`

z. Click **Shared** to allow this cache to be shared by multiple domain partitions.



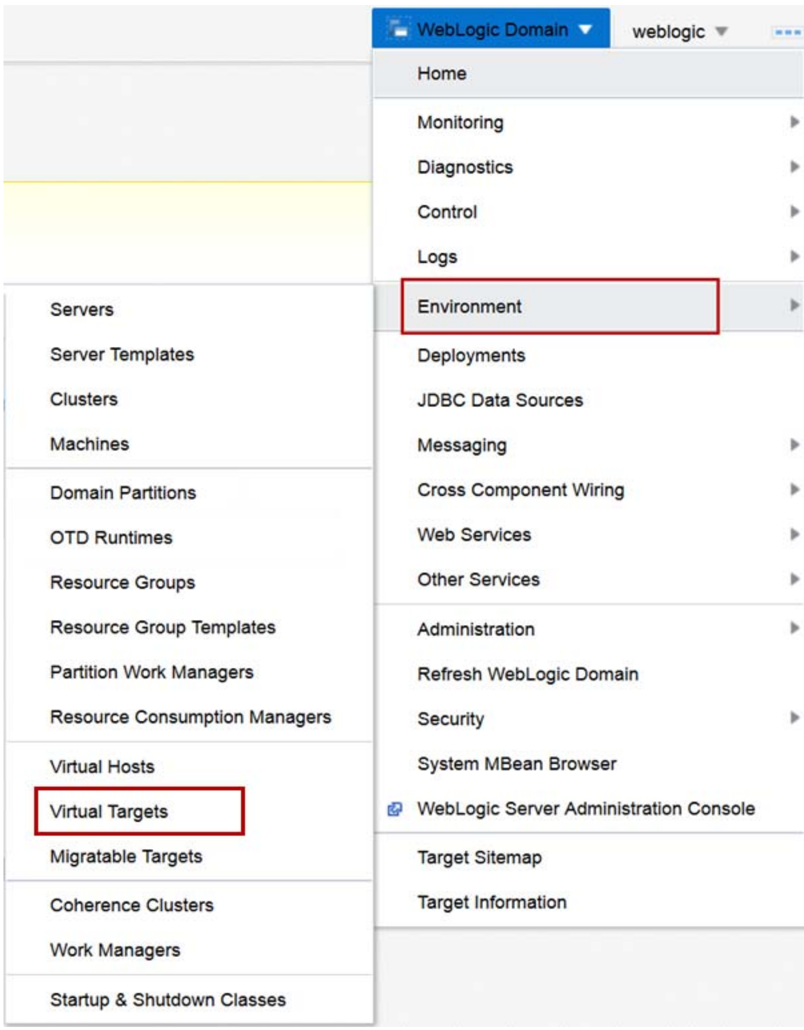
aa. Click **Save.**

The baylandcontact domain partition is created.

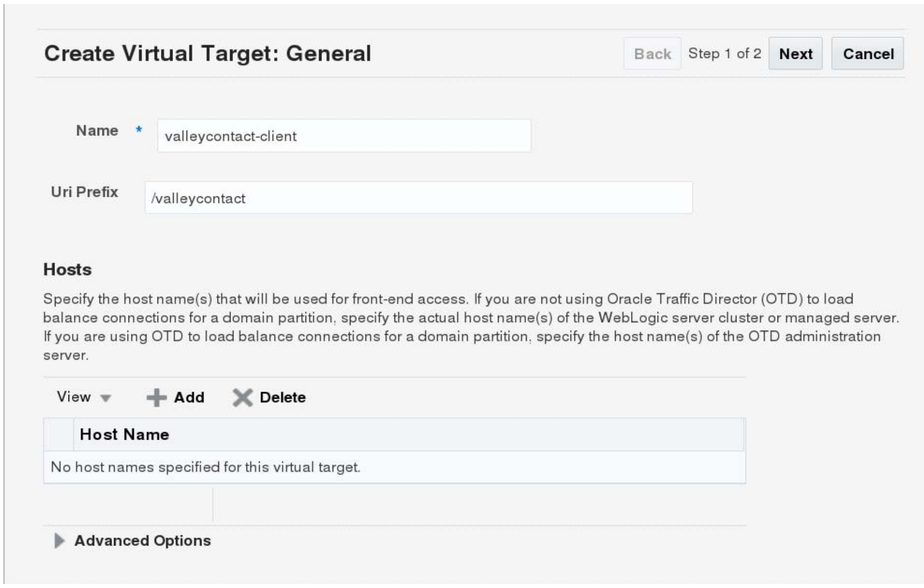
5. Start the baylandcontact domain partition.
   a. In the **baylandcontact** domain partition details page, click **Start Up**.
   b. When the Confirmation dialog box indicates the Start Operation is successful, click **Close.**

      The baylandcontact domain partition successfully starts.
6. Test the baylandcontact domain partition.
   a. In the Browser window, navigate to **http://fmw2:8080/baylandcontact/example-web-app**.

      Note that when you created the domain partition and started it, the Lifecycle Manager performed the necessary configuration for OTD for the baylandcontact deployment.
7. Create the valleycontact-client virtual target.
   a. From the **WebLogic Domain** menu, select **Environment** and click **Virtual Targets.**



   b. Click **Create.**
   c. For **Name,** enter `valleycontact-client`.
   d. For **Uri Prefix,** enter `/valleycontact`.



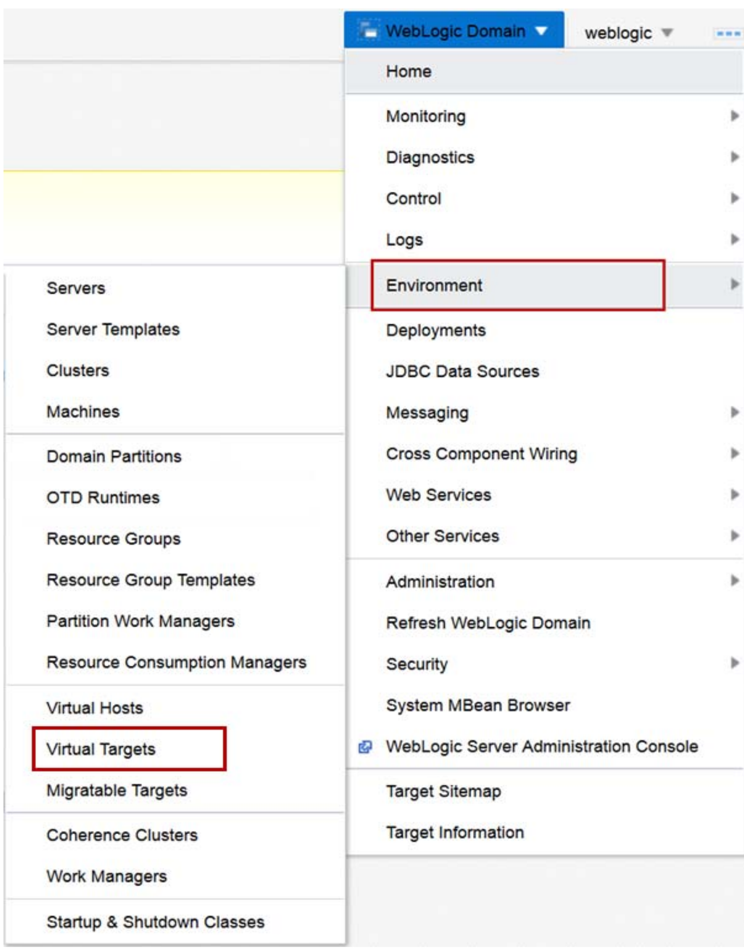
e. Click **Next.**

f. Select **appcluster** from the **Cluster** menu.

g. Click **Create.**

8. Create the valleycontact-storage virtual target.

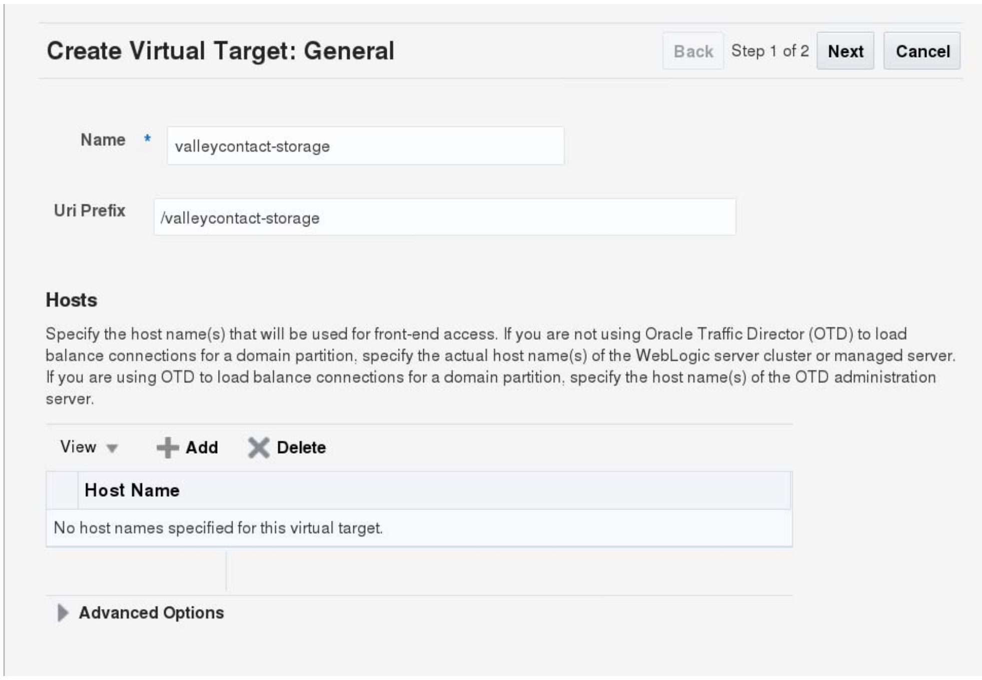
a. From the **WebLogic Domain** menu, select **Environment** and click **Virtual Targets.**



b. Click **Create.**

c. For **Name,** enter `valleycontact-storage`.

d. For **Uri Prefix,** enter `/valleycontact-storage.`

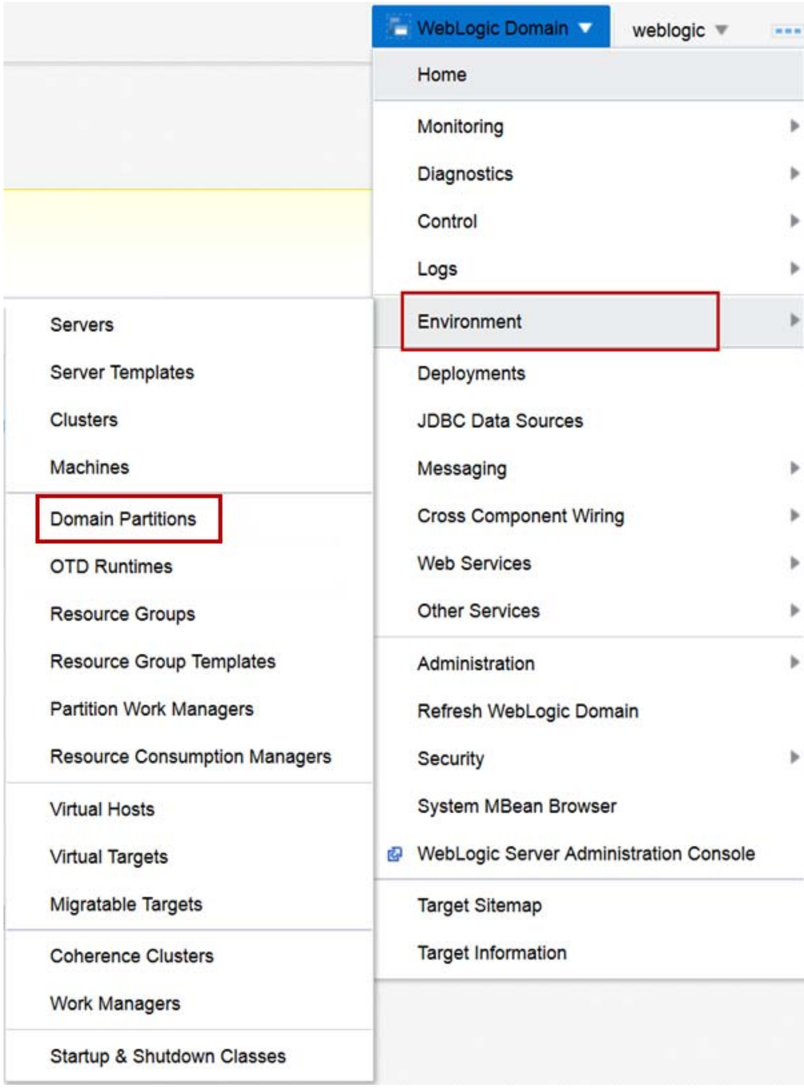


e. Click **Next.**

f. Select **cachecluster** from the **Cluster** menu.

g. Click **Create.**

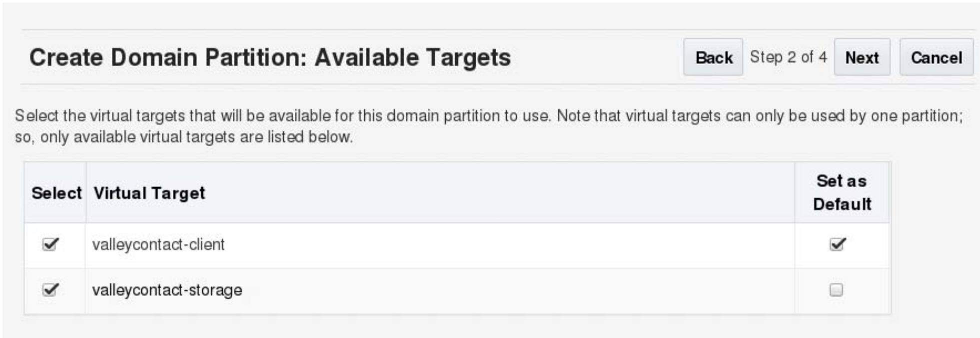
9. Create the valleycontact domain partition.
   a. From the **WebLogic Domain** menu, select **Environment** and then click **Domain Partitions.**



   b. Click **Create.**
   c. For **Name,** enter `valleycontact`.
   d. For **Security Realm,** select valley.
   e. In the **Load Balancer Configuration** section of the page, select **Use OTD for load balancing.**
   f. For **OTD Runtime,** select **medrec.**
   g. Click **Next.**

h. For the Virtual Target **valleycontact-client,** choose both **Select** and **Set as Default.**

i. For the Virtual Target **valleycontact-storage,** choose **Select** and click **Next.**

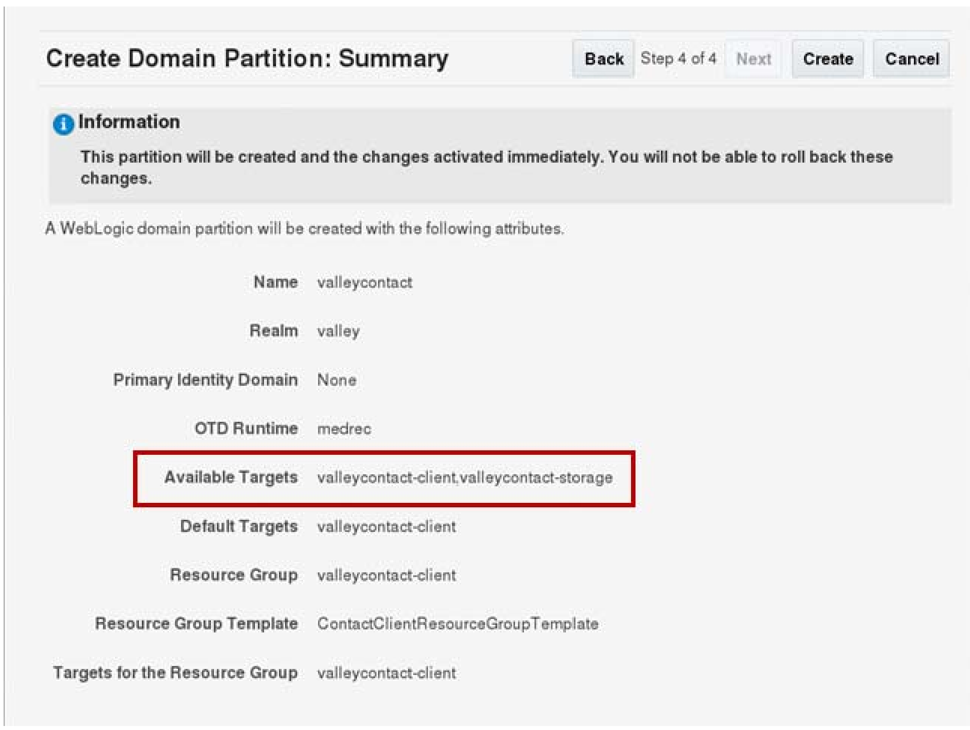


j. For **Resource Group Name,** enter `valleycontact-client`.

k. For **Resource Group Template,** select **ContactClientResourceGroupTemplate.**

l. Move the **valleycontact-client** target to **Selected Targets.**

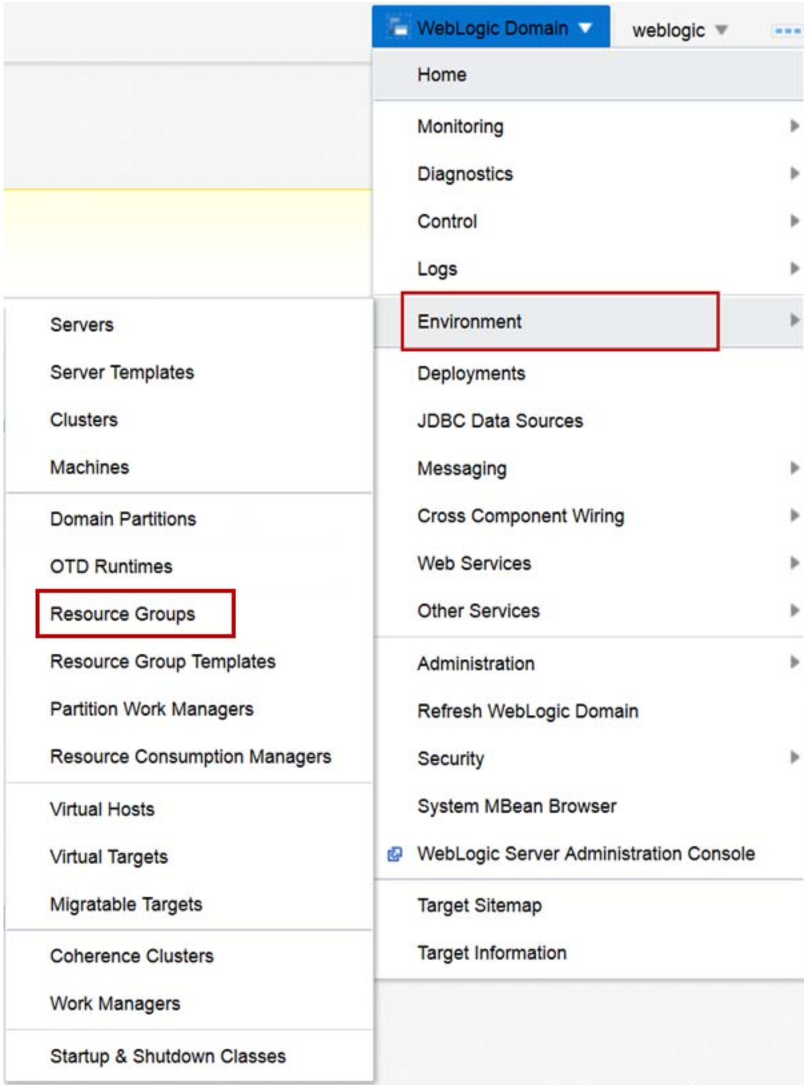
m. Click **Next.**



n. Click **Create.**

o. Click the **valleycontact** domain partition.

p. From the **Domain Partition** menu, select **Administration** and select **Resource Groups.**

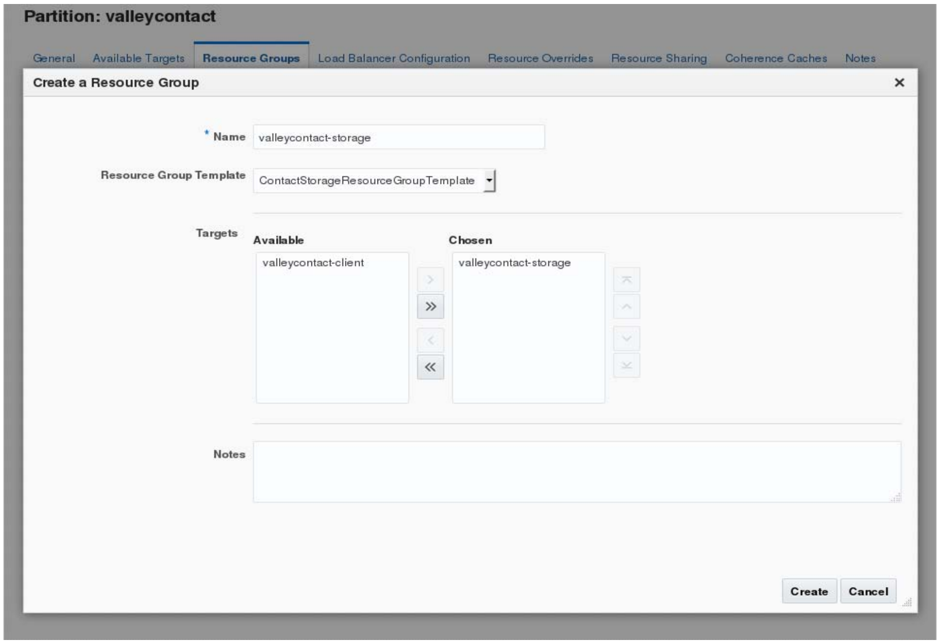


q. Click **Create.**

r. For **Name,** enter `valleycontact-storage.`

s. For **Resource Group Template,** select **ContactStorageResourceGroupTemplate.**

t. For **Targets,** move **valleycontact-storage** from **Available** to **Chosen** and click **Create.**



u. Click the **Coherence Caches** tab.

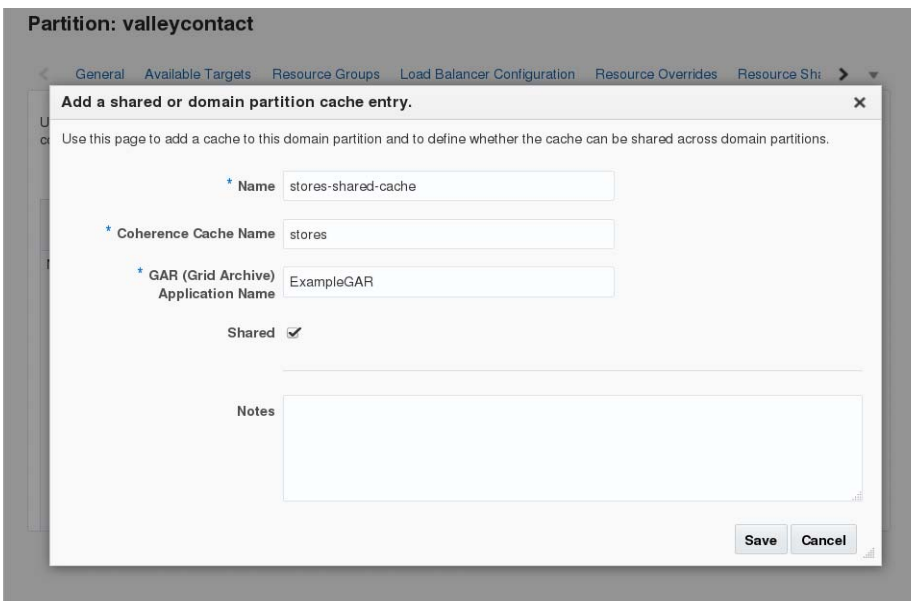
v. Click **Add.**

w. For **Name,** enter `stores-shared-cache`.

x. For **Coherence Cache Name,** enter `stores`.

y. For **GAR (Grid Archive) Application Name,** enter `ExampleGAR`.

z. Click **Shared** to allow this cache to be shared by multiple domain partitions.



aa. Click **Save.**

The valleycontact domain partition is created with two resource groups – the valleycontact-client for client applications and the valleycontact-storage for coherence storage.

10. Start the valleycontact domain partition.
    a. In the **valleycontact** domain partition details page, click **Start Up**.
    b. When the Confirmation dialog box indicates that the Start Operation is successful, click **Close.**

       The valleycontact domain partition successfully starts.
11. Test the valleycontact domain partition.
    a. In the Browser window, navigate to **http://fmw2:8080/valleycontact/example-web-app.**

       Note that when you created the domain partition and started it, the Lifecycle Manager performed the necessary configuration for OTD for the valleycontact deployment.
12. Test the stores shared coherence cache by viewing the contact sample application and creating more store cache entries, which should display in both application domain partitions.
    a. In the browser window, navigate to **http://fmw2:8080/valleycontact/example-web-app/faces/StoreList.jsp.**

       Note that the current cache size is 20.
    b. In another browser window, navigate to **http://fmw2:8080/baylandcontact/example-web-app/faces/StoreList.jsp.**

       Note that the current cache size is 20.
    c. Click **Insert 20 Random Stores.**
    d. Note that the current cache size is 40.
    e. Refresh browser window that is displaying **http://fmw2:8080/baylandcontact/example-web-app/faces/StoreList.jsp.**

       Note that the current cache size is 40. The applications in both baylandcontact and valleycontact are using the stores' shared coherence cache.